MAgent Theory 2.0

A Physicist's Companion

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Chapter 1

What an Agent Carries

A classical statistical description distinguishes a phase-space point from a distribution over phase space. The same distinction begins this theory: $k \in \mathsf{K}$ is a represented possibility, while q is a probability law over such possibilities. The agent carries the law and mechanisms for changing it. Its contents need not be positions and momenta, and the statistical law need not describe an ensemble of physically separate agents.

1.1 Beliefs, models, and predictions

There are two sectors. The belief law q_i describes state contents k. The model law s_i describes model presentations $m \in \mathsf{M}$. Their most general joint description is

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm). \tag{1.1}$$

The notation $q_i(dk \mid m)$ means a conditional law, and $q_i(dk)$ without the condition denotes its marginal. Choosing $\zeta_i = q_i \otimes s_i$ removes dependence between the sectors in that description. It still permits coupled equations for q_i and s_i, just as coupled equations for two coordinates do not require a statistical ensemble over those coordinates.

A model presentation becomes predictive through a normalized evaluator $G_i(m, dk)$. For each m, this assigns a probability law to the state contents. Averaging the evaluator over the model law gives

$$(s_i G_i)(A) = \int G_i(m, A) s_i(dm). \tag{1.2}$$

For example, let k, m be binary, with $s_i(m = 1) = 0.6$, $G_i(1, \{1\}) = 0.9$, and $G_i(0, \{1\}) = 0.2$. The model mixture assigns probability 0.62 to $k = 1$. A present belief $q_i(k = 1) = 0.8$ is a different object. An interaction rule can bring these descriptions into agreement, revise the model law, or change both. Those dynamics must be specified.

An evaluator is a Markov kernel: a function whose output is a probability measure. Ordinary functions are included as the case in which the output measure is concentrated at one value. This language accommodates discrete, continuous, singular, and multimodal laws without selecting a Gaussian family. When differentiating laws later, we will restrict to a regular statistical family or positive finite tables.

1.2 A belief is not a generative model

The evaluator is half of a generative model. The other half is a record likelihood $\ell_i(k, m; do)$, which says how represented contents would show up in the agent's own registers. Together with a prior

$r_i(dm)$ over presentations they form one normalized joint law,

$$P_i(dm, dk, do) = r_i(dm)G_i(m, dk)\ell_i(k, m; do). \tag{1.3}$$

Each presentation m labels a complete forward model: a prior over contents and a likelihood for records. The agent's laws q_i and s_i are something else. They are its current beliefs, the recognition law that approximates the posterior $P_i(dm, dk \mid o)$ once a record o is in hand, and their quality is scored by the variational free energy $D_{\mathrm{KL}}(\zeta_i \| P_i(\cdot \mid o)) - \log P_i(o)$. The analogy is a detector with a calibration curve. The calibration curve is the generative model; the current estimate of the signal is the belief. Two detectors of one design share the calibration curve and can still hold different estimates; two detectors with the same estimate predict the same counts only if their calibrations also agree.

The same distinction organizes populations. Agents whose generative models agree up to a relabeling of presentations and contents form a species. Sharing a model is a fact about mechanisms. Agreement of beliefs is a fact about states, and a record registered by one member changes its beliefs without changing those of a member that did not register it.

Without records the best belief under a model is the prior itself, with the conditional $q_i(dk \mid m)$ equal to $G_i(m, dk)$. Under the product restriction $\zeta_i = q_i \otimes s_i$ this is no longer reachable: the best belief at fixed s_i is the geometric average $q_i \propto \exp \int \log g_i(m, \cdot) s_i(dm)$ of the evaluator densities, not the arithmetic mixture $s_i G_i$. The product description therefore already separates what an agent believes from what its models predict.

1.3 A record has an address inside the agent

Let $x_i \in \mathsf{S}_i$ denote the complete internal state, including its laws, interfaces, and any memory. A record readout is an ordinary measurable function

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i. \tag{1.4}$$

The output alphabet O_i specifies the distinctions the agent can read. A binary register gives $\mathsf{O}_i = \{0, 1\}$; a real-valued comparison can use $\mathbb{R}^r$. The detector analogy is precise at this level: an internal state contains a readable distinction. It does not require an outside observer to inspect it. A detector reading must also not depend on the coordinates in which it is written down. A readout of the first coordinate of a displayed mean would change under a mere rotation of the agent's frame, so readouts are required to be invariant, or to transform by a declared rule, under such passive changes.

Write $x \in \mathsf{S}$ for the complete population configuration. A deterministic interaction $F_{ij} : \mathsf{S} \to \mathsf{S}$ produces the record

$$o_{ij} = \rho_i\big(\mathrm{pr}_i\, F_{ij}(x)\big), \tag{1.5}$$

where pr_i selects the receiver's state. For a stochastic transition $\Gamma_{ij}(x, dx')$, the probability of a record in A is

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\big(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\big). \tag{1.6}$$

This is a pushforward: collect the probability of every next state that displays an outcome in A. The transition specifies what happens; the readout specifies what is accessible. A persistent record requires that some later readout still recover the distinction. A transient comparison and a memory trace therefore have different dynamical requirements.

A likelihood on represented contents has another role. For instance, the expression $o = A_i k_i - B_{ij} k_j + \epsilon$ gives a comparison value once the contents and noise are given. Their joint law predicts possible comparisons. A registration rule must still place the outcome in an internal state and specify the response to it.

1.4 Which probability space is being used?

The configuration x contains the agents' laws. A tuple of represented contents $(k_i, m_i)_i$ belongs to a different space. A probability law over possible configurations is a third construction, useful when a preparation or stochastic dynamics has been declared. Deterministic motion of x requires neither sampling the contents nor drawing an ensemble of configurations at each step.

This distinction also clarifies correlations. Two independent fair bits have probability $1/4$ for each pair. Condition on a shared record that the bits agree. Their marginals remain fair, but the conditional joint law has probability $1/2$ on each of $(0, 0)$ and $(1, 1)$. A calculation of the next joint comparison needs this dependence; the marginals do not contain it. A coupled product-family evolution remains a possible model, but it will not represent that exact conditional joint law. It does not even keep the fair marginals. The best product description of the hard agreement record is a point mass on $(0, 0)$ or on $(1, 1)$. For a soft record $\propto e^{J\sigma_1\sigma_2}$ in spin variables, the symmetric product description is the mean-field magnet $m = \tanh(Jm)$: unmagnetized for $J \leq 1$ and spontaneously magnetized above $J = 1$, although the exact posterior keeps zero magnetization for every J. Whether a correlation is needed is decided by the response being predicted.

A third kind of law joins the contents of several agents, or of one agent at several contexts, into one joint law of fields. Only such a law can predict a record that compares the same sample along two routes. A comparison of the two route laws cannot see a loop that leaves the transported law unchanged, such as a rotation of an isotropic Gaussian, whereas the joint law of a sample and its image along the loop determines the loop map.

The laws can be attached to local domains of a context base $\mathcal{C}$. This is the section picture: a context selects the law applicable there, and a frame specifies its coordinates. Most interactions below occur at a common context. Moving between contexts introduces the connection problem in chapter 3.

Chapter 2

Frames, Messages, and Interactions

An agent's probability law and the coordinates used to describe that law are different objects. Communication introduces a third object: a rule for translating, filtering, or modifying what another agent supplies. Keeping these distinctions explicit prevents a coordinate change from being mistaken for an interaction, or a selective message from being mistaken for an invertible transport.

2.1 Reading a link from source to receiver

There are two sectors. Write $\nu_i^q = q_i$ for the belief marginal and $\nu_i^s = s_i$ for the model marginal. In a finite-dimensional realization, a sector $a \in \{q, s\}$ has a reference vector space V_a, an agent space E_i^a, and an invertible presentation

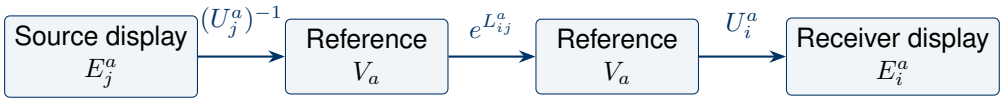
$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{\nu}_i^a = (U_i^a)_{\#}^{-1} \nu_i^a. \tag{2.1}$$

The pushforward notation expresses the represented possibilities in reference coordinates. The space E_i^a is agent i's displayed copy of V_a: at one context all agents' laws lie in one fiber of the statistical bundle, and the presentations are frames used to display them. The two sectors carry two representations of one frame group, with different dimensions, laws, and interaction rules: a belief vector rotates by $\varrho_q(g)$ and a model presentation by $\varrho_s(g)$ under the same change of frame g. Think of one set of axes carried by the agent, with beliefs and models as two different kinds of object written on those axes. Each edge can carry a belief link and a model link, both valued in the same group. When they agree the edge transports beliefs and models the same way; when they differ, the mismatch $\tilde{\theta}_{ij}\theta_{ij}^{-1}$ is a property of the edge that no change of frame removes. The collective geometry of chapter 3 uses this single group.

A law-dependent reference generator and its displayed link are

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a e^{L_{ij}^a} (U_j^a)^{-1}. \tag{2.2}$$

The index order is receiver first, source second. Reading from right to left, the source is expressed in reference coordinates, the generator acts there, and the result is expressed at the receiver. The function $\mathcal{L}_{ij}^a$ is a constitutive choice. The exponential parameterization does not derive it or equate the belief and model generators. It is also only one kind of interaction. The general interaction is a normalized Markov channel, possibly noisy or information-discarding; the exponential link is the special deterministic channel $y \mapsto \Theta_{ij}^a y$. Its values e^L lie in $\exp(\mathfrak{gl}(V_a))$, which in dimension at least two misses some determinant-one matrices, such as $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$, and every reflection [12].



$$\Theta_{ij}^a = U_i^a e^{L_{ij}^a} (U_j^a)^{-1}$$

$a = q$ or s: two separately specified interactions

Figure 2.1: The source presentation is undone by $(U_j^a)^{-1}$, the interaction $e^{L_{ij}^a}$ acts within the reference space, and U_i^a presents the result at the receiver. Endpoint frames and the intervening interaction have separate roles.

2.2 Changing coordinates

For a passive change $y_i' = g_i^a y_i$, both the law and its presentation change: $\nu_i^{a\prime} = (g_i^a)_\# \nu_i^a$ and $U_i^{a\prime} = g_i^a U_i^a$. The reference law stays fixed, giving

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.3}$$

This endpoint covariance is familiar from gauge link variables [48], here in display coordinates. It holds for every generator rule, because the reference laws do not change, and so it constrains nothing: the frames act as a dressing field [15]. Every interaction cost can be computed entirely in reference coordinates. Metrics and additional model data must also be re-expressed when forming scalar predictions. Covariance alone supplies no physical symmetry or evolution law.

Changing the common reference presentation is a separate operation. Under $U_i^a \mapsto U_i^a h_a$, the reference laws become $(h_a^{-1})_\# \bar{\nu}_i^a$. Reference consistency requires the constitutive rule, including any interface data, to give $L_{ij}^{a\prime} = h_a^{-1} L_{ij}^a h_a$. The identity $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$ then leaves the displayed link unchanged. This is the substantive symmetry requirement. It is not automatic, and the main text imposes it as an explicit standing hypothesis wherever a conclusion should not depend on the choice of reference frame.

2.3 Selective interfaces and an explicit pair

Sending and receiving can be represented by

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a, \qquad L_{ij}^a = O_i^a T_j^a. \tag{2.4}$$

The sender T_j^a extracts a message; the receiver O_i^a inserts its effect. A direct content map $O_i^a T_j^a$ can lose dimensions. Its exponential is always invertible and acts as identity on $\ker T_j^a$. Thus exponentiating a low-rank generator retains unread directions, and the link compares complete laws. Sending only the extracted message is a different, information-discarding operation. General normalized Markov kernels allow communication without any linear realization.

One explicit law-dependent choice uses finite second moments and positive-definite covariance. Let μ_i^a, Σ_i^a be the reference mean and covariance. On nonself edges choose a scalar λ_{ij}^a and set

$$L_{ij}^a = \lambda_{ij}^a \mu_i^a (\mu_j^a)^T (\Sigma_j^a)^{-1}. \tag{2.5}$$

The row $(\mu_j^a)^T(\Sigma_j^a)^{-1}$ sends a scalar message; the column $\lambda_{ij}^a\mu_i^a$ receives it. Under a reference change, the mean transforms by h_a^{-1} and covariance by congruence. Their product consequently transforms by conjugation. These assumptions concern moments, not Gaussian shape. For these links the primitive self link is set to the identity separately. In general the self channel is constitutive data, and some later constructions use a nonidentity one.

For a concrete non-Gaussian example, let Z have independent centered Laplace coordinates of variance one in two dimensions. Give two agents the laws of $e_1 + Z$ and $e_2 + Z$, with identity frames and unit coefficients. Then $\Sigma_1 = \Sigma_2 = I$, $L_{12} = E_{12}$, and $L_{21} = E_{21}$. Since each generator squares to zero,

$$\Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.6}$$

Here E_{ab} has its only nonzero entry, one, in row a, column b. The directed communication loop is nontrivial. It establishes neither a connection over contexts nor a physical curvature. Writing $\Theta_{12} = \Omega_{12}N_2^{1/2}$ with $N_2 = \Theta_{21}\Theta_{12}$, the reciprocal part Ω_{12} is here a rotation with $\Omega_{12}\Omega_{21} = I$, so the original two-cycle is carried entirely by the backtrack factor.

Chapter 16 turns smooth contextual fields of such laws, interfaces, and links into displacement transport under stated rank or identifiability assumptions. The additional comparison rule and contextual derivatives distinguish that construction from a link at one context.

A further limit concerns a comparison that reads a loop map H only through the transported law $H_{\#}\nu$. If H fixes the law, as a rotation fixes a centered isotropic Gaussian, that comparison cannot register it. The joint law of a sample Y and its image HY does determine H almost everywhere. A record comparing the same sample along two routes can therefore resolve distinctions that the marginal laws omit. Grounded messages offer another route: sufficiently informative referents and codes can identify each edge dictionary and hence its loop product. A covariance-dressed matrix defect can also penalize every nonidentity loop, including those that fix the law. Its use as a prior is an additional modeling choice.

Chapter 3

From Statistical Change to Base Geometry

The agent graph and the context base describe different directions of comparison. An edge $j \to i$ compares agents at one context. A path in the base $\mathcal{C}$ compares states at successive contexts. A product of communication links can be nontrivial without defining curvature on $\mathcal{C}$. Conversely, one agent can carry a curved connection over contexts; collective averaging is not required. The context-local and formation flows use the connection to describe evolved or supplied sections. Chapter 15 adds a contextual energy and shows how its auxiliary alignment enters the law velocity.

3.1 Relating neighboring contexts

To describe such a connection, suppress the sector label and fix one agent. On a coordinate neighborhood, take a twice continuously differentiable presentation $U(c)$ and a supplied, continuously differentiable matrix-valued one-form $a = a_\mu(c)dc^\mu$. The calculation here takes a as given. Chapter 16 constructs it from regular contextual laws, interfaces, and links after the comparison rule is declared. Declare directional links whose first-order behavior is

$$\Theta(c + \delta c, c) = U(c + \delta c)e^{-a_\mu(c)\delta c^\mu}U(c)^{-1} + O(|\delta c|^2). \tag{3.1}$$

The displacement enters linearly at first order, and zero displacement gives identity. These are additional properties beyond having a finite interaction matrix at each context. For a principal G-connection, the generators must belong to the represented Lie algebra; unrestricted matrices instead define general linear bundle connections.

Using $D = d + B$, expansion identifies

$$B = UaU^{-1} - dUU^{-1}. \tag{3.2}$$

The first term carries the declared directional interaction into displayed coordinates. The second compensates for the change of presentation along the path. Their relationship is exactly the operator identity

$$d + B = U(d + a)U^{-1}. \tag{3.3}$$

Indeed, acting on a vector field f gives $Ud(U^{-1}f) = df - dUU^{-1}f$; the remaining term is $UaU^{-1}f$. Squaring the operator therefore gives

$$F = dB + B \wedge B = U(da + a \wedge a)U^{-1}. \tag{3.4}$$

This is the mathematical connection to gauge theory, with the sign convention fixed by the displayed covariant derivative [47].

3.2 Curvature and paths

Curvature need not come from spatially varying coefficients. On a two-dimensional context patch choose $U = I$, $a_x = E_{12}$, and $a_y = E_{21}$. Both directional generators have rank one, but

$$F_{xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.5}$$

The matrix E_{ab} has one unit entry in position (a, b). Applying the two infinitesimal operations in opposite orders gives different results. This example supplies directional data; it does not derive them from the agents' laws. A presentation change alone, $B = -dUU^{-1}$, instead has zero curvature.

Finite path transport solves

$$\frac{dT}{dt} = -B(\dot{\gamma}(t))T(t), \qquad T(0) = I. \tag{3.6}$$

Its solution is a path-ordered exponential. Concatenating paths composes their transports; reversing the same path gives the inverse. Two independently specified attending edges can have unrelated generators, so their opposite labels do not impose this inverse relation. The distinction is necessary when borrowing Wilson-loop language from gauge theory [48].

3.3 Local descriptions and global compatibility

Endpoint covariance requires

$$\Theta'(c_2, c_1) = g(c_2)\Theta(c_2, c_1)g(c_1)^{-1}. \tag{3.7}$$

Differentiating the endpoints produces the affine law

$$B' = gBg^{-1} - dgg^{-1}. \tag{3.8}$$

Likewise, keeping the intrinsic transport fixed under $U' = Uh$ requires $a' = h^{-1}ah + h^{-1}dh$. A directional receiving–sending product that transforms only by conjugation is a tensorial one-form. It can supply the difference between two connections, but does not by itself supply the full affine transformation law.

Global compatibility adds another requirement. If patch coordinates obey $v_\alpha = g_{\alpha\beta}v_\beta$, then $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps, and local connection forms must satisfy the affine law there. A nontrivial bundle may have no single global presentation. Even a trivial bundle can carry flat transport with global memory: on a circle, the scalar connection $B = bd\theta$ has $F = 0$ but holonomy $e^{-2\pi b}$ around one turn. For real $b \neq 0$, this differs from identity.

3.4 Distinguishability supplies a fiber metric

A connection tells us how to compare laws at neighboring contexts. To measure the change remaining after that comparison, use the Fisher metric on a declared regular statistical family. Let ζ_θ be a family of probability laws, with a centered square-integrable score ℓ_v in parameter direction v. In a dominated differentiable realization,

$$\ell_v = \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} \log p_{\theta+\epsilon v}, \qquad g^F_\zeta(v,w) = \mathbb{E}_\zeta[\ell_v \ell_w]. \tag{3.9}$$

Quadratic-mean differentiability supplies a more general score definition. Positive definiteness requires that the admitted tangent directions be statistically distinguishable. These hypotheses apply to the chosen family, not to every element of the measurable space of laws. The laws of the family may have different null sets [2]; a finite-dimensional family is still dominated by a countable mixture of its members. Up to a constant factor, the Fisher metric is the only metric on finite statistical models invariant under Markov congruent embeddings [9], which is the reason for choosing it.

The relevant law is the full $\zeta(dk, dm) = q(dk \mid m)s(dm)$. Its log density has a marginal part and a conditional part. The conditional score averages to zero for each m, so the cross term disappears:

$$g^F_\zeta = g^F_s + \int g^F_{q(\cdot|m)} s(dm). \tag{3.10}$$

For the product family this is $g^F_q + g^F_s$. For a general joint family the second term is conditional geometry. It cannot be replaced by the geometry of the state marginal.

For example, take $k, m \in \{-1, 1\}$ and

$$\zeta_\rho(k,m) = \frac{1+\rho km}{4}, \qquad |\rho| < 1. \tag{3.11}$$

Both marginals are uniform for every ρ. The score is $km/(1+\rho km)$, giving

$$g^F_\zeta(\partial_\rho, \partial_\rho) = \frac{1}{2}\left(\frac{1}{1+\rho} + \frac{1}{1-\rho}\right) = \frac{1}{1-\rho^2}. \tag{3.12}$$

The law changes through dependence even though neither marginal changes. This is why the statistical metric must follow the actual probabilistic description.

3.5 Pulling distinguishability onto the context base

Let $\zeta(c)$ be an admitted smooth section and let ω be its connection. Subtract the horizontal motion from the derivative of the section:

$$D^\omega \zeta = \mathrm{ver}^\omega\, T\zeta, \qquad h^\omega_\zeta(X,Y) = g^F_\zeta(D^\omega \zeta X, D^\omega \zeta Y). \tag{3.13}$$

The tangent vectors X, Y now lie in the context base. The tensor h^ω_ζ measures how much the represented law changes when the context changes, after the declared transport accounts for the

change of frame. In a local chart it uses $D^A\zeta = d\zeta + J_\zeta A$, where $J_\zeta\xi$ is the infinitesimal action of the Lie algebra element ξ on the law.

Under second-order relative-entropy regularity, move a small distance ϵX, transport the neighboring law back, and compare it with the original:

$$D_{\mathrm{KL}}\left(\zeta(c)\middle\|T^{\omega}_{c+\epsilon X\to c\#}\zeta(c+\epsilon X)\right) = \frac{\epsilon^2}{2}h^{\omega}_{\zeta}(X,X) + o(\epsilon^2). \tag{3.14}$$

The familiar local quadratic behavior of KL therefore measures contextual distinguishability. The transport is part of the comparison.

For a simple case with identity transport, let the law at real context c be $\mathrm{Ber}(p(c))$, with $0 < p(c) < 1$. Direct differentiation of its two log probabilities gives

$$h = \frac{p'(c)^2}{p(c)(1-p(c))}dc^2. \tag{3.15}$$

The same change of success probability is more distinguishable near a deterministic outcome. If $p'(c) = 0$, this family records no first-order distinction in that context direction. The example uses no Gaussian assumption.

A passive gauge change transforms both the section and connection, leaving the tensor unchanged. Choosing a different connection changes what is subtracted. If A changes by a Lie-algebra-valued one-form α, the statistical derivative changes by $J_\zeta\alpha$; its squared Fisher norm acquires both cross terms and the squared norm of that added vector. Gauge invariance is thus a statement about presentation, not a claim that all transport rules give the same geometry.

Positive-definite fiber geometry makes $\ker h^{\omega}_{\zeta} = \ker D^{\omega}\zeta$. Constant rank gives a metric on the vector spaces obtained by quotienting these null directions. To obtain a lower-dimensional context manifold, the null directions must also fit together into leaves and the metric must descend along them. Both requirements are measured by the curvature F of the connection: for null directions X, Y, the derivative of the section along $[X, Y]$ is $-J_\zeta F(X, Y)$, so the null directions integrate exactly when $J_\zeta F$ vanishes on them, and a similar curvature expression decides whether the metric descends. A flat connection always passes both tests. The supplied connection $dz - xdy$ for a translation family illustrates failure: its square has rank one everywhere, but its kernel is not integrable because $(dz - xdy) \wedge d(dz - xdy)$ is nonzero. Counting statistical directions does not by itself construct a quotient space of contexts.

3.6 Choosing a collective connection

Suppose several laws have been placed in compatible fibers by declared comparison maps. Give their product tangent space the positive metric

$$\mathsf{G} = \sum_i w_i g^F_{p_i}, \qquad w_i > 0. \tag{3.16}$$

The weights specify how the individual statistical changes are compared. A common infinitesimal frame motion ξ changes the tuple by $J_{\boldsymbol{p}}\xi$. For each contextual direction X, select the correction a

that minimizes

$$\|d\boldsymbol{p}(X) + J_{\boldsymbol{p}}a\|^2_{\mathsf{G}}. \tag{3.17}$$

Differentiation with respect to a gives the normal equation

$$\mathsf{G}(JA^F(X), J\xi) = -\mathsf{G}(d\boldsymbol{p}(X), J\xi) \quad \text{for every } \xi. \tag{3.18}$$

The residual is orthogonal to all common frame motions. This is the mechanical connection of Smale [44], used to separate rotation from shape change in molecules [19] and the n-body problem [32], with the Fisher metric as the kinetic metric. The residual is the shape change: under a free proper action, $h^{\min}$ is the pullback of the quotient metric on the space of orbits [38]. Its null directions are those along which only the orbit moves, and they always integrate.

For an explicit calculation, take two unit-variance normal location laws with means $\mu_1 = m + d$ and $\mu_2 = m - d$, and give them equal weight. Common translation acts as $Ja = (a, a)$. The cost is

$$(dm + dd + a)^2 + (dm - dd + a)^2 = 2(dm + a)^2 + 2(dd)^2. \tag{3.19}$$

It is minimized at $A^F = -dm$, leaving

$$h^{\min} = 2(dd)^2. \tag{3.20}$$

Common location change is removed while change in their relative separation remains. This calculation is a realization of the general construction; the construction itself applies to any admitted smooth isometric group action.

If some nonzero ξ moves none of the laws, then $J\xi = 0$ and the normal equation cannot determine that connection component. At constant orbit rank it determines a smooth affine class modulo this infinitesimal stabilizer. When the infinitesimal stabilizer is zero the solution is unique and obeys the affine patch law. A free proper action additionally gives the usual smooth statistical orbit space. These conclusions concern different hypotheses [31]. Agreement alone need not change the stabilizer: several planar normal laws with the same nonzero mean still have a free common rotation action. The obstruction occurs where infinitesimal isotropy increases. For one unit-covariance normal law in the plane with mean (ϵ, c) under rotations, the selected connection is $-\epsilon/(c^2 + \epsilon^2)dc$; its integral is $-\pi$ for every $\epsilon > 0$, so it concentrates at the rotation-invariant point as $\epsilon \to 0$. The minimized tensor stays bounded while the connection concentrates. This one-dimensional example makes no curvature claim.

Orthogonality also measures the cost of choosing another connection:

$$h^A = h^{\min} + \mathfrak{k}(A - A^F, A - A^F), \qquad \mathfrak{k}(\xi, \eta) = \mathsf{G}(J\xi, J\eta). \tag{3.21}$$

The minimized tensor is positive semidefinite, and its rank is at most the number of shape directions. For a single law whose group acts transitively, such as one Gaussian under the affine group, there are none; a single agent's contextual tensor then records only the supplied connection. Whether a collective tensor has enough independent directions, and whether inference preserves those directions, depends on the actual section and its dynamics. The least-squares rule alone supplies neither persistence nor a physical interpretation of the contextual base.

3.7 What compression does to this geometry

A fixed normalized channel K transforms a family P_θ into $P_\theta K$. The output score is the input score averaged over possibilities compatible with the output. The law of total covariance gives

$$I(P_\theta) - I(P_\theta K) = \mathbb{E}_\theta \operatorname{Cov}_\theta(\ell_\theta(X) \mid Y). \tag{3.22}$$

The missing Fisher information is exactly the uncertainty remaining about the score. A zero value recovers that local score, not necessarily the full family of laws [40]. If the channel itself varies with θ, its own score must be added before conditioning. The output can then acquire information through the channel parameter, so contraction relative to the input alone need not hold.

Even a fixed equivariant channel need not preserve the collectively selected connection. In the two-location example, take $(\mu_1, \mu_2) = (x, -x)$. The fine metric is $\operatorname{diag}(1, 1)$ and its collective connection is zero. Add independent unit-variance normal noise to the first observation. The first variance becomes two, so the output Fisher metric is $\operatorname{diag}(1/2, 1)$. Its connection minimizes

$$\frac{1}{2}(1 + a)^2 + (-1 + a)^2, \tag{3.23}$$

whose derivative is $3a - 1$. The new minimum is $a = 1/3$. The channel has reduced distinguishability and commutes with translation, yet the weighting of shape change against translation has changed.

In general, let a smooth equivariant retention R map the fine family to the coarse family, with metrics g, g_c. Set $\Delta = g - R^* g_c$, and let r be the fine horizontal residual. Because $dRJ = J^c$, the fine connection also minimizes the coarse cost exactly when

$$\Delta(r(X), J\xi) = 0 \quad \text{for every } X, \xi. \tag{3.24}$$

This follows by inserting the fine connection in the coarse normal equation and using fine orthogonality. It is a condition on the mixed shape–orbit part of information loss. Preserving all orbit scores is sufficient for a fixed channel, but contraction alone is not. The minimized tensor itself always contracts under a fixed channel, with an exact account of the loss:

$$h^{\min} - h_c^{\min} = \Delta(r, r) + \mathfrak{k}_c(A^F - A_c^F, A^F - A_c^F). \tag{3.25}$$

The first term is information discarded along the shape residual; the second is the shift of the selected connection, weighted by the coarse locked inertia. In the example above, $2 = \frac{4}{3} + \frac{1}{2} + \frac{1}{6}$.

For independently chosen fine and coarse connections there is another contribution. A bundle map may send a fine horizontal vector to a vector that is not coarse-horizontal. If a denotes that vertical difference and $u = D^\omega \zeta$, the coarse derivative is $T^V \Psi u + a$. Expanding its squared norm gives the fine information loss minus two cross terms and minus $\|a\|^2$. Zero horizontal mismatch is sufficient for contraction of the base tensor; a nonzero mismatch can still be dominated by information loss. The actual derivative and both connections must therefore be specified.

3.8 Geometric structure and physical interpretation

The pullback construction is already a mathematical geometry of contextual statistical change. Its physical interpretation requires a model of the contexts, a mechanism selecting the comparison data, and observable responses to transport. Inherited evolution and inherited responses impose further conditions on a parent; connection preservation is an additional requirement.

Fisher geometry is positive semidefinite. An indefinite metric can be introduced through a separate construction. For example, a Hermitian 2×2 matrix $H = tI + \boldsymbol{x} \cdot \boldsymbol{\sigma}$ has determinant $t^2 - |\boldsymbol{x}|^2$, where $\boldsymbol{\sigma}$ are the Pauli matrices. Pulling its determinant bilinear form back by an invertible soldering field, and dividing by a positive scale that transforms as $|\det g|^2$, gives a Lorentzian metric invariant under the full complex frame group. The natural scale is the determinant of the aggregate second moment; the metric is then one half of a complex-Gaussian Fisher metric minus the square of the entropy rate, so time-like directions are those in which the entropy changes faster than the law's Fisher speed. Without a scale the form is invariant only under frames with $|\det g| = 1$. A circle extension then requires a circle connection and a positive radius. Those choices do not follow from Fisher positivity, and their existence does not establish spacetime dynamics.

Chapter 4

Free Energies, Coupled Relaxation, and Source Selection

The dynamical variables are probability laws. Moving an agent therefore means changing its expectations, uncertainties, and dependencies, rather than moving a point through its sample space. The physical analogue developed here is overdamped motion on a probability manifold: an objective supplies the driving covector, and the Fisher metric converts that covector into a velocity. Neither ingredient determines the other.

4.1 Three objectives

Before any equation, one must say what is being minimized, and for interacting agents there are three different answers. In the global picture all agents are parts of one normalized joint law over their contents and records, as the spins of one lattice share one Boltzmann weight. Each agent's own free energy is then that single functional seen as a function of its own factor, peers act on it only through expectations of their contents, and every local update lowers the same total: coordinate descent, like a sequence of single-spin heat-bath moves.

In the per-agent picture each agent has its own generative model, in which its own evaluator and the laws its peers communicate compete as sources of prior. The peers' laws are data for the agent, like an external field it does not control. Each agent lowers its own free energy, but the free energies of the agents that listen to it change through terms it does not control. Such non-reciprocal forces need have no global Lyapunov function: on a directed ring of three Bernoulli agents, each pulled toward its successor, the linearized motion spirals, with complex eigenvalues that no gradient flow can have.

The third object, developed in this chapter, is the reciprocal law potential. It adds the divergences between agent laws directly, so each edge pulls its receiver toward the source and also pulls the source toward its receivers, like a spring that acts on both ends. All agents then descend one shared function, and the dynamics dissipates that function and cannot spiral into an equilibrium. The price is that this function is not the free energy of any fixed normalized model in which the agent laws are the beliefs, and records enter it only as separate assimilation events. The constructions of the later chapters use this third object; their conclusions are not claims about the other two.

4.2 The potential and its source rows

At a fixed context, let q_i describe agent i's represented state k, and let s_i describe its model presentation m. For this construction their joint recognition law is the explicit product $q_i \otimes s_i$.

Choose a model reference $r_i(dm)$ and a normalized evaluator $G_i(m, dk)$. With the reference joint $R_i(dk, dm) = r_i(dm)G_i(m, dk)$, their local contribution is

$$\mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = \sum_i \left[D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot)) s_i(dm) \right]. \tag{4.1}$$

The equality separates the model marginal from the conditional state law. Thus a product recognition family still has coupled dynamics: changing s_i changes the evaluator against which q_i relaxes, and changing q_i changes the model costs.

The channels of chapter 2 send peer laws into comparable spaces. Write $\nu^q_{ij} = q_j \mathcal{K}^q_{ij}[\mathbf{q}]$ and $D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij})$, where $\mathbf{q}$ collects the belief laws. Define ν^s_{ij} and D^s_{ij} similarly for model laws. With normalized source rows β_i, γ_i, positive reference rows π^q_i, π^s_i, and $T_q, T_s > 0$, the potential is

$$\begin{aligned} \mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_j \beta_{ij} D^q_{ij} + T_q D_{\mathrm{KL}}(\beta_i \| \pi^q_i) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ij} D^s_{ij} + T_s D_{\mathrm{KL}}(\gamma_i \| \pi^s_i) \right]. \end{aligned} \tag{4.2}$$

Each sum uses its declared source set. The row penalty balances preference for compatible peers against departure from the reference routing.

Hold the laws fixed. For either sector, minimize $f(\eta) = \sum_j \eta_j D_j + T D_{\mathrm{KL}}(\eta \| \pi)$ subject to $\sum_j \eta_j = 1$. Stationarity gives $D_j + T[\log(\eta_j/\pi_j) + 1] = \lambda$, hence

$$\eta^*_j = \frac{\pi_j e^{-D_j/T}}{Z}, \qquad Z = \sum_j \pi_j e^{-D_j/T}, \qquad f(\eta) = -T \log Z + T D_{\mathrm{KL}}(\eta \| \eta^*). \tag{4.3}$$

The last identity proves that this stationary row is the unique minimum. Eliminating both sets of rows gives $\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z^q_i - T_s \sum_i \log Z^s_i$. At fixed temperatures and reference rows,

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta^*_{ij} dD^q_{ij} + \sum_{ij} \gamma^*_{ij} dD^s_{ij}. \tag{4.4}$$

This follows by differentiating each log partition sum. Optimized row derivatives disappear; derivatives inside their costs remain.

4.3 Every interaction role contributes to motion

For normalized densities a, b, variation of $\int a \log(a/b)$ gives

$$dD_{\mathrm{KL}}(a \| b) = \int \log(a/b) da - \int (a/b) db, \qquad d\nu^q_{ij} = (dq_j)\mathcal{K}^q_{ij} + q_j D\mathcal{K}^q_{ij}(d\mathbf{q}). \tag{4.5}$$

Here da, db denote mass-preserving variations, and differentiation assumes a regular dominated family. An agent moves through its receiver role, its source role, and any dependence of the

channels on its law. Directed routing does not remove incoming contributions to source motion. The source is pulled toward the pooled law of the agents that listen to it, so a source more confident than its listeners' spread is widened; a communicating population realizes this only if listeners send a reverse message back to their sources. For an identity self-channel, the two laws remain identical and their divergence stays zero. Holding an old self snapshot fixed instead creates a different potential after the current law changes.

For positive finite probability tables, collect all dynamical tables as u^a, including the source rows when they are not eliminated. The Fisher metric and its unit-mobility descent are

$$\begin{aligned} g_u(v,v) &= \sum_{a,z} \frac{(v_z^a)^2}{u_z^a}, \\ \dot{u}_z^a &= -u_z^a(h_z^a - \mathbb{E}_{u^a} h^a), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}, \\ \dot{\mathcal{E}} &= -\sum_a \mathrm{Var}_{u^a}(h^a). \end{aligned} \tag{4.6}$$

Centering preserves each normalization. Substituting the velocity into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ proves the dissipation identity. The same construction uses the Fisher tensor on a regular smooth family [1]. Both sectors have equal mobility; a slower model response must arise from the resulting dynamics. Equal mobility is what one gets by treating all tables as one product law of all represented variables and source labels, since Fisher information adds over independent factors. It has a cost for the source rows: under unit mobility a row's log odds relax at a rate equal to its temperature, so a row that selects sharply moves slowly, and replacing it by its instantaneous Gibbs optimum is a fast-row limit rather than the same flow.

A Bernoulli law makes the metric concrete. Its success probability p has metric $dp^2/[p(1-p)]$, so the mobility is $p(1-p)$. For $\mathcal{E}(p) = D_{\mathrm{KL}}(\mathrm{Ber}(p) \| \mathrm{Ber}(p_0))$, with $p, p_0 \in (0,1)$, write $l = \log[p/(1-p)]$ and $l_0 = \log[p_0/(1-p_0)]$. Then

$$\dot{p} = -p(1-p)(l - l_0), \qquad \dot{l} = -(l - l_0). \tag{4.7}$$

Log odds relax exponentially, although probability does not follow a linear restoring equation. Here t parametrizes the chosen relaxation, and the dimensionless T_q, T_s weight routing penalties. They have not been identified with physical time or bath temperature.

4.4 A record likelihood defines a different objective

A fixed normalized prior P_0 and record likelihood ℓ_o give posterior $\Pi_o = \ell_o P_0 / Z_o$ and the exact identity

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o. \tag{4.8}$$

Source inference in a product recognition family then uses expected negative log likelihood as its cost. Peer coupling instead uses divergence between laws, whose source template can move during relaxation. Both yield Gibbs rows because both contain a categorical relative-entropy penalty, with coefficients T_q and 1 respectively; their costs and state derivatives differ. There is one setting

in which the two meet exactly. If each agent receives n messages from a source chosen by its prior and registers whether their empirical frequencies match its own law, the posterior over which source was heard is the Gibbs row at temperature $1/n$. The row temperature is then the reciprocal number of exchanged messages, as in a counting experiment in which a sharper verdict needs more counts. Identical independent Gaussian laws have zero peer divergence but nonzero expected squared sample difference. A normalized likelihood for that difference therefore describes an interaction distinct from direct agreement of the laws.

Chapter 5

What a Message Lets the Receiver Infer

Sending a few observables can change an agent's description of many variables. The extra structure comes from the receiver's conditional model. This is familiar in physics: measuring one coordinate of a correlated system can change predictions for another coordinate without measuring it directly. The distinction between transmitted information and inferred structure is explicit here.

5.1 Conditional prediction within an agent

An agent can use its model to infer a result from a different, correlated cue. Let $A \in \{1, \dots, M\}$ be a finite model presentation with prior weights $\pi_a > 0$ summing to one. On standard Borel cue and result spaces, presentation a supplies a cue law $\mu_a(du)$ and a normalized result kernel $K_a(u, dv)$. A normalized record kernel $\mathcal{L}(u, do)$ measures the cue. The joint law is

$$P(a, du, dv, do) = \pi_a \mu_a(du) K_a(u, dv) \mathcal{L}(u, do). \tag{5.1}$$

Successive integration of the kernels and summation over a give total mass one. In the agent description, a is the model presentation, (u, v) is the represented content, and $\mu_a(du)K_a(u, dv)$ is its evaluator. The kernels and prior weights constitute the stored model; the posterior after a record describes the agent's current beliefs.

Suppose $\mathcal{L}(u, do) = l(o \mid u)\lambda(do)$ and the observed o has finite positive evidence $Z_a(o) = \int l(o \mid u)\mu_a(du)$ in every component. Bayes' rule gives

$$\begin{aligned} w_a &= \frac{\pi_a Z_a(o)}{\sum_b \pi_b Z_b(o)}, \\ \Pi_a^o(du, dv) &= \frac{l(o \mid u)\mu_a(du)K_a(u, dv)}{Z_a(o)}. \end{aligned} \tag{5.2}$$

The exact recognition law is $w_a \Pi_a^o$, so the model belief $s(a) = w_a$ is generally correlated with the contents. A product restriction would replace the component-dependent content laws by one common law. Conditioning revises these beliefs without changing the stored kernels. This construction uses general probability laws; Gaussian conditioning makes a particular realization explicit.

For a scalar example, choose one presentation and independent variables

$$X \sim \mathcal{N}(0, 1), \qquad \epsilon \sim \mathcal{N}(0, 1/4), \qquad \eta \sim \mathcal{N}(0, 1), \tag{5.3}$$

with result $Y = -2X + \epsilon$ and record $O = X + \eta$. Observe $O = \xi$. The cue posterior follows by completing the square:

$$p(x \mid \xi) \propto \exp\left[-\frac{x^2 + (\xi - x)^2}{2}\right] \propto \exp[-(x - \xi/2)^2], \tag{5.4}$$

hence $X \mid O = \xi \sim \mathcal{N}(\xi/2, 1/2)$. Since the result noise is independent of the cue and record,

$$Y \mid O = \xi \sim \mathcal{N}(-\xi, 9/4). \tag{5.5}$$

The variance is $4(1/2) + 1/4 = 9/4$. Increasing the observed cue increases the inferred cue but decreases the predicted result. The negative response comes from the model's conditional relation between two variables. Estimating the cue itself in this example would instead have positive slope $1/2$. Conditional prediction therefore gives a statistical account of a transformation that reverses a direction. Conditioning a joint Gaussian mixture to predict one group of variables from another is Gaussian mixture regression [7].

With several presentations, uncertainty also concerns which relation applies. For vector-valued results with finite second moments, let $m_a = \mathbb{E}[Y \mid o, A = a]$ and $S_a = \mathrm{Cov}(Y \mid o, A = a)$. The result marginal p^o_Y has moments

$$\begin{aligned} \bar{m} &= \sum_a w_a m_a, \\ S &= \sum_a w_a \left[S_a + (m_a - \bar{m})(m_a - \bar{m})^\top\right]. \end{aligned} \tag{5.6}$$

The first covariance contribution is uncertainty within a presentation. The second is uncertainty from the disagreement of its predictions with those of other presentations. These terms follow by writing $Y - \bar{m} = (Y - m_a) + (m_a - \bar{m})$ and taking the conditional expectation; the cross terms vanish.

If the result is a finite mixture of nonsingular Gaussians, one possible carrier is $q_G = \mathcal{N}(\bar{m}, S)$. It minimizes $D_{\mathrm{KL}}(p^o_Y \| q_G)$ over Gaussian laws. Indeed, the expected negative Gaussian log density depends on p^o_Y only through its first two moments, and its minimum occurs at their matching mean and covariance. This is the moment projection used in assumed-density filtering [35]. The variational free energy instead contains the divergence with recognition in the first slot. Minimizing $D_{\mathrm{KL}}(q \| p^o_Y)$ over Gaussians is a different problem and need not return the moment carrier.

Retaining w alongside q_G still does not preserve their original dependence. If the retained description is the product $w \otimes q_G$, the exact label/result posterior $\Pi^o_{A,Y}$ obeys

$$D_{\mathrm{KL}}(\Pi^o_{A,Y} \| w \otimes q_G) = I_{\Pi^o}(A; Y) + D_{\mathrm{KL}}(p^o_Y \| q_G). \tag{5.7}$$

Here $I_{\Pi^o}(A; Y) = D_{\mathrm{KL}}(\Pi^o_{A,Y} \| w \otimes p^o_Y)$ is mutual information. Insert $w \otimes p^o_Y$ into the logarithmic density ratio to obtain the identity. It separates the loss of model/content dependence from the loss of non-Gaussian result information. Neither loss is determined by the accuracy of the mean alone.

A later step may carry the component belief forward, for example as wT under a declared stochastic transition matrix T. An interpretation as Bayesian filtering also needs the conditional law of the next

record given the retained history and presentation. A deterministic function of a record already available to the agent supplies no further evidence conditional on that record; treating it as an independent observation would change the model. Communication can supply information new to the receiver, but its record kernel must be specified. The local conditioning construction therefore provides an agent computation, while repeated assimilation, model learning, and population dynamics require their own declared mechanisms.

5.2 Selective sending and conditional reconstruction

A sender uses a measurable map $T_j : E_j \to \mathsf{H}$ into a message space. If its content has law q_j, possible messages have law

$$\mu_j = (T_j)_{\#} q_j. \tag{5.8}$$

The pushforward symbol means $\mu_j(A) = q_j(T_j^{-1}A)$. Sampling a message and transmitting the whole law μ_j are different communication protocols. Messages run from sender to receiver. The reciprocal potential of the dynamics chapter also moves each sender toward its listeners, which a communicating population can only realize if the listeners send something back. A linear receiving map $O_i : \mathsf{H} \to E_i$ embeds a message, but its composition with T_j cannot have higher rank than T_j.

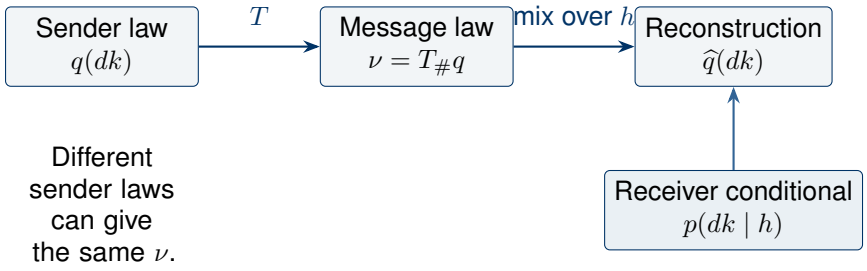


Figure 5.1: Selective communication and receiver structure. The sender fixes the transmitted marginal. The receiver supplies a conditional law for distinctions that the message does not specify. This distributional reconstruction presupposes that the message law is available; one received sample supplies a value instead.

To see exactly what is lost, work in one common presentation and write $T : E \to \mathsf{H}$. Let q be the sender's law and p the receiver's reference law, with finite $D_{\mathrm{KL}}(q\|p)$. The receiver can combine the transmitted marginal $\mu = T_{\#}q$ with its own conditional law:

$$\widehat{q}(dk) = \int p(dk \mid h)\mu(dh). \tag{5.9}$$

Disintegration on standard Borel spaces provides these conditional laws [25]. The reconstruction has exactly the right message marginal. Its remaining discrepancy is

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\mu(dh). \tag{5.10}$$

Both laws assign the same probability to h, so that factor cancels in their density ratio. The remaining logarithm compares the two conditional laws, which gives the equality by integration. This reconstruction is Jeffrey's rule of conditioning [13], and it is the law closest to the receiver's prior among all laws with the transmitted marginal. The divergence then splits like a right triangle,

$D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\|\widehat{q}) + D_{\mathrm{KL}}(\widehat{q}\|p)$, with the second leg equal to the divergence of the message marginals [11].

For a calculation on four points, let $k = (H, V)$ be two bits. The sender gives probability $1/2$ to each of 00 and 11, and transmits only H. A receiver whose reference makes both bits independent and fair reconstructs the uniform law on all four points. The message marginal is correct, while the relative entropy is $\log 2$. A receiver whose conditional model has $V = H$ reconstructs the sender's full law. The missing correlation was supplied by the receiver.

One reference conditional can recover a whole family only under a stronger condition. Suppose $q_\alpha = L_\alpha p$ and use the same $p(dk \mid h)$ for every α. Its reconstruction has density

$$\frac{d\widehat{q}_\alpha}{dp}(k) = \mathbb{E}_p[L_\alpha \mid T](k). \tag{5.11}$$

To obtain this identity, weight the reference message marginal by its conditional mean likelihood and then apply the reference conditional. The result equals q_α precisely when L_α is already measurable from the message, which is the classical statement that the message is a sufficient statistic for the family [20]. Reconstructing one posterior with its own conditional does not establish this family-wide property. If a local reconstruction must also ignore retained exterior information, the reference needs conditional independence from that exterior given the message.

For a single received value h, Bayesian assimilation instead takes the form

$$q_i^+(dk) = \frac{\ell_i(h \mid k) q_i^-(dk)}{\int \ell_i(h \mid k') q_i^-(dk')}, \tag{5.12}$$

with a positive finite denominator. In the perfectly correlated two-bit example, a noiseless observation $H = 1$ also fixes $V = 1$. This is inference through an existing relation, not an inversion of an arbitrary lossy map. Under the reciprocal potential such an update is a kick: the flow afterwards relaxes toward states that do not depend on the record, unless the record's likelihood is kept in the objective.

5.3 Agreement needs a common prediction rule

Suppose a model law s and a normalized record evaluator $K(m, do)$ predict the record law sK. If two agents compare model laws in one frame, they must also compare their evaluators in that frame. Equal model laws can predict opposite bits when their evaluators differ.

For one common evaluator and finite $D_{\mathrm{KL}}(s\|t)$, an exact information balance is

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do), \tag{5.13}$$

where s^o, t^o are the conditional model laws after record o. The joint laws $s(dm)K(m, do)$ and $t(dm)K(m, do)$ have divergence $D_{\mathrm{KL}}(s\|t)$ because their common kernel cancels. Factoring the same joint laws by their record marginal gives the displayed split. The last term is the distinction left unresolved by the record. A constant evaluator discards every distinction; a sufficient record leaves none.

For a model link Θ^s_{ij}, compatibility means $K_i(\Theta^s_{ij} m, A) = K_j(m, A)$ after identifying record alphabets. Only then does transporting the model law also transport its predictive meaning. If the link changes with the laws, this condition must continue to hold at the updated state.

5.4 Compression should name what matters

Rate–distortion theory makes the desired fidelity part of the communication problem [42]. A low-dimensional real vector alone is not a finite-bit channel. Noise, precision, or an encoding constraint must specify the communication cost. Wyner–Ziv coding additionally gives the receiver a realized variable correlated with the source [49]; possession of a prior by itself is not that realized side information.

For a predictive criterion, name the source X, target response Y, and accessible receiver information Z. An encoder generates U from X alone, so $U - X - (Y, Z)$ is a conditional-independence chain. The optimal decoder is $P(Y \mid U, Z)$. Its expected relative-entropy distortion is

$$\mathcal{D} = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{5.14}$$

The equality follows by expanding $I(Y; X, U \mid Z)$ in both orders. It is the information about the chosen response that compression discards. An information-bottleneck criterion trades this quantity against a cost such as $I(X; U \mid Z)$ [46]; an operational bit rate requires additional coding assumptions.

If $X = (A, B)$ consists of independent fair bits and $U = A$, the distortion is zero for target $Y = A$ and $\log 2$ for target $Y = B$. For a prospective parent, the target can be a future boundary response. Retaining one such response does not ensure that every later interaction can be inherited. That stronger requirement concerns dynamics.

Chapter 6

When a Collective Can Act as an Agent

A collective can have a simple description without being able to evolve from that description alone. Knowing the total energy of two bodies, for example, does not usually determine their next interaction. A parent of law-valued agents faces the same question: does its retained state determine its future responses?

There are two operations to distinguish. A channel C compresses represented contents and gives a parent law $Q_x C$. A map R retains a complete dynamical state $R(x)$. Besides belief and model laws, that state can include the model evaluator, source information, boundary channels, and memory. A distribution over possible contents does not automatically contain these additional data.

6.1 Two routes to the same parent

Suppose the fine configuration advances by F. Exact inherited evolution requires a parent update F^c such that

$$R(F(x)) = F^c(R(x)). \tag{6.1}$$

We can first evolve the children and then form their parent, or first form the parent and evolve it. Both routes must give the same retained state. Repeating the identity gives agreement at every later step. Boundary response mechanisms must likewise depend only on the retained state.

The immediate test is whether configurations with the same parent have the same next parent. If $R(x) = R(x')$ but $R(F(x)) \neq R(F(x'))$, no single F^c can satisfy the equation. Conversely, in a finite state space, this test is sufficient: evolve any representative of each parent state. Agreement within each fiber of R makes the choice irrelevant. On general measurable spaces the resulting map must also be measurable.

For stochastic dynamics, replace the next state by its distribution. If T is the fine transition matrix, the probability

$$\sum_{y:R(y)=z'} T(x, y) \tag{6.2}$$

must depend only on $R(x)$. These sums then define normalized parent rows. Conditioning on any parent history merely averages identical rows, so the parent reproduces the whole retained history law. This is strong lumpability [27]. Agreement for one equilibrium preparation is a weaker statement.

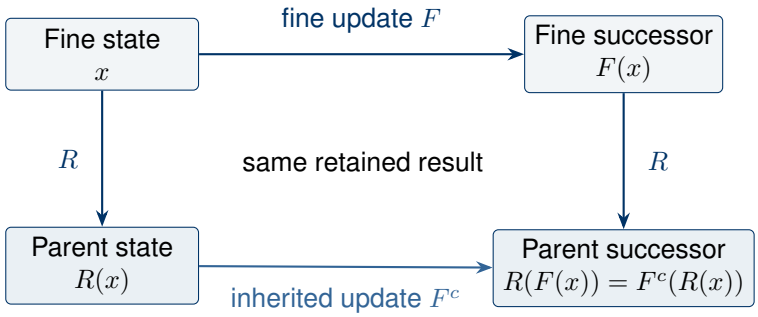


Figure 6.1: The two routes to the next parent configuration must agree. Vertical maps retain information; horizontal maps advance the dynamics. A compressed law alone need not make this square commute.

6.2 When a gradient flow descends

Suppose the fine configuration lies on a Riemannian manifold with metric g and evolves by $-\operatorname{grad}_g E$. Let a smooth surjective submersion R retain the parent, whose metric is g_c. One sufficient condition is that the energy depend only on the parent,

$$E = E_c \circ R, \tag{6.3}$$

and that R preserve lengths of horizontal vectors, meaning vectors orthogonal to $\ker DR$. Then the gradient is horizontal because the energy has no derivative along a fiber of R. For a horizontal u,

$$g_c(DR \operatorname{grad}_g E, DRu) = g(\operatorname{grad}_g E, u) = dE_c(DRu). \tag{6.4}$$

Every parent tangent has a horizontal lift, so the pushed gradient is exactly $\operatorname{grad}_{g_c} E_c$. The dynamics descends in the same time.

If instead $E = \chi(E_c \circ R)$ and the horizontal metric is scaled by $d(x)^2$, the pushed vector field acquires the factor $\chi'(E_c(Rx))d(x)^2$. A positive factor preserves oriented trajectories after a time change. A factor that depends on discarded state does not specify an autonomous parent clock. Taking the minimum of E over each fiber does not supply the first hypothesis: it gives the value at the minimizing states, not the fine energy everywhere on that fiber.

The requirement $E = E_c \circ R$ is stronger than descent needs, and it has a physical cost: it makes the energy blind to every motion along a fiber, so the discarded coordinates cannot move relative to the retained ones. Only the horizontal derivatives matter. The computation above uses $dE(u)$ for horizontal u alone, so descent holds exactly when

$$dE(u) = dE_c(DR(u)) \qquad \text{for every horizontal } u. \tag{6.5}$$

Two particles on a line with unit masses, harmonic self-energies $\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$, and any pair interaction $W(x_1 - x_2)$ show the difference. The center of mass $(x_1 + x_2)/\sqrt{2}$ relaxes as if the interaction were absent, because a force that depends only on the separation pushes the two particles in opposite directions and cancels in their sum. Meanwhile the separation evolves under W. The parent is exact, the discarded coordinate has its own dynamics, and yet the energy is not a function of the center of mass.

The opposite case is a retention that simply drops some agents. If the retained and discarded agents exchange any peer cost, the retained velocity depends on the discarded state, because each agent pulls on every agent it attends to and is pulled back. Dropping whole agents is exact only for blocks that do not interact at all. Exact compression of interacting agents therefore comes from symmetry, as when every member of a block can be rotated together without changing any cost, or from special energy structure such as the center-of-mass cancellation. A related consequence is that exact inheritance cannot create oscillation: the linearization of a gradient flow at an equilibrium has real eigenvalues, and an exact parent inherits a subset of them.

6.3 A missing role can prevent closure

Let two independent represented bits have laws $\mathrm{Ber}(p_1)$ and $\mathrm{Ber}(p_2)$. Their probabilities evolve according to

$$\begin{aligned} \dot{p}_1 &= -p_1 + \lambda(p_2 - p_1), \\ \dot{p}_2 &= -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \end{aligned} \tag{6.6}$$

The square $[0,1]^2$ is preserved: the vector field never points outward on its boundary. It is tangent to a face when $\lambda = 0$ and vanishes at the corner $(0,0)$. Retain only the law of the sum of the two bits. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for sums $0, 1, 2$.

Their subsequent changes differ. The coupling terms cancel when we add the two equations, giving

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2). \tag{6.7}$$

The derivatives are -1.2 and -1.8. Thus the same retained law has two different possible derivatives. Compression erased which probability belonged to the faster-decaying role. The fine recognition law remains a product throughout. What must be retained is the information this evolution uses; cross-agent covariance is not automatically required.

6.4 An exact supplied hierarchy

A finite construction shows that two successive reductions can work. Take $x = (a, b, c, r) \in \{0,1\}^4$, with internal register r and readout $\rho(x) = r$. Let the agent laws be

$$q_a = \mathrm{Ber}\left(\frac{1+2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2+a}{4}\right). \tag{6.8}$$

The model evaluator assigns probability $1/4 + m/2$ to record value one when the model presentation is $m \in \{0,1\}$. Averaging over s_a gives the predictive law $\nu_a = \mathrm{Ber}(1/2 + a/8)$.

At each step draw independent bits ξ, η, υ with probabilities $1/4, 1/3, 1/5$ of being one, independently of previous draws, and set

$$\begin{aligned} a' &= a \oplus \xi, \\ b' &= b \oplus a \oplus \eta, \\ c' &= c \oplus b \oplus \upsilon. \end{aligned} \tag{6.9}$$

Here $\oplus$ means addition modulo two. Independently of these draws, conditional on the current state, give r' law ν_a. The record is therefore an internal state change. Its generation rule is supplied, rather than deduced from the existence of an evaluator.

First retain (a, b, r), then retain (a, r). Summing the fine transition over c' removes a normalized factor and leaves a row independent of c. Summing that row over b' removes another normalized factor and leaves a row independent of b. The configuration spaces shrink from sixteen to eight to four states. Both parents retain q_a, s_a, their evaluator, and the register. They reproduce the complete retained state and record histories for every initial law. The coupling is real, since a affects b' and b affects c'; its triangular form permits these reductions. This is a drive and its response: a drives b, which drives c, and nothing acts back. Each retained block is autonomous because the variables it keeps never read the variables it discards. The configuration describes one agent with internal marks, so neither level groups two agents, and the hierarchy itself was prescribed.

6.5 What a parent must remember

When closure fails, a parent can retain more state or, under a declared stochastic process, the conditional law of hidden states given its accessible history. In a finite model this is a probability vector: propagate it with the fine transition, restrict to the new retained value, and normalize. That vector determines every future response even when the current summary does not. Keeping only an equilibrium hidden distribution generally changes transient predictions. The vector lives on the simplex of all fine states, so it saves nothing unless noise or poor observability keeps the history from pinning down the fine state.

There is a middle case in which the parent needs no memory although the current summary is not autonomous for every preparation. Suppose that for each retained value z there is a fixed way $\Lambda(z, \cdot)$ of spreading probability over the fine states with that value, and that spreading and then evolving gives the same result as evolving the parent and then spreading, $\Lambda T = T^c \Lambda$. A child population prepared by spreading a parent law in this way stays spread in this way forever: the hidden distribution is always $\Lambda(Z_n, \cdot)$, and the parent is an ordinary Markov chain with kernel T^c [41]. A handoff that prepares children differently can break the parent, even though the kernels are unchanged.

Given the fine update and a list of what the parent must reproduce (its laws, the responses to a declared set of probes, and the channels it reads), there is a smallest exact parent. Two configurations are lumped together exactly when the listed quantities agree for them now and at every later time. Nothing coarser can be exact, and anything exact remembers at least this much. A longer list of probes gives a finer parent; if outside agents may probe the children without restriction, they can tell every exposed child law apart and the parent cannot forget any of them. How much a parent forgets is therefore decided by the probes declared before the parent is chosen.

There is an exact way to separate insufficient memory from inaccurate prediction. Let P be a declared law of a finite record sequence $Z_0, \ldots, Z_{T-1}$ on standard Borel spaces. A retained statistic is $M_t = f_t(Z_{<t})$. Let S use normalized prediction rows $s_t(\cdot \mid M_t)$. The path-law chain rule and

conditional disintegration give

$$D_{\mathrm{KL}}(P\|S) = \sum_{t=0}^{T-1} I_P(Z_t; Z_{<t} \mid M_t) + \sum_{t=0}^{T-1} \mathbb{E}_P D_{\mathrm{KL}}\big(P(Z_t \in \cdot \mid M_t)\|s_t(\cdot \mid M_t)\big). \tag{6.10}$$

The first sum is information about the next record that was discarded with the history. The second is error in the chosen prediction given the retained information. Both are nonnegative, including when the total is infinite. This extends the finite-window Markov decomposition [18, Thm. 7.3.1]. Zero first sum gives predictive sufficiency for the declared process; a compact recursively updated memory still needs a realization $M_{t+1} = U(M_t, Z_t)$. Changing the allowed controls changes which histories must be predicted.

Statistical dependence alone also does not select an agent hierarchy. A declared finite cross-agent joint law has a finest partition into independent blocks, but arbitrarily weak couplings can merge those blocks. A deterministic selector reading only a fully permutation-symmetric law cannot choose a nontrivial intermediate hard partition while respecting relabeling: swapping one member with an outside agent would create a different block that overlaps the original. More generally, such a selector can only output a partition that every symmetry of the law preserves; on a ring of four agents with the symmetries of a square, it can pair opposite agents but never neighbors. Realized histories, additional geometry, or a distribution over partitions can fall outside that restriction. The result identifies what a selector must read; it does not identify independence with agency.

6.6 The formation standard

The theory aims to show that interacting agents can produce further agents. A construction that groups agents and hands their laws to parents does not yet show this, because the grouping and the handoff can be arranged without the interaction doing anything. The formation standard of the main text lists six clauses, each closing one such loophole.

The first clause asks for a set of starting configurations that is open, or a family declared in advance and preserved by the dynamics. A single trajectory is not enough. Along one path, almost any summary of the state traces out a curve, and every curve is the solution of some differential equation, so "the parent evolves autonomously" would hold for nearly any choice of parent. Exactness means something only when it is tested on configurations that the parent cannot tell apart. The second clause asks the dynamics to pick nontrivial groups that persist by a declared criterion, with the moment of handoff read off from the state or its accessible history. The third asks that each group carry inherited laws, evaluators, responses, and dynamics, with the probes that outsiders may use fixed before the parent is chosen.

The fourth clause is compression. A parent that simply keeps every child's state is always exact and says nothing. Four agents with Gaussian laws pulled together by symmetric peer costs make the point: a threshold rule first pairs the two closest agents, then the other two, then the two pairs, and keeping the full tuple of child states at each stage reproduces the dynamics exactly. Nothing

has been forgotten, so nothing has formed. The clause therefore asks each parent to lump together configurations that some admitted probe of the children could still distinguish.

The fifth clause is the control experiment. Delete every cost that couples one agent to another, freeze the population dependence of the channels at its initial value, keep everything else, and run the same selector. This is the null flow. If it produces the same groups, the interaction did no selection work. Agents that do not interact can be grouped by any rule that reads their initial data and a common relaxation clock, much as runners in a race can be grouped by finishing order even though they never touch. The standard asks that the sequence of selected groups differ from the null one on an open set of starting configurations. Only the groups count: if the interaction merely makes the same joins happen sooner or later, it has changed a clock, not the outcome. A stronger version asks that the final grouping not be predictable from any summary of the initial data that evolves the same way with and without interaction, and an amplifying version asks that groups be tighter relative to their separations than they would be without interaction.

A whole class of architectures fails this control. If every receiver compares itself with the same target law, and the channels see the population only through additive moments, then each pair of agents relaxes toward each other at a common rate: the logarithm of the ratio of their laws shrinks by the factor e^{-2t}, up to a constant. Without the peer costs the same quantity shrinks by e^{-t}. The interaction then only speeds up a clock that every pair shares, and any selector that compares pairs in this way produces the same sequence of groups with or without it, at different times. The same happens whenever summing two agents' parameters is an exact parent for every pair: the parameter equation must then be affine, all differences shrink by a common factor, and the order of joins is fixed by the initial distances. Selection that depends on the interaction needs receivers that weigh different sources differently.

The sixth clause is recursion. The parents must themselves be grouped, by the inherited interactions, without resetting their data or adding a new attraction tuned for the purpose, and this second grouping must depend on what the first stage produced. If it would occur from every possible handoff in the inherited class, the second stage is a sequence of scales supplied in the initial data, not a parent built from parents.

A construction that meets the selection, inheritance, and recursion clauses but fails compression or the null-flow control is called a two-stage selection with exact inheritance. The constructions in the later chapters are measured against this standard. The nonsmooth fusion example meets the null-flow control, since without coupling its agents never collide, at the cost of changing the interaction. The common-precision matrix-law construction of chapter 11 selects two parents and then their union, but its schedule does not depend on the coupling: it is a two-stage selection with exact inheritance. The six-agent construction of chapter 12 works on a full open neighborhood of laws with selective channels, but in the regime of its proof the null flow reproduces its membership history. Both smooth constructions supply their compression architectures in advance. A construction that meets every clause, with a selector and a compression interface derived rather than declared, remains open.

Chapter 7

A Nonsmooth Route to Finite Coalescence

Smooth attraction can bring two laws progressively closer without making them equal at any finite time. An absolute-difference interaction behaves differently: its force does not vanish as the gap closes. This chapter explains a nonsmooth modification of the reciprocal law potential in which agents coincide at finite times and their coincident blocks inherit laws, evaluator, metric, and responses exactly. Four agents can fuse into two parents that subsequently fuse under the inherited interaction. The grouping follows the evolving state; the equations contain no prescribed pairs. The outcome is always a finite coalescence tree that ends in one block.

Pairwise convex fusion penalties and finite-time nonsmooth consensus have established precedents [10, 23], and on the restricted family used here each sector is a weighted particle solution of the one-dimensional aggregation equation with kernel $\lambda|x|$, whose sticky collisions and finite-time collapse are known [5, 8]. The additional task here is to inherit the probability laws, model evaluator, metric, and subsequent responses when fusion occurs.

7.1 Laws, a model, and inherited multiplicity

Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of a positive random variable whose logarithm is normal with mean μ and variance σ^2. A component containing n primitive agents carries

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad n \geq 1. \tag{7.1}$$

The variances are held fixed; this is a constraint, not a family that a variance-inclusive flow would preserve, and releasing the variance would destroy the encoding of n. Recognition is the product $q_n \otimes s_n$. Its normalized reference model p_n is jointly Gaussian in the logarithms, with zero mean and precision

$$n \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}, \qquad a > c > 0. \tag{7.2}$$

Thus the model couples the two sectors even though recognition is factorized. Its normalized conditional evaluator is

$$G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{7.3}$$

Direct Gaussian relative entropy gives

$$D_{\mathrm{KL}}(q_n \otimes s_n \| p_n) = C + nU, \qquad U = \frac{a}{2}\left[(\mu^q)^2 + (\mu^s)^2\right] - c\mu^q\mu^s, \tag{7.4}$$

where $C = a - 1 - \frac{1}{2}\log(a^2 - c^2)$. The mixed term makes the belief update depend on the model coordinate and conversely. The Fisher metric of this recognition family is

$$ds^2 = n\big[(d\mu^q)^2 + (d\mu^s)^2\big]. \tag{7.5}$$

Both the metric and the local potential therefore carry the same multiplicity, which cancels from an agent's own velocity. The multiplicity acts as a friction weight and as an interaction charge, not as an inertia: the dynamics below is overdamped one-dimensional motion in a harmonic trap under the pair potential $n_I n_J |x_I - x_J|$.

A parent communicates at primitive resolution through the normalized channel

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw). \tag{7.6}$$

At $n = 1$ this means the identity channel. Adding the independent log variances shows that $q_n B_n = \mathrm{LN}(\mu^q, 1)$, with the analogous identity for s_n. This restores a primitive message law while retaining the narrower parent law internally. Statistically, B_n is the law of one child sample given the mean of n children, which is sufficient and does not depend on the unknown mean; that parameter independence is why fresh messages descend exactly.

7.2 The interaction that permits contact

All means here use common reference coordinates. In sector $\alpha \in \{q, s\}$, transform the source's restored message by

$$S^\alpha_{IJ}(w) = \exp\big[\chi_\alpha(\mu^\alpha_I - \mu^\alpha_J)\big]w, \qquad 0 < \chi_\alpha < 1. \tag{7.7}$$

Its log mean moves toward the receiver. Including endpoint frames gives the link $U_I \exp[\chi_\alpha(\mu^\alpha_I - \mu^\alpha_J)]U_J^{-1}$, with the usual endpoint covariance under passive re-expression. This link is pure gauge: it equals $V_I V_J^{-1}$ with $V_I = U_I e^{\chi_\alpha \mu^\alpha_I}$, so every loop is trivial, and it is a law-dependent change of frame rather than a curved transport.

The compared message variances are both one. Consequently,

$$\sqrt{2D_{\mathrm{KL}}\left(\widehat{\nu}^\alpha_I \| (S^\alpha_{IJ})_\# \widehat{\nu}^\alpha_J\right)} = (1 - \chi_\alpha)|\mu^\alpha_I - \mu^\alpha_J|, \tag{7.8}$$

where the displayed comparison is in a common reference presentation and $\widehat{\nu}$ denotes a restored law. Rescaling the coefficient by $1/(1-\chi_\alpha)$ gives an absolute-difference penalty with strength $\lambda > 0$, and χ_α then disappears from the dynamics altogether. The square root is the new constitutive choice. It changes the smooth quadratic peer penalty of the reciprocal potential into a convex function with a corner at equality.

On the complete contact graph, the energy apart from constants is

$$\mathcal{E} = \sum_I n_I U_I + \lambda \sum_{I<J} n_I n_J \left(|\mu^q_I - \mu^q_J| + |\mu^s_I - \mu^s_J|\right). \tag{7.9}$$

The coefficient $n_I n_J$ counts primitive contacts. With unit Fisher mobility the evolution is $n_I \dot{\mu}_I \in -\partial_{\mu_I} \mathcal{E}$. The subgradient of $|x|$ is its sign away from zero and the interval $[-1, 1]$ at zero; this inclusion is the same as the Filippov regularization of the almost-everywhere gradient.

Convexity makes this evolution well posed. For every initial configuration there is exactly one solution for all time, by the theory of maximal monotone operators [6]. For two solutions, the derivative of their squared distance in the fixed multiplicity metric is the negative pairing of the state difference with a subgradient difference, hence is at most $-2(a - c)$ times their squared distance. Coincident primitive agents are interchangeable in the complete-graph equations. Uniqueness then keeps them equal after contact. Sticking follows from the same dynamics that brought them together.

The corner also forces complete consensus. The largest and smallest means in a sector are pushed toward each other with a force at least λ times the total multiplicity, while the smooth part shrinks exponentially. The spread in each sector therefore reaches zero at a finite time, for every initial configuration and every set of multiplicities, and every coalescence tree ends in one block.

7.3 Two successive fusion events

Consider the invariant family

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{7.10}$$

Both sectors have the same interagent gaps, so their interaction forces cancel in the equation for v. Set $\kappa = a - c > 0$ and introduce

$$u_i(t) = e^{-\kappa t} z_i(\tau), \qquad \tau = \frac{e^{\kappa t} - 1}{\kappa}. \tag{7.11}$$

This removes the common linear contraction. Before collisions,

$$\frac{dz_I}{d\tau} = -\lambda \sum_{J \neq I} n_J \operatorname{sign}(z_I - z_J). \tag{7.12}$$

Start four primitive agents at $z = (-3d, -2d, 2d, 3d)$, with $d > 0$. Their slopes are $(3\lambda, \lambda, -\lambda, -3\lambda)$. Each neighboring outer pair closes a gap d at relative speed 2λ. The resulting stages are

Time	Positions in z	Multiplicities
0	$-3d, -2d, 2d, 3d$	$1, 1, 1, 1$
$\tau_1 = d/(2\lambda)$	$-3d/2, 3d/2$	$2, 2$
$\tau_2 = 5d/(4\lambda)$	0	4

After the first event, the parents have slopes 2λ and -2λ. Their remaining gap $3d$ closes in another $3d/(4\lambda)$. Physical event times are $t_r = \kappa^{-1} \log(1 + \kappa\tau_r)$. No coupling or mobility is retuned at the second level. The pattern survives small perturbations off the invariant family, and with gaps separated enough in scale the same mechanism gives coalescence trees of any finite depth.

7.4 What the parent retains

At synchronization, independent log samples with a common mean have an average of variance $1/n$. Exponentiating that average gives the geometric-mean retention and the stated parent law. Restricting the primitive Fisher metric to equal location coordinates gives the factor n. Summing contact forces gives $n_I n_J$, so the reduced dynamics use exactly the inherited metric and contacts. Redundant location coordinates are removed after synchronization; an average of separated children is not claimed to close their dynamics.

Individual sample distinctions require care. If X is the retained log average, the entire primitive vector is reconstructed as

$$(X_1, \ldots, X_n) = X\mathbf{1} + R, \qquad R \sim N\left(0, I - \frac{\mathbf{1}\mathbf{1}^T}{n}\right), \tag{7.13}$$

with R independent of X. The residuals sum to zero and are correlated. Independent scalar applications of B_n to one reused parent sample preserve individual marginals but give the wrong joint law.

Fresh record sampling also closes explicitly. Applying G_n to s_n gives

$$\mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right), \tag{7.14}$$

the law of the geometric mean of n fresh primitive model records. A primitive record instead uses a fresh restored model sample followed by G_1. These are exact statements for the declared fresh sampling protocols; they do not establish closure for sampled-record feedback into the dynamics.

7.5 How this measures against formation

Switching the interaction off is the decisive control. With $\lambda = 0$ each agent follows a linear contraction toward zero, distinct agents never meet, and no event occurs; with $\lambda > 0$ every configuration fuses. The interaction is therefore necessary for the grouping. Two other requirements fail. A fused block consists of identical copies, so retaining its common law forgets nothing that a declared probe could detect: the noninjective geometric mean acts on samples, and the forgotten sample detail is reconstructed by the parameter-free kernel above. And although the parents do select their union under the inherited interaction, that second fusion happens from every possible pair of parents, because every configuration ends in consensus; the counterfactual part of the recursion requirement fails. The construction is a two-stage selection with exact inheritance, not a recursive formation.

The comparison with the Wishart construction in chapter 11 is instructive. That construction keeps the smooth KL potential, its children remain distinct at every finite time, and its agents also approach a common limit, so the difference in eventual outcome is small. The decisive difference is the role of the interaction. There the membership schedule is the centroid dendrogram of the initial shapes and does not depend on the coupling, so switching the interaction off leaves it unchanged; here switching it off removes every event. Both are two-stage selections with exact inheritance, for

different reasons. The square-root penalty is a constitutive addition, and this example derives no geometry of the context base.

Chapter 8

Hidden Conditional Information Under Smooth Dynamics

This chapter concerns the reciprocal law potential and its Fisher flow. An exact reduction need not identify agents with one another. The requirement is that discarded distinctions have no effect on the future quantities the reduced description promises to reproduce. Under smooth KL dynamics this requirement has a concrete content: hidden conditional information can contribute to the force on a retained marginal. A per-agent reduced state must retain that contribution, or have a reason why it cancels. The first part of the chapter builds such a state completion for a single agent's private information; parents, in the sense of grouped agents, appear only in the erasure example at the end.

8.1 A hidden conditional law can move a visible marginal

Write a law as $q(dz, dv) = \mu(dz)\kappa_z(dv)$. Here z is retained and v is internal. Against a reference $\nu(dz)A(dv)$, the KL chain rule is

$$D_{\mathrm{KL}}(q\|\nu \otimes A) = D_{\mathrm{KL}}(\mu\|\nu) + \int H(z)\mu(dz), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z\|A). \tag{8.1}$$

If one public value has a more costly hidden conditional law, changing its probability changes the full energy. The public Fisher velocity therefore contains $H(z)$ as well as the ordinary public force. A single average hidden entropy does not tell the reduced state which public values carry that cost.

There is an exact architecture in which this is the only hidden contribution. A sender transmits its public coordinate through a normalized, possibly law-dependent channel, and its private coordinate is not transmitted. The receiver reconstructs its private coordinate from its own conditional $\kappa_{i,z}$. The receiver law and compared source law then have the same private conditional, so that conditional cancels from their peer KL. With a fixed reference whose private factors are independent, the two-sector local model still couples the public belief and model laws. The original potential and Fisher metric give

$$\begin{aligned} \dot{\mu} &= -\mu[h^c + H - \mathbb{E}_\mu(h^c + H)], \\ \dot{\ell}_z(v) &= H(z) - \ell_z(v), \qquad \ell_z = \log(d\kappa_z/dA). \end{aligned} \tag{8.2}$$

The public score h^c includes every incoming interaction and every channel derivative. Cancellation is a functional identity, so it does not require freezing a live channel while differentiating.

The second equation contains no public law and no other agent. Each private conditional simply relaxes toward its reference along an explicit path, whatever the rest of the population does, and the public flow feels it only as a time-dependent forcing fixed at the start. In this architecture private

information never changes an interaction. That makes the construction solvable, and it is also its main limitation: the interesting case, in which private variables later influence interactions, is not covered.

8.2 Retain the distribution of density values

The conditional equation in eq. (8.2) moves a log-density value according to its current value and the conditional KL. It does not need the name of the private point carrying that value. This suggests retaining the probability law

$$\xi_z = (\ell_z)_{\#}A, \qquad \int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{8.3}$$

For example, use a uniform reference on two private states. The laws $(3/4, 1/4)$ and $(1/4, 3/4)$ have the same spectrum: half the reference mass sees $\log(3/2)$ and half sees $\log(1/2)$. Their private preferences differ, but their contribution to the public force agrees.

What this forgets depends entirely on the declared reference. With the reference $(0.6, 0.4)$ the same two laws have different spectra, and on a finite alphabet whose atoms have no coincidences among subset masses the spectrum identifies the conditional completely. The compression is the symmetry of the reference, not something the dynamics creates. In statistical language the spectrum is the law of the log-likelihood ratio between the conditional and the reference, so equal spectra mean that the two comparisons are equally informative in Blackwell's sense.

The complete spectrum evolves exactly. With $a = e^{-t}$,

$$\xi_z(t) = \left(r \mapsto ar - \log \int e^{ar'} \xi_z(0, dr') \right)_{\#} \xi_z(0). \tag{8.4}$$

This follows by solving $d\kappa_z(t)/dA \propto (d\kappa_z(0)/dA)^a$ and normalizing. The formula applies to arbitrary private conditionals whose log densities are bounded uniformly in the public coordinate; on general public spaces the public flow also needs bounded log densities for its reference and channels whose forces are Lipschitz in those charts. It does not fit a Gaussian, and the retained state can remain infinite-dimensional. When the initial conditionals belong to an exponential family based at the reference, however, only their natural parameters change, shrinking by the factor a, so the private part then closes on finitely many numbers.

If all public values within an agent have the same conditional spectrum, $H(z)$ is constant and disappears under Fisher centering. The public laws then follow the original coarse KL dynamics exactly. The private conditionals can still differ: a public bit can select between the two swapped binary laws above. This gives internal correlation without changing the public force. Other agents may have different public laws and different spectra; equality between agents is unnecessary. For the public flow alone even less is needed: only how the Rényi divergences of the conditionals from the reference vary with z, up to a part common to all z.

This property is not created at a finite time by the conditional relaxation. At every finite time $a > 0$, the power transformation and its normalization can be inverted. Different spectra therefore remain

different until an asymptotic limit, unless some additional operation changes them. Retaining unequal spectra is an exact alternative, but their existence alone does not select groups of agents.

The architecture matters. If the channel carried the sender's private coordinate, two agents with the same spectra could sit at different peer costs: with a uniform two-point reference, the laws $(3/4, 1/4)$ and $(1/4, 3/4)$ compared through an identity private channel cost $\frac{1}{2}\log 3$, while two copies of the first cost nothing. Separate spectra per agent then no longer determine the forces.

8.3 What a future record is allowed to inspect

The same two swapped private laws show why a boundary protocol matters. Let record 1 have probabilities $(3/4, 1/4)$ at the two private points. The record probabilities are respectively $5/8$ and $3/8$. Equal unmarked spectra therefore do not preserve this response.

If all permitted private readouts use a fixed mark $T(v)$, retain the joint reference distribution of $(\ell_z(v), T(v))$. A likelihood $L(o \mid z, T(v))$ then gives the conditional record probability

$$N_z(o) = \int e^r L(o \mid z, d)\eta_z(dr, dd), \qquad \eta_z = (\ell_z, T)_{\#}A. \tag{8.5}$$

After the record, the log-density coordinate changes by

$$r^+ = r + \log L(o \mid z, d) - \log N_z(o). \tag{8.6}$$

The public marginal is multiplied by $N_z(o)$ and normalized. Between records the same smooth conditional relaxation resumes, so the record's imprint on the private sector fades: the potential has no record term, and records act only at the moments they arrive. This gives an exact inherited record-and-update sequence for the declared protocols. If the marks identify every private point, however, the supposed compression may retain everything. Sufficiency is relative to what the future interaction can ask.

8.4 A smooth channel can erase a dependence in finite time

A separate four-agent example shows that exact independence can become available during the original smooth flow. Each belief and model law has a public bit, a private bit, and an internal gate bit. A fixed normalized local evaluator couples the public belief and model means. The gate means relax toward a fixed reference value. The private part of a peer channel depends on the source's private mean multiplied by a smooth function α of the population's gate means.

The function α has an exactly flat zero region. Its derivative is included in the Fisher equations. The gate is in effect an internal clock: it relaxes on its own, perturbed only by a term that a small private gain keeps dominated, and it enters the flat region at a finite time and stays there. Once inside, every peer's private cost in a receiver row is the same for all sources. It cancels from the Gibbs source comparison. The reduced energy and Fisher metric then split into a public part and an independent private-and-gate part. Both the finite-time erasure and the change of preferred source follow from a declared ordering of the private gain, the public peer strength, and the gate scale; the example does not show that the KL dynamics produces either without that ordering.

The public states start in two invariant sign regions, with two agents in each. No labels are assigned to groups by the equations. Initially, private similarities can select crossed pairs, each containing one member from each public region. After private erasure, the preferred sources change: each agent prefers the other member of its public region, and that preference persists. All cross-pair source weights remain positive. The two members of a preferred pair are distinct at every finite time, but this heterogeneity is transient: for a fixed temperature and a small enough public peer strength, their public states converge exponentially to each other, so each pair approaches a common state.

The interaction is not what produces this sequence. If every peer cost is deleted, the same rule for choosing preferred sources reads the crossed pairs first and the regional pairs afterward, because the private advantage and the public sign regions are supplied by the initial states; only the moment of the switch moves, since the gates then close more slowly. The example is therefore a single-stage selection with exact inheritance, not a formation.

For each pair, retain the empirical measure of its two paired public belief and model means. The original public vector field gives its exact evolution, and fresh public responses are computed by selecting and sampling a child from that measure. The private and gate laws can be forgotten after erasure, because they form a factor of the dynamics that has decoupled and that no admitted response reads. This is a noninjective reduction, but the empirical measure keeps every public child state, up to its label, and it is not itself a law pair of the children's type.

The example is an existence construction with an engineered channel, a product preparation on finite alphabets, and a specified class of boundary responses. It adds no fusion penalty and imposes no synchronization. It does not show that arbitrary smooth channels acquire a sufficient interface, that the distinct public child states admit a smaller aggregate description, or that the two resulting groups form another parent. Those are the remaining formation obligations.

Chapter 9

A collective trajectory is not a complete interface

Everything in this chapter is a statement about the reciprocal law potential, the cooperative consensus potential whose Fisher flow moves each source to reduce the costs of the receivers that attend to it. It is not a statement about the free energy of any agent's generative model. Within that potential there is a way to compress distinct public laws under the smooth interaction. It uses their collective motion. Near a common stable reference, differences between agents decay faster than their shared evolution. One can then assign each fine configuration a virtual collective state whose future follows exactly the same reduced trajectory. The fine agents need not coincide at any finite time, but they all approach the reference, so what is compressed is the decaying part of the approach. The construction is the asymptotic phase familiar from the theory of isochrons and isostables, carried out in charts of bounded log densities.

For an explicit general-law construction, take probability references A, B with balanced binary features φ, ψ, and the fixed normalized evaluator

$$G(m, dk) = \frac{e^{J\varphi(k)\psi(m)}}{\cosh J} A(dk), \qquad |J| < 1. \tag{9.1}$$

The bound $|J| < 1$ is exactly the condition for the common reference to be stable. The other sample coordinates can be arbitrary. Write $q_i = e^{u_i} A / \mathbb{E}_A e^{u_i}$ and $s_i = e^{v_i} B / \mathbb{E}_B e^{v_i}$, with bounded reference-centered log densities. No finite-dimensional family is imposed. Use identity channels, uniform nonself source priors, and the Gibbs-reduced reciprocal potential.

Set $z_i = (u_i, v_i)$. On the virtual diagonal the common law follows

$$\dot{u} = -u + J\varphi \mathbb{E}_{s_v} \psi, \qquad \dot{v} = -v + J\psi \mathbb{E}_{q_u} \varphi. \tag{9.2}$$

At the reference state, the linearized fine dynamics is

$$\dot{z}_i = \mathcal{A} z_i - \frac{2N}{N-1}(z_i - \bar{z}), \qquad \bar{z} = \frac{1}{N} \sum_i z_i. \tag{9.3}$$

Here $\mathcal{A}$ is the linearization of eq. (9.2). Its slowest common decay rate is $1 - |J|$, whereas the slowest difference rate is $1 + 2N/(N-1) - |J|$. The factor two in the peer term includes both receiving and being used as a source. This gap is what the interaction contributes. If the peer costs are switched off, each agent relaxes on its own with the common rates, the differences decay no faster than the average, and the construction below loses its foundation.

The arithmetic average of the log densities is not itself an exact closed state. Away from the diagonal, its equation contains a correction quadratic in the child differences. That correction decays fast enough to define a nonlinear phase. Let Φ_t be the fine flow and ϕ_t the flow of eq. (9.2). The exact parent coordinate is

$$\Pi(x) = \lim_{t\to\infty} \phi_{-t}\big(\overline{\Phi_t x}\big), \qquad \Pi(\Phi_t x) = \phi_t(\Pi x). \tag{9.4}$$

The backward common flow transports the late collective state back to the initial time. The fast decay of child differences makes the limit converge, and the limit is the same whichever smooth symmetric average is used. Its local fibers contain continuously many distinct child configurations, so this construction discards public degrees of freedom rather than merely their labels. The parent evolves exactly once its initial phase is known. The phase can be computed to exponential accuracy by integrating forward for a time t_0, averaging, and integrating the common flow backward for the same time. It is also the only choice: any Lipschitz parent map that is the identity on equal children and intertwines the two flows equals Π.

The response of the group is a separate question. For two binary agents with equal belief and model means within each agent, put $x_1 = m + a$ and $x_2 = m - a$. Their phase-law mean is

$$\bar{p} = m + \frac{3(2-J)}{2(5-J)} m a^2 + O(|m|a^2(|m| + |a|)^2). \tag{9.5}$$

A uniform choice of child followed by one bit sample has mean m. The diagonal configuration with both child means equal to $\bar{p}$ has the same phase parent, but its corresponding response has mean $\bar{p}$. For small $m = a > 0$, these means differ. The parent predicts its own inherited trajectory while failing to reproduce this ordinary group response. Sampling the phase law would change the interface. Because the phase is unique, no exact single-law parent of this kind can do better, although the defect is small: it is second order in the child differences.

An unrestricted identity-channel Gibbs receiver can distinguish even more. For child densities f_j and positive weights w_j, its block normalizer is

$$Z(r) = \sum_j w_j e^{-D_{\mathrm{KL}}(r\lambda \| f_j \lambda)/T}. \tag{9.6}$$

Knowing this response for all nearby receiver laws determines every distinct exposed child density and its total weight. To see the mechanism, cancel the known receiver entropy. The remaining function is a finite sum of exponentials of the linear probes $\int r \log f_j$. A direction that separates these probes gives a Vandermonde system, which recovers their coefficients. Distinct child laws therefore cannot be forgotten while retaining this entire interface. A fixed comparison channel changes the conclusion in a precise way: the receiver then sees only the channelled laws, and children with equal channelled laws are indistinguishable to it. For a deterministic coarse-graining channel even the force that the receiver exerts on its sources depends only on the channelled laws.

Restricting the possible receivers may permit compression. It must be accompanied by an exact check of the incoming forces and of repeated-response memory. For a second formation step, joining parents must also respect the information already discarded. Measured against the formation standard, the collective phase is a retention of a supplied block rather than a formation: nothing is

selected, the ordinary sampling response is not inherited, and the fine differences it compresses decay. It supplies a continuous autonomous reduction without an erasure gate, but not the recursive interface properties or a dynamically selected hierarchy.

Chapter 10

A Parent That Keeps Its Fresh-Sample Responses

A parent can now be constructed that preserves evolution and the fresh uniform-descendant response interface, through any finite number of supplied binary levels. The dynamics is the reciprocal law potential with its Fisher flow: the original smooth identity-channel KL interaction, with an evaluator that couples belief and model. The children keep distinct public laws. The construction works because a particular class of public patterns is invisible to fresh uniform-descendant sampling, and remains invisible under the inherited evolution. The tree is supplied, and nothing here shows the dynamics selecting it.

10.1 Four children and three public signs

Take a general base variable r with reference law A, and three independent public angles. Each angle determines a balanced sign, for example

$$\sigma_L = \operatorname{sign}\cos(\theta_L - \alpha_L(r)), \tag{10.1}$$

and similarly σ_R, σ_0 for the right node and the root. The phase $\alpha_L(r)$ can be any measurable function. Rotating it changes which actual public samples carry positive or negative contrast. A fair coin with a base-dependent orientation would serve equally well for every exact statement below; the angles are needed only to rotate a phase continuously.

Choose positive profiles $m_+(r), m_-(r)$ and contrast magnitudes $0 < a_+(r), a_-(r) < 1$, with $\int (m_+ + m_-)dA/2 = 1$. Four child densities are

$$\begin{aligned} q_{L\pm} &= m_{\sigma_0}(1 \pm a_{\sigma_0}\sigma_L), \\ q_{R\pm} &= m_{-\sigma_0}(1 \pm a_{-\sigma_0}\sigma_R). \end{aligned} \tag{10.2}$$

They are laws on the full base-and-angle space. There is a separate copy for the model laws. A fixed joint reference $\rho(r, w)$ couples the belief and model bases and is independent of the angles. Every child still compares its full law with every other child through identity channels.

If a response freshly chooses $L+$ or $L-$ with equal probability and then samples that child, its density is m_{σ_0}. The sign σ_L cancels pointwise. Its phase therefore changes the two individual public laws without changing this pair response. The pair nevertheless retains $a_{\sigma_0}^2$: its magnitude influences the future mixture through the KL forces.

The two first parents have response densities m_{σ_0} and $m_{-\sigma_0}$. They are distinct when $m_+ \neq m_-$. Choosing one parent uniformly cancels σ_0 and gives the full-coordinate response density

$$\bar{m}(r) = \frac{m_+(r) + m_-(r)}{2}. \tag{10.3}$$

The root also retains the unordered joint pair

$$\left\{(m_+(r), a_+(r)^2), (m_-(r), a_-(r)^2)\right\}. \tag{10.4}$$

It can discard the phase of σ_0 while retaining the magnitudes that determine its future. A density value and its contrast must remain paired.

10.2 Why the original dynamics closes

For a tree of depth d, assign a balanced public sign to every internal node. A leaf reads one sign on each step of its path. Write the resulting bit string as $u \in \{-1, 1\}^d$, and let its density be $f(r, u)$. This profile is an arbitrary positive bounded function; it is not required to remain a product of independent factors.

Two leaves that separate immediately after their first k common edges have the same first k bits and opposite next bits. Their later signs come from separate branches. Define P_k to keep the shared bits, flip the next bit, and average the remaining bits. The exact original peer cost is

$$D_k(f) = \int f[\log f - P_k \log f] dA dU_d, \tag{10.5}$$

where U_d is the uniform bit law. The source classes have evolving total weights b_k. Differentiating the complete energy gives

$$h = 2\log f + V - \sum_k b_k \left[P_k \log f + \frac{P_k f}{f}\right], \qquad \dot{f} = -f(h - \mathbb{E}_f h). \tag{10.6}$$

Here $V(r) = -\int g_0(w) \log \rho(r, w) dB(w)$ carries the other sector's base marginal. The ratio $P_k f / f$ comes from being a source in other agents' costs. Omitting it would give a different flow.

All quantities in eq. (10.6) use the profile values and their nested grouping. A branch exchange preserves them. At every physical sample point, the source leaves in each class run through every possible suffix bit string, so each leaf's original force equals the profile force eq. (10.6) evaluated at that leaf's own path bits. Nothing is averaged over receivers. Both the original energy and the original Fisher metric carry the factor 2^d; it cancels in the inherited equation. There is no inserted coarse timescale.

The reason this works is symmetry. The fine energy and the Fisher metric do not change when the public angles are relabeled in any way that keeps the base fixed and preserves the angle laws, or when agents are permuted. The family above is exactly the set of configurations left fixed by those relabelings that are compensated by a tree automorphism, so the flow cannot leave it. The phases, and the order of the two branches at each node, are therefore invisible to the energy, the metric, and the admitted records. This is the classical principle of symmetric criticality applied to a gradient

flow. What symmetry does not provide is the explicit cost D_k, its incoming term, and the formula for the fresh response.

The parent therefore keeps an unordered tree of profile values. Joining siblings places their two retained trees inside one unordered pair. Subsequent joins can be evaluated in the same way, and the result agrees with retaining the entire subtree directly. The grouping must remain: a flat multiset would forget which profiles enter each D_k. The parent state can still grow with the number of children.

This joining changes the description of one already coupled fine system. It does not switch on new edges or prove that the agents have selected that tree. Original uniform source rows are allowed; their class totals evolve under the original row equations.

10.3 What the inherited interface permits

At every level, choose a descendant freshly and uniformly, draw its full original sample, and conceal its label. The retained subtree mean gives the exact response density. Choosing a parent and then a descendant gives the same response as direct descendant selection. Repeat with fresh labels and fresh samples, using independent labels for belief and model responses.

Feedback also closes for record likelihoods that depend only on the base and are applied to every agent. In the reciprocal potential a record is an assimilation event: it updates the laws, and the potential itself is unchanged. Every child has the same base marginal $f_0(r)$, so a record with likelihood $L_e(o \mid r)$ has the common normalization $Z_o = \int f_0 L_e dA$. Its exact component update is

$$f^+(r,u) = \frac{L_e(o \mid r) f(r,u)}{Z_o}. \tag{10.7}$$

Every value at that base point receives the same multiplier, which commutes with every tree retention. The next experiment may depend on the previous record history. Induction then preserves the full finite history law, with the original continuous dynamics between updates. These admitted records see only the root marginal f_0. A record that reads a public sign, or a record delivered to only one subtree, multiplies the two branches by different factors and exposes the phase that the parent discarded.

Reusing one hidden child for two samples changes the response. For a pair $q_\pm = p(1 \pm a\sigma)$ it gives

$$p(\xi)p(\xi')[1 + a(\xi)a(\xi')\sigma(\xi)\sigma(\xi')]. \tag{10.8}$$

The cross term reads the discarded public pattern. Unrestricted external probes can also expose it. Exact inheritance is a statement about the declared interface, not every experiment one could perform on the children.

The discarded contrast is also transient. If the profile is close to one that does not depend on the bits, then to first order each contrast mode decays like $e^{-\lambda t}$ with $\lambda \geq 1$, and at the uniform source rows every mode decays at the common rate $3 + 2/(N-1)$. The same reciprocal flow that preserves the compression drives the distinctions it hides toward zero.

Finally, a phase rotation changes the public laws continuously in total variation: the angular region whose sign changes is proportional to the rotation angle. The corresponding squared Hellinger distance has the same linear dependence, so the Fisher–Rao distance grows like the square root of the angle. A phase path is therefore Hölder of order one half and has infinite Fisher–Rao length; it has no finite Fisher tangent. The original time evolution remains smooth on each fixed-partition profile chart. This construction establishes recursive boundary inheritance on a prepared family. Chapter 11 gives a different prepared family with selected membership and inherited Fisher geometry. It uses moment-dependent channels and an extensive interface; it does not establish selection of the identity-channel profile tree used here.

Chapter 11

Two-Stage Selection on a Wishart Family

This chapter concerns the reciprocal law potential, the direct interaction potential between agent laws whose Fisher gradient flow moves each source toward the receivers that attend to it. It is neither the global free energy nor the per-agent free energy of the dynamics chapter, and nothing here is transferred to those objectives. Four distinct agents are grouped into two parents and then into a parent of those parents, while each parent keeps the original smooth KL/Fisher dynamics, a declared response interface, and the collective connection of its children.

The example is best read as exact inheritance along a prepared ordering. The rule that groups the agents reads their belief shapes, and every belief shape relaxes as $\dot{\alpha}_i = -2(\alpha_i - \alpha_0)$, with no trace of the coupling, of the precisions, of the model sector, or of any other agent. The grouping is therefore the centroid dendrogram of the initial shapes, read on a clock whose origin the threshold sets, and the same grouping occurs when the agents do not interact at all. Under the formation standard of the inheritance chapter this fails interaction necessity, so the construction is a two-stage selection with exact inheritance, not a recursive formation. The architecture assumes common matrix precisions, a constant model-independent evaluator, a particular moment-dependent interaction, fresh sum responses, and a fixed membership threshold. All block-sum descriptions already exist mathematically at the initial time, and they are exact for every partition; the threshold selects when one becomes operative, and the dynamics does not first create the redundancy that permits it. The initial class is open within this statistical architecture.

11.1 Random matrices as represented contents

Let both represented-content spaces be $\mathsf{K} = \mathsf{M} = \mathrm{SPD}(2)$, the cone of real symmetric positive-definite two-by-two matrices. A sample k or m is a matrix content. It is distinct from a contextual coordinate and need not be the covariance of another random variable. Use the Wishart density

$$\begin{aligned} W_2(\nu, \mathsf{P})(dk) &= \frac{\det(\mathsf{P})^{\nu}}{\Gamma_2(\nu)} \det(k)^{\nu - 3/2} e^{-\operatorname{tr}(\mathsf{P}k)} dk, \\ \Gamma_2(\nu) &= \sqrt{\pi}\,\Gamma(\nu)\Gamma(\nu - \tfrac{1}{2}), \end{aligned} \tag{11.1}$$

where $\nu > 1/2$, $\mathsf{P} \succ 0$, and dk is Lebesgue measure on the three independent entries. The shape ν is real; a Gaussian sampling representation is not assumed. Γ_2 is the multivariate gamma function [36], and $\nu > 1/2$ is exactly the range of real shapes for which this density is a probability law [17]; Letac and Massam [30] discuss the harder noncentral case.

Normalization can be checked by writing $k_{11} = u$, $k_{12} = \sqrt{u}v$, and $k_{22} = w + v^2$, with $u, w > 0$. At $\mathsf{P} = I_2$, the determinant is uw, the trace is $u + w + v^2$, and the Jacobian is $\sqrt{u}$. Two gamma integrals and one scalar Gaussian integral give $\Gamma_2(\nu)$. Congruence by $\mathsf{P}^{-1/2}$ gives the general normalizer. The same calculation yields

$$\mathbb{E}e^{-\operatorname{tr}(Tk)} = \left(\frac{\det \mathsf{P}}{\det(\mathsf{P}+T)}\right)^{\nu}, \qquad \mathbb{E}k = \nu\mathsf{P}^{-1}, \qquad \mathsf{P}+T \succ 0. \tag{11.2}$$

Multiplying these transforms proves that independent matrices with common precision add their shapes. Transform uniqueness applies because the transforms are finite near $T = 0$. Thus $k_i \sim W_2(\alpha_i, \mathsf{P})$ independently implies $\sum_{i\in I} k_i \sim W_2(\sum_{i\in I}\alpha_i, \mathsf{P})$. This is an exact convolution identity for laws of matrices.

Take four agents with $q_i = W_2(\alpha_i, \mathsf{P})$, $s_i = W_2(\eta_i, \mathsf{Q})$, and product recognition $\zeta_i = q_i \otimes s_i$. Shapes exceed $1/2$. They may differ across agents and preparations, but are constant over the context base. They will evolve in inference time. The common precisions are preparation assumptions.

11.2 An interaction that moves every precision and no shape

Fix $\alpha_0, \eta_0 > 1/2$ and $\kappa > 0$. The proper local reference is the product of $W_2(\alpha_0, I_2)$ and $W_2(\eta_0, I_2)$; its normalized evaluator is the constant kernel $G_i(m, dk) = W_2(\alpha_0, I_2)(dk)$. The constancy matters. A belief evaluator whose precision depended on the model sample m would add the average of that precision over s_i to the force on q_i; since the model laws differ across agents, their precisions would separate and the sum identity above would no longer produce Wishart parents. With the constant evaluator the two sectors are uncoupled. Let $N = 4$ remain the primitive population count. Define

$$\begin{aligned} M_q &= \sum_i \mathbb{E}_{q_i} k, \qquad \Lambda_q = I_2 + \frac{\kappa}{N} M_q, \\ \mathcal{K}^q_{ij}[\boldsymbol{q}](k', dk) &= W_2(\alpha_0, \Lambda_q)(dk). \end{aligned} \tag{11.3}$$

The model sector exchanges (q, α_0, k) for (s, η_0, m). Every source, including self, is admitted with prior $\pi_{ij} = 1/N$; the self channel is deliberately not the identity, since an identity self channel would have zero cost and would tilt every row away from its prior. These normalized channels ignore the individual input sample k', but read the population's current moment. Each receiver's costs are identical across source labels, so exact Gibbs reduction at any positive source temperature gives the original prior row. The reduced belief potential is

$$\mathcal{E}_q = \sum_i \left[D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_q)) \right]. \tag{11.4}$$

Changing one law changes every target in eq. (11.3). The moment-dependent peer terms are $\operatorname{tr}(\Lambda_q M_q) - N\alpha_0 \log\det\Lambda_q$. Their full differential is $\operatorname{tr}(C_q dM_q)$, where

$$C_q = \Lambda_q + \frac{\kappa}{N}(M_q - N\alpha_0\Lambda_q^{-1}) = I_2 + \frac{2\kappa}{N} M_q - \kappa\alpha_0\Lambda_q^{-1}. \tag{11.5}$$

The second term in the first expression comes from differentiating the channel. Freezing the target would remove it. Up to a constant, the density force and its ordinary unit-Fisher flow are

$$\begin{aligned} h_i(k) &= 2\log q_i(k) - 2(\alpha_0 - \tfrac{3}{2})\log\det k + \operatorname{tr}[(I_2 + C_q)k], \\ \dot{q}_i &= -q_i(h_i - \mathbb{E}_{q_i} h_i), \\ \dot{\alpha}_i &= -2(\alpha_i - \alpha_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q. \end{aligned} \tag{11.6}$$

The last two equations follow by substituting eq. (11.1) into the density equation and matching the coefficients of $\log\det k$ and k. They establish tangency of the unrestricted density field. No parameter projection or mobility factor is inserted. The model equations use $(\eta_i, \eta_0, \mathsf{Q}, C_s)$.

The matching shows where the interaction acts. Substituting $\log q_i = (\alpha_i - \frac{3}{2})\log\det k - \operatorname{tr}(\mathsf{P}k) + \text{const}$ turns the force into $2(\alpha_i - \alpha_0)\log\det k$ plus a term linear in k. The coupling κ and the other agents live only in the linear term, so they move the precision and never the shape. The precisions are genuinely coupled: a moment perturbation H gives $DC_q[H] = (2\kappa/N)H + (\kappa^2\alpha_0/N)\Lambda_q^{-1}H\Lambda_q^{-1}$, whose trace pairing with $H \neq 0$ is positive. The product evaluator supplies no coupling between sectors.

A general picture lies behind this. When every receiver sees the same evaluator and the same channel target, and the target reads the laws only through an additive moment, the force on each agent is $2\log q_i$ plus one function common to all agents. Every log-likelihood ratio between two agents then decays as e^{-2t} whatever the interaction, and in an exponential family every difference of natural parameters does too. The precision difference $\mathsf{P}_i(t) - \mathsf{P}_j(t) = e^{-2t}(\mathsf{P}_i(0) - \mathsf{P}_j(0))$ and the shape law are two instances. A grouping rule that compares agents with each other can only see a shrinking copy of the starting configuration. Making the interaction matter for grouping requires targets that depend on the receiver.

These equations remain regular for all finite times. Put $\bar{\alpha} = \sum_i \alpha_i/N$. Then $M_q = N\bar{\alpha}\mathsf{P}^{-1}$ and each precision eigenvalue satisfies

$$\dot{p} = f(p, \bar{\alpha}(t)) = -2p + 2 + \frac{2\kappa\bar{\alpha}(t)}{p} - \frac{\kappa\alpha_0 p}{p + \kappa\bar{\alpha}(t)}. \tag{11.7}$$

The mean shape relaxes to α_0 and stays above $1/2$. The right side is positive for sufficiently small p and negative for sufficiently large p, uniformly along this shape trajectory. Each eigenvalue therefore remains in a compact positive interval. Shapes also remain in compact intervals above $1/2$, ensuring finite KL values and square-integrable scores. Common precision is preserved, and starting with unequal precisions they never become equal at a finite time.

11.3 The same clock after joining

For a current block I, retain $n_I = |I|$, $A_I = \sum_{i\in I}\alpha_i$, $B_I = \sum_{i\in I}\eta_i$, and the common precisions. Fresh independent addition gives $q_I = W_2(A_I, \mathsf{P})$ and $s_I = W_2(B_I, \mathsf{Q})$. The parent references and evaluator are the convolutions of the children's references and evaluator outputs; for the evaluator this is a kernel from the parent's model sum only because the child evaluators ignore the model sample. Their shapes are $n_I\alpha_0, n_I\eta_0$; the parent peer target has belief shape $n_I\alpha_0$ and precision Λ_q. Its source prior is $\pi^c_{IJ} = n_J/N$, the sum of the original source probabilities over sender block J.

The total moment is still $M_q = \sum_I \mathbb{E}_{q_I} k_I$, and $N = \sum_I n_I$ does not become the number of current parents. The inherited original-form potential is

$$\mathcal{E}_q^c = \sum_I \left[D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, I_2)) + D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, \Lambda_q)) \right]. \tag{11.8}$$

Its target normalizers still contribute $-N\alpha_0 \log\det \Lambda_q$. The complete derivative is consequently the same C_q, giving the unit-Fisher equations

$$\dot{A}_I = -2(A_I - n_I \alpha_0), \qquad \dot{B}_I = -2(B_I - n_I \eta_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q, \tag{11.9}$$

and the exchanged equation for Q. These are exactly the summed fine shape equations and original precision flow, in the same time. Fine and parent energies need not be equal. Joining adds (n, A, B) and keeps the common precisions, so joining is associative and commutes with evolution. Independent addition also commutes with the evaluator and channel outputs. All of this holds for every partition, so inheritance alone does not single out the blocks that the rule below picks. Even the final single parent keeps its inherited nonidentity self channel and second KL term; deleting it would change the flow.

11.4 Two selected parents and an uncoupled lost allocation

Fix a threshold $\delta > 0$ and start with singletons whose belief shapes are pairwise farther than δ apart. A block's score is $b_I = A_I/n_I - \alpha_0$. It obeys $\dot{b}_I = -2b_I$; joining replaces two scores by their count-weighted average. Join current blocks when $|b_I - b_J|$ first reaches δ from above. Simultaneous threshold edges join their connected components. This is a declared, relabeling-equivariant rule applied to current states. Two shapes that start closer than δ would never be joined, because their distance never reaches δ from above; the separation assumption is part of the rule.

Because every score is the initial centroid times e^{-2t}, a pair of blocks reaches the threshold at $\frac{1}{2}\log(D/\delta)$, where D is the distance between their initial centroids. The first join is the closest pair. After it, the merged centroid lies between its two parts, and every other block lies outside that segment, so its new distance exceeds the merge distance. Heights therefore increase, nothing cascades, and the rule reproduces count-weighted centroid clustering of the initial shapes, the UPGMC method of numerical taxonomy [28, 45], with join k at time $\frac{1}{2}\log(D_k/\delta)$. It is not single linkage: for shapes $0, 1, 2.05, 3.5$ single linkage attaches the third point to the first pair, while the rule forms the two pairs first. Neither κ, nor the precisions, nor the model shapes enter, and changing δ only shifts every time by $-\frac{1}{2}\log\delta$. With the peer interaction switched off, the same events occur at the same times.

Sort the initial belief shapes as $u_1 < u_2 < u_3 < u_4$, and let d_1, d_2, d_3 be successive gaps. The lower pair and the upper pair are joined first, followed by their union, exactly when

$$\delta < \min(d_1, d_3) < d_2, \qquad \max(d_1, d_3) < d_2 + \tfrac{1}{2}\min(d_1, d_3), \tag{11.10}$$

up to ties. The first two times and the union time are

$$t_L = \tfrac{1}{2}\log(d_1/\delta), \qquad t_R = \tfrac{1}{2}\log(d_3/\delta),$$
$$t_2 = \tfrac{1}{2}\log\frac{d_2 + (d_1+d_3)/2}{\delta}. \tag{11.11}$$

To see why no other join intervenes, suppose the lower pair joins first. Its closest remaining singleton has distance $e^{-2t}(d_2 + d_1/2)$, larger than the upper pair's distance $e^{-2t}d_3$ by the second condition. Thus the upper pair joins next. The reversed ordering has the same argument. The resulting parent separation is $e^{-2t}[d_2 + (d_1+d_3)/2]$, the centroid distance rather than the nearest-neighbor gap d_2, so both first parents coexist for a positive interval. Each selected child-block distance remains below threshold afterward. All primitive shape differences remain nonzero at every finite time, but every law converges to one common limit: the parents are distinct at finite times and the population approaches consensus asymptotically.

For explicit values choose $\alpha_0 = \eta_0 = \kappa = 1$, $\delta = 0.1$, and initial belief shapes $(1, 1.2, 1.8, 2.1)$. The gaps are $(0.2, 0.6, 0.3)$ and the times are $\frac{1}{2}\log 2$, $\frac{1}{2}\log 3$, and $\frac{1}{2}\log(17/2)$; the first parents coexist for $\frac{1}{2}\log(17/6) \approx 0.52$. Model shapes can vary independently near $(0.8, 1.1, 1.4, 1.7)$. A small open neighborhood and all its relabelings preserve the strict inequalities. No binary tree was supplied; all additive block descriptions were available before any threshold crossing.

A formation time can reveal a belief-shape gap: at a join the two child scores differ by exactly δ, so the parent sum, the counts, and the time recover the children's belief shapes up to exchange. To exhibit information that remains lost even with all times recorded, fix the belief shapes and change two model shapes inside a joining pair by $(\eta_i, \eta_j) \mapsto (\eta_i + \epsilon, \eta_j - \epsilon)$. The parent model shape, population moments, subsequent precision flow, and all selection times stay unchanged. The individual public model laws change continuously, with a nonzero Fisher tangent because their shape scores contain the nonconstant statistic $\log\det m$. Distinct fixed belief labels exclude a mere permutation. At the second join, transfer model shape from one first-parent sum to the other while fixing the total. The final parent loses a further continuous distinction between its children. The lost allocation is, however, a spectator even before the join: each model shape relaxes on its own, the model precision sees only the total, the belief sector never reads the model laws, and the rule reads only belief shapes. The loss shows that the parent description is many-to-one, not that it discards something any agent responded to.

11.5 What a parent can be asked

At each query, draw fresh independent samples $k_i \sim q_i$, $m_i \sim s_i$ across agents and sectors. For every current block report $k_I = \sum_{i\in I} k_i$, $m_I = \sum_{i\in I} m_i$. At partition Π, the whole response vector has the conditional law

$$K_\Pi = \bigotimes_{I\in\Pi} \left[W_2(A_I, \mathsf{P}) \otimes W_2(B_I, \mathsf{Q})\right]. \tag{11.12}$$

Allow any normalized readout kernel on this vector. This includes overlapping unions of whole current blocks using the same batch of samples, so their correlations are retained. A controller may choose future readouts and finite query times from its carried record history and current retained state.

Condition on that history. Fine and parent waiting-time evolution agree by eq. (11.9), joins give the same retained state, and eq. (11.12) gives the same next-record kernel. Induction therefore gives the same finite joint history of records, actions, times, and retained states for each initial preparation. The argument is short for a reason: the interface sees only block sums, the retained state is a deterministic function of preparation and time, and records never feed back into the laws, so adaptivity only chooses when and what to read. Any expected cost or record entropy of an adaptive policy is therefore the same in both descriptions. Earlier records are carried forward; different preparations need not have identical record laws from time zero. Recording a response does not reset any belief. A query cutting through a retained block can reveal an unretained intersection shape. Reusing a hidden child's samples, applying child-specific Bayesian updates, or letting records update the laws requires another interface and another closure proof.

11.6 Prepared curvature that survives both joins and then fades

Use the contextual chart $(a, \chi) \in (1/5, 2/5) \times (-\chi_0, \chi_0)$, with $\chi_0 > 0$, and one $SO(2)$ acting by matrix conjugation on belief and model samples together. Let R_χ be rotation through angle χ, and prepare

$$\mathsf{P}(0; a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a}) R_\chi^T, \qquad \mathsf{Q}(0) = \operatorname{diag}(e^{3/10}, e^{-3/10}). \tag{11.13}$$

These coordinates describe how the laws depend on context, whereas k, m remain represented matrix samples. Context-constant shapes make all event times independent of context. The flow acts separately at each context, and nothing in it reads a connection: the connection below is computed from the evolved sections as a diagnostic. At fixed shape, a precision variation U has score and Fisher metric

$$\begin{aligned} \ell_U(k) &= \nu \operatorname{tr}(\mathsf{P}^{-1} U) - \operatorname{tr}(Uk), \\ g^F(U, V) &= \nu \operatorname{tr}(\mathsf{P}^{-1} U \mathsf{P}^{-1} V). \end{aligned} \tag{11.14}$$

The joint score for common contextual changes depends on the children only through their matrix sums. The same holds for common rotation scores. Addition therefore retains every contextual and orbit direction used here, with its exact Fisher inner products. Fine and parent infinitesimal stabilizers also agree. With unit child weights, the collective minimizing connection and residual contextual tensor are identical before and after a join.

Write $p_+ > p_-$ for the belief precision eigenvalues, and $\tilde{p}_+ > \tilde{p}_-$ for the model precision eigenvalues. The derivative f_p in eq. (11.7) is negative, and the scalar solution's derivative with respect to its initial value is $\exp(\int_0^t f_p du) > 0$ at finite time. Thus $\partial_a p_+ > 0$, $\partial_a p_- < 0$, and neither eigenvalue pair meets in finite time. For a block I, define its two rotational inertias and radial coefficient by

$$\begin{aligned} \mathcal{I}_1 &= 2A_I \frac{(p_+ - p_-)^2}{p_+ p_-}, & \mathcal{I}_2 &= 2B_I \frac{(\tilde{p}_+ - \tilde{p}_-)^2}{\tilde{p}_+ \tilde{p}_-}, \\ g_{aa} &= A_I \left[(\partial_a p_+ / p_+)^2 + (\partial_a p_- / p_-)^2 \right]. \end{aligned} \tag{11.15}$$

All three are positive, $\partial_a \mathcal{I}_1 > 0$, and $\mathcal{I}_2$ is independent of a. An angular connection correction ξ has Fisher cost $\mathcal{I}_1(d\chi + \xi)^2 + \mathcal{I}_2\xi^2$. Minimization, with the convention $D^{\mathcal{A}}p = dp + J_p\mathcal{A}$, gives

$$\begin{aligned} \mathcal{A}^F &= -\frac{\mathcal{I}_1}{\mathcal{I}_1 + \mathcal{I}_2} d\chi, & h^{\min} &= g_{aa}da^2 + \frac{\mathcal{I}_1\mathcal{I}_2}{\mathcal{I}_1 + \mathcal{I}_2} d\chi^2, \\ \mathcal{F} &= -\frac{(\partial_a \mathcal{I}_1)\mathcal{I}_2}{(\mathcal{I}_1 + \mathcal{I}_2)^2} da \wedge d\chi \neq 0. \end{aligned} \tag{11.16}$$

The scalar connection form represents the coefficient of the standard $SO(2)$ generator. Its curvature is its exterior derivative. The tensor has rank two at every finite time. Applying the same formulas to each block's fine children or retained parent gives equality through both joins. A passive frame rotation by ϑ sends $\mathcal{A}^F$ to $\mathcal{A}^F - d\vartheta$, with the compensating change in the section derivative. The fixed addition map contributes no moving-channel term to this geometric comparison. Physically, this is the gauge field of a two-part rotor whose belief part has an inertia that varies along the context while the model part does not; it is the mechanical connection of rotating bodies in a statistical setting [44].

Three qualifications keep this picture honest. First, the curvature exists because one rotation is declared to act on beliefs and models together. Nothing in the potential ties them; with separate rotations for the two sectors, the minimizing connection is $-d\chi$ on the belief factor and zero on the model factor, which is flat, and the tensor has rank one. Second, the contextual dependence was put into the initial section, and none of the formulas uses $\kappa > 0$: at zero coupling the same curvature appears. Third, the flow erases it. All contexts relax to the same isotropic precision $p_* I_2$, where p_* is the positive zero of $f(\cdot, \alpha_0)$, and eigenvalue splittings and their context derivatives shrink like $e^{-\lambda t}$ with $\lambda = -f_p(p_*, \alpha_0)$. For $\alpha_0 = \kappa = 1$, $p_* = \sqrt{2}$ and the tensor decays at rate $2\lambda = 12 - 4\sqrt{2} \approx 6.34$; by the root join it has lost more than three orders of magnitude. In the isotropic limit every rotation stabilizes the laws and the connection is undetermined. Keeping the geometry alive needs anisotropic, context-dependent local data in both sectors, for example a belief evaluator $W_2(\alpha_0, \Gamma(a, \chi))$ with Γ the prepared section and an anisotropic model reference; the precisions then relax to anisotropic fixed points and the curvature tends to a nonzero constant. The curvature is then a property of those declared data rather than of the interaction.

Context-constant shape allocations are substantive: an opposite contextual variation of two model shapes has positive fine Fisher length but zero derivative of their sum law. It would destroy the claimed tensor equality. Common precision, a constant model-independent evaluator, global moment channels, product recognition, and fresh extensive responses likewise delimit this construction. The threshold is constitutive, and selection does not alter the underlying fine flow. Chapter 12 removes the common-parameter preparation through a different selective architecture with native preferential routing and retained contextual profiles. Grouping that depends on the interaction, a derived interface or membership criterion, and unrestricted child probes remain separate problems. Neither a thermodynamic limit nor spacetime follows from these finite constructions.

Chapter 12

Preferential Groups from Open Initial Laws

This chapter works with the reciprocal law potential, the cooperative potential whose Fisher flow moves each source toward the receivers that attend to it. A prepared common-precision family makes exact matrix addition possible. A different construction starts from an open set of independently variable laws and lets native source preferences, read by a fixed threshold, determine which groups are selected. It uses two represented coordinates with different roles: a bit that communication transmits, and an angle whose relative constant offsets can be discarded. The parent retains enough angular profile information to preserve geometry. The selective architecture is supplied once; no interaction changes when a group is selected.

Three facts frame what the construction shows. The selection reads only the bits, while the compression is a symmetry of the angles that is available for every grouping at every time; the groups and times come from the bits, and the loss of angular information at a handoff comes from a declared response interface that becomes coarser when groups merge. In the regime of the proof, integrations indicate that switching off every peer interaction leaves the sequence of groups unchanged. Every group relation is temporary, and the same rule eventually merges everything. Measured against the formation standard, this is a two-stage selection with exact inheritance, not a recursive formation.

12.1 The transmitted bit and the retained angle

Take each belief and model possibility to be a pair $(z, \theta) \in \{-1, 1\} \times S^1$. Densities are relative to the uniform bit and circle probability measure. Every belief law q_i and model law s_i can have its own positive smooth bit-angle correlation. Recognition uses $q_i \otimes s_i$. Local reference laws and evaluators are uniform, source priors are fixed and uniform, all sources including self are admitted, and the temperature T is fixed and positive.

The comparison channel transmits z and generates a new uniform angle. It is used on every edge, the self edge included. If $\mu_i(z)$ is the binary marginal density, then

$$D_{\mathrm{KL}}(q_i \| \mu_j \otimes \mathrm{Haar}) = D_{\mathrm{KL}}(q_i \| \mu_i \otimes \mathrm{Haar}) + D_{\mathrm{KL}}(\mu_i \| \mu_j). \tag{12.1}$$

The first term depends on the receiver but not on the source. It cancels when the receiver normalizes its source probabilities:

$$\beta_{ij} = \frac{e^{-D_{\mathrm{KL}}(\mu_i \| \mu_j)/T}}{\sum_k e^{-D_{\mathrm{KL}}(\mu_i \| \mu_k)/T}}. \tag{12.2}$$

The model source row is the same expression using its own binary marginals. The cancellation needs the selective self channel: with an identity self channel the self cost would be zero while every peer cost kept the receiver term, and at low temperature the peers would be suppressed. In the language of the smooth-closure chapter, the bit is the public coordinate, the one the channels transmit, and the angle is private. The angle is absent from this selection equation but is still part of each agent's law, of the local force, and of an aggregate response defined below. Because every channel randomizes the angle, no agent ever holds a common angular frame with another; this is the familiar situation of parties without a shared reference frame.

Cancellation from a row is not cancellation from the variational potential. The complete belief force is

$$\begin{aligned} h_i^q &= 2\log q_i - \sum_j \beta_{ij}\log\mu_j - \sum_k \beta_{ki}\frac{\mu_k}{\mu_i}, \\ \dot q_i &= -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q). \end{aligned} \tag{12.3}$$

One entropy comes from the local KL and the other from the normalized receiver row. The final sum is the incoming source derivative. The self ratio is one, but a self channel still replaces the angle, so its full divergence is generally nonzero. All these terms are required by the original smooth potential.

The formula is well defined on a positive smooth-law neighborhood. Logarithms and ratios are bounded there, and the force contains no angular derivative, so local solutions and their dependence on initial densities follow from an ordinary differential equation on a Banach chart. No Gaussian approximation or finite-dimensional moment closure is needed.

12.2 Why pairs appear before a larger group

The force splits exactly. Write $q_i(z,\theta) = \mu_i(z) f_i(\theta \mid z)$. The angular conditional at each bit value relaxes on its own toward the uniform law,

$$f_i(\theta \mid z, t) = \frac{f_i(\theta \mid z, 0)^{e^{-2t}}}{\int f_i(\psi \mid z, 0)^{e^{-2t}} d\psi/(2\pi)}, \tag{12.4}$$

and the binary means obey their identity-channel Fisher equations plus a small known forcing proportional to the difference of the two conditional entropies. For product laws the forcing vanishes. Thus an angular factor $e^{\kappa\cos(\theta-\chi)}/I_0(\kappa)$ keeps its phase and has concentration κe^{-2t}. The decay rate two is calculated from eq. (12.3); it is not a new slow mobility. Because the bit equations see angles only through rotation-invariant entropies, nothing that decides the groups can depend on an angle.

The binary routing mechanism is easiest to see at small amplitude. Write the mean as $x_i = h u_i$ and choose one temperature $T = h^2$. Without rescaling time, the limiting equation is

$$\dot u_i = -u_i + \sum_j (w_{ij} + w_{ji})(u_j - u_i), \qquad w_{ij} = \frac{e^{-(u_i-u_j)^2/2}}{\sum_k e^{-(u_i-u_k)^2/2}}. \tag{12.5}$$

The local term moves every mean toward zero. The peer terms act strongly on nearby means and weakly on well-separated means. Self mass remains in the denominator. This is the gradient flow of a Gaussian log-sum-exp energy, the same clustering mechanism as Gaussian mean shift and self-attention particle dynamics.

Place four reference coordinates at

$$-A-2, \quad -A+2, \quad A-2, \quad A+2 \tag{12.6}$$

with large finite A. Their internal gaps shrink while their centers are still far apart. Initially even a partner's native weight is at most e^{-8}, so the pairs do not begin as activated attention cliques. In the exact symmetric calculation, the half-distance a between pair centers and the half-gap b obey, while $a \geq 4, b \leq 2$,

$$-1.0014a \leq \dot{a} \leq -a, \qquad -3.03b \leq \dot{b} \leq -0.977b. \tag{12.7}$$

A partner weight crosses $1/10$ while exterior weights remain tiny. When the centers reach $a = 4$, the residual gap satisfies

$$b \leq 2(A/4)^{-0.975}. \tag{12.8}$$

It can therefore be made arbitrarily small without setting the children equal.

In the zero-gap comparison, the inherited source mass from one pair to the other is

$$B = \frac{e^{-2a^2}}{1+e^{-2a^2}}, \qquad \dot{a} = -a(1+4B). \tag{12.9}$$

It grows through $1/10$ at $a = \sqrt{\log 9/2}$. This crossing occurs while the original partner weights still dominate their complements. A small nonzero residual gap preserves the strict inequalities and the crossing. The second stage therefore happens when the supplied separation A has shrunk to order one, at a time close to $\log A$. The same formulas hold for two collapsed clusters of any common size, which suggests deeper hierarchies by repeating the scale separation; that extension is not proved.

Two pairs alone would leave no exterior at this second event. Add a third pair initially near a much larger finite coordinate R. Over the finite observation interval its center is close to Re^{-t}, and every force between it and the core is bounded by a constant times $(1+D)e^{-D^2/2}$, where D is their separation. Choosing R sufficiently large keeps every mass between this pair and the four-agent core below 10^{-3}. It forms internally but remains a genuine exterior while the two core parents join.

For a current partition, define the inherited row by averaging receivers and summing source columns:

$$B^q_{IJ} = \frac{1}{|I|}\sum_{i\in I}\sum_{j\in J}\beta_{ij}, \tag{12.10}$$

and similarly for the model sector. A proposed pair must satisfy, in both directions and sectors,

$$B^a_{IJ} > \frac{1}{10}, \qquad B^a_{IJ} - \sum_{K\notin\{I,J\}} B^a_{IK} > \frac{1}{20}. \tag{12.11}$$

Self loops are retained in normalization. The same condition selects primitive pairs and parent pairs. It has no competing partners at one vertex, because two partners could not each exceed the other by the positive contrast. At later joins, receiver averages use the inherited primitive counts. Neither temperature nor source data is refitted.

The limiting equation is a proof device. First choose finite A, then finite R, and finally one sufficiently small positive h. Smooth convergence transfers all strict finite-time events to the exact binary KL field, through one lemma that controls both trajectories and the event observables. The symmetry and product form are also proof devices: the exact split above transfers the event windows to a full open set of independent positive belief and model densities, with bit-angle correlations of size comparable to h. The initial class is open, not universal. Equal uniform laws, for example, do not produce the required contrast. The constants are not explicit. With the proof's choices the far pair sits near $u \approx 2 \times 10^7$, and the temperature implied by the written estimates is below 10^{-15}, or about 10^{-21} once the slower decay of distant means in the exact field is accounted for.

12.3 What switching off the interaction does

The formation standard asks whether a grouping would also occur with the interaction switched off. Deleting every peer cost leaves $\dot{u} = -u$, so $u(t) = e^{-t}u(0)$, and the rows become a Gaussian kernel of growing width applied to the initial positions. Integrations from the proof's initial data, at $A = 8$, 64, and 4096, give the same three pairs, the same parent join, and the same exterior under both flows; the events move by less than ten percent. These integrations use the limiting equation, not the exact field at the theorem's temperature, and they are not a theorem. What is proved is that peer forces contract the pairs faster than the null flow does. The groups themselves are fixed by the scale hierarchy written into the initial data.

The groups are also temporary. Every mean tends to zero, so every row tends to $1/6$. A first-stage contrast tends to $1/6 - 4/6 < 0$, and once only the merged core and the third pair remain, the contrast condition has no exterior and the two remaining rows tend to $2/6$ and $4/6$. The same rule therefore merges everything at a later finite time. The windows in the theorem are finite by construction.

12.4 What the parent actually forgets

An active rotation turns both laws of one agent by an angle in fixed sample coordinates. It leaves the agent's binary marginals, entropy, and evaluator unchanged, and the full dynamics is equivariant: a rotated initial law gives the rotated trajectory. The parent of I discards rotations whose angles sum to zero:

$$\mathsf{H}_I = \{(a_i)_{i\in I} : \sum_i a_i = 0 \pmod{2\pi}\}. \tag{12.12}$$

A two-child parent loses one continuous relative orientation. Joining two parents loses one further relative block orientation. These are changes of probability laws, not passive changes of coordinates. Whether they matter depends on who can see them. No agent, channel, row, or evaluator of the population distinguishes them; every one of these quantities is unchanged by independent rotations

of all belief and model angles. They are distinguished only by an external querier, declared with the construction, who reads angles in a fixed frame that no agent holds.

To represent the retained state by actual normalized laws, fix a contextual anchor c_*. Let ϑ_i be the phase of the belief first circular moment there and $\phi_I = |I|^{-1} \sum_i \vartheta_i$, using a local lifted-angle chart. Rotate the entire contextual section of child i by $\phi_I - \vartheta_i$ and take products of the rotated laws:

$$q_I(c) = \bigotimes_{i \in I} \tau_{\phi_I - \vartheta_i} q_i(c), \qquad s_I(c) = \bigotimes_{i \in I} \tau_{\phi_I - \vartheta_i} s_i(c). \tag{12.13}$$

This is a slice in the sense of the method of slices for symmetric evolution equations, using the first Fourier mode. The parent still contains all centered profiles and all changes of phase away from the anchor. It loses the constant relative offsets. Its state size grows with the number of children, and every child's bit marginal is kept, so the parent does not compress the public laws into a smaller collective state.

The equal-phase choice cannot be made globally: no rule that treats all children alike and turns with a common rotation can pick a phase from n phases, because rotating $(0, 2\pi/n, \dots)$ by $2\pi/n$ merely relabels it. A global representative exists if one child is singled out, and nothing observable depends on the choice.

For a subsequent join,

$$\phi_{I \cup J} = \frac{|I|\phi_I + |J|\phi_J}{|I| + |J|}. \tag{12.14}$$

The resulting primitive rotations equal those obtained by joining all children directly. Native routing is invariant under the discarded rotations, so even a complete record of the event times does not identify their offsets.

The parent has exact dynamics at the original time, but its moving representation must be differentiated. If ω_i is the current angular velocity of the first moment at the anchor, then

$$\partial_t \widetilde{q}_i = V_i^q(\widetilde{x}) - (\overline{\omega}_I - \omega_i) \partial_\theta \widetilde{q}_i. \tag{12.15}$$

Every quantity is determined by retained factors. The second term keeps the representative aligned as it evolves; it does not add a force to the fine system. Omitting it would generally change the parent trajectory.

An exact inherited vector field need not be the ordinary Fisher gradient of a new single KL model. For two phase inertias I_1, I_2, the quotient metric for total phase F is $dF^2/(I_1^{-1} + I_2^{-1})$, while the equal-phase representative has metric $(I_1 + I_2)dF^2/4$. These differ when the inertias differ. The fine velocity is orthogonal to the discarded rotations, so the parent loses its potential at the rate given by the quotient metric, and on finite-dimensional von Mises families the parent flow is exactly the gradient flow for that metric. The correction term in eq. (12.15) is the part of the representative's motion along the discarded rotations. The construction uses the derived flow in eq. (12.15) and retains the primitive interaction ports.

12.5 Which responses survive

Before any selection, declare that a query to a current block returns the bit vector and the sum modulo 2π of independent constituent angles. A model query uses the analogous product of model laws. The same rule applies at every block size. Native source choices and transmitted bits are also available, and future query choices may depend on the entire recorded history.

The sum of canonicalizing rotations is zero, so the response distribution calculated from eq. (12.13) is exactly the original one. It has a nontrivial angular signal: its first moment is the product of the constituent first moments. At the next join, bit vectors concatenate and angular sums add.

Equality of the waiting-time flow and the next joint response kernel proves equality of every finite adaptive history by conditioning successively on its records. This concerns the full joint history, not just its one-time marginals. Earlier finite sample records are carried forward. Positivity means that they do not exactly reconstruct the discarded continuous offsets. They do carry information about them: each angle sample of a singleton carries Fisher information equal to its rotation inertia, so many pre-merge samples estimate the offsets well. The loss is that no finite record set pins them down exactly. An exact archive of the child laws would remove the advertised loss.

Individual absolute-angle probes of a child inside a compressed block are not allowed by this interface after the merge. They change under a discarded relative rotation and cannot be recovered. The construction therefore preserves specified experiments, not every possible microscopic measurement.

12.6 Selection and inheritance are independent

Nothing in the last two sections used the selected groups. The rows and event times are unchanged by independent rotations; the slice, the moving flow, the response descent, and the geometric equality below hold for any grouping at any time, including at the start. The bits choose the groups and the times. The response rule decides that, from those times on, only angle sums are asked for. The orientation information lost at a handoff is exactly what the coarsened interface stops asking about. This is why the theorem is stated as a conjunction of selection, interface coarsening, and symmetry reduction.

12.7 Why the collective geometry remains

At any fixed time, the canonicalizing angle in eq. (12.13) is constant across evaluation context. The contextual derivative of a rotated law is therefore just the rotated original derivative. Rotation preserves Fisher inner products. Independent product scores have zero mean, so their cross terms vanish. The actual parent Fisher metric equals the fine sum on these contextual directions.

The common collective rotation acts on every primitive belief and model coordinate at once and commutes with the active rotations. This tie of belief and model angles to one group is a declared choice: if they were rotated independently, the minimizing connection below would be flat and the tensor would have rank one. Consequently the full cost

$$g(dp + J\xi, dp + J\xi) \tag{12.16}$$

is unchanged for every common angular velocity ξ. Its minimizer, minimized tensor, and local curvature are the same before and after either join. Nothing in the dynamics reads this connection; it describes the evolved sections.

This statement has nonzero examples throughout the same finite interval. Let belief angles have von Mises concentration κ and phase χ, and model angles have fixed positive concentration λ and phase zero. Use (κ, χ) as contexts and keep the reference binary trajectory independent of them. Define

$$\mathcal{I}(v) = v\frac{I_1(v)}{I_0(v)}, \qquad a_t = \mathcal{I}(\kappa e^{-2t}), \quad b_t = \mathcal{I}(\lambda e^{-2t}). \tag{12.17}$$

The function $\mathcal{I}$ is the rotation Fisher inertia. Integration by parts gives $\mathcal{I}(v) = \mathbb{E}_{p_v}[v^2 \sin^2\theta]$, and $\mathcal{I}'(v) > 0$ for $v > 0$. Minimizing the cost $a_t(d\chi + \xi)^2 + b_t\xi^2$ gives

$$A_t^F = -\frac{a_t}{a_t + b_t}d\chi, \qquad F_t^F = -\frac{b_t e^{-2t}\mathcal{I}'(\kappa e^{-2t})}{(a_t + b_t)^2}d\kappa \wedge d\chi. \tag{12.18}$$

For a block of size n, the two positive tensor components are

$$h_{\chi\chi} = n\frac{a_t b_t}{a_t + b_t}, \qquad h_{\kappa\kappa} = ne^{-4t}\operatorname{Var}_{p_{\kappa e^{-2t}}}(\cos\theta). \tag{12.19}$$

The tensor has rank two and the curvature is nonzero at every finite time. As time grows the tensor shrinks like e^{-4t}, while the curvature approaches $-2\lambda^2\kappa/(\kappa^2 + \lambda^2)^2$, which is not zero: what degenerates at long times is rank, not curvature. The curvature is present from the start and does not depend on which groups form; the retention keeps it because it is an isometry for every grouping. The geometry is supplied by this declared family. Repetition is the reference example, not a restriction on all admitted laws. To extend it through arbitrary nearby law values, tilt each belief by $e^{(\kappa-\kappa_*)\cos(\theta-\vartheta_i)}$, normalize, and rotate by χ; keep its model law constant over context. The tilt follows the law's own phase, so an active rotation rotates the whole assigned section. It cannot reveal an erased offset through a fixed external template. Smooth dependence then preserves the positive geometric bounds on a full open neighborhood.

The handoff is selected at one fixed anchor and applied to the entire section at the same time. If different contexts use different event times $\tau(c)$, differentiating their states adds $V d\tau$ and gives a different comparison. The phase roots are also local: a global block phase has a finite root ambiguity. Neither issue can be removed by treating it as notation.

Chapter 13

Per-Agent Inference and Conditional Inheritance

An agent that changes its beliefs to fit a peer need not make that peer change in return. This chapter uses that per-agent picture: each agent minimizes the free energy of its own normalized model, taking communicated peer laws as data. A persistent record can then favor one sign locally while communication brings the agent into a group with the opposite sign. Three agents give an explicit example. Their selected group has an exact compressed parent for a declared interface, including its record histories and a nonflat collective geometry. The compression interface is supplied before selection, so this is a conditional inheritance construction rather than a completion of recursive formation.

13.1 Whose free energy supplies the force?

Let $k \in \mathsf{K}$ be a represented content, $m \in \mathsf{M}$ a model presentation, and j, r their respective source labels. Agent i has a normalized evaluator $G_i(m, dk)$, a presentation reference $r_i(dm)$, and fixed positive source priors π_i^q, π_i^s. Its content sources are $T_{ii} = G_i$ and communicated peer laws $T_{ij} = \nu_j^q$ for $j \neq i$; its presentation sources are $R_{ii} = r_i$ and $R_{ir} = \nu_r^s$ for $r \neq i$. With a normalized record likelihood $\ell_i(do \mid k, m)$, its own model is

$$P_i(dm, dk, j, r, do) = \pi_{ir}^s R_{ir}(dm) \pi_{ij}^q T_{ij}(m, dk) \ell_i(do \mid k, m). \tag{13.1}$$

Integrating the record and content, summing j, and then integrating the presentation and summing r gives one. The model stays fixed during an own-block variation with peers held fixed. It does not specify a common generative law for the population.

Restrict recognition to $Q_i = q_i \otimes s_i \otimes \beta_i \otimes \gamma_i$. For a retained record o_i, define $D_{ij} = \mathbb{E}_{q_i \otimes s_i} \log(q_i / T_{ij})$ and $E_{ir} = D_{\mathrm{KL}}(s_i \| R_{ir})$. All densities are taken against their declared common reference measures. The negative evidence lower bound is exactly

$$\begin{aligned} F_i = &\sum_j \beta_{ij} D_{ij} + D_{\mathrm{KL}}(\beta_i \| \pi_i^q) + \sum_r \gamma_{ir} E_{ir} + D_{\mathrm{KL}}(\gamma_i \| \pi_i^s) \\ &- \mathbb{E}_{q_i \otimes s_i} \log \ell_i(o_i \mid K, M). \end{aligned} \tag{13.2}$$

The likelihood remains in this objective after observation. Exact source minimization at unit temperature gives

$$\beta_{ij} = \frac{\pi_{ij}^q e^{-D_{ij}}}{\sum_k \pi_{ik}^q e^{-D_{ik}}}, \qquad \gamma_{ir} = \frac{\pi_{ir}^s e^{-E_{ir}}}{\sum_k \pi_{ik}^s e^{-E_{ik}}}. \tag{13.3}$$

Indeed, differentiating the row objective subject to normalization gives $D_j + \log(\beta_j/\pi_j) + 1 + \eta = 0$; its positive Hessian gives the unique minimum. The reduced content term is $-\log \sum_j \pi^q_{ij} e^{-D_{ij}}$, whose differential is $\sum_j \beta_{ij} dD_{ij}$.

With the own-law Fisher metric and unit mobility, write $h^q_i = \delta F_i/\delta q_i$ and evolve by

$$\dot{q}_i = -q_i(h^q_i - \mathbb{E}_{q_i} h^q_i), \qquad \dot{s}_i = -s_i(h^s_i - \mathbb{E}_{s_i} h^s_i). \tag{13.4}$$

This is natural-gradient descent in each agent's own block [1]. There is no incoming derivative from the objectives of agents listening to i. Such a derivative belongs to the reciprocal potential of chapter 4. If G_i couples belief and presentation, both derivatives of its expected log density carry β_{ii}, because the evaluator is a content self source; the presentation reference carries γ_{ii}. Even an exact channel identity $pK[p] = pW$ only simplifies its full derivative to $D(pK[p])[v] = vW$. It does not convert the per-agent force into the reciprocal force or transfer reciprocal equilibria.

The null control deletes nonself costs while retaining every label, prior, and row entropy. Its content self weight is therefore

$$\beta^0_{ii} = \frac{\pi^q_{ii} e^{-D_{ii}}}{\pi^q_{ii} e^{-D_{ii}} + \sum_{j \neq i} \pi^q_{ij}}, \tag{13.5}$$

with the corresponding formula for γ^0_{ii}. Setting either self weight to one would change this control. The example below uses fixed kernels, so no hidden population dependence remains to freeze.

13.2 A bit and two different circles

Take three agents of one species with $\mathsf{K} = \{-1, 1\} \times S^1 \times S^1$. Write $k = (z, \phi, \psi)$, $e_\phi = (\cos\phi, \sin\phi)$, and let H be Haar probability measure. Belief densities are relative to the product of the uniform bit measure and two copies of H. Define $I_0(v) = \int e^{v\cos\phi} H(d\phi)$ and $h(A) = \log I_0(|A|)$. The invariant family is

$$\begin{aligned} q_i(z, \phi, \psi) &= \frac{\exp[l_i z/2 + A_i \cdot e_\phi + z C_i \cdot e_\phi + b_i \cos(\psi - \theta_i)]}{Z_i}, \\ Z_i &= \frac{I_0(b_i)}{2}\left[e^{l_i/2} I_0(|A_i + C_i|) + e^{-l_i/2} I_0(|A_i - C_i|)\right], \end{aligned} \tag{13.6}$$

where $A_i, C_i \in \mathbb{R}^2$ and $b_i > 0$. The presentation law is $s_i(w) = e^{n_i w/2}/[2\cosh(n_i/2)]$ on $w \in \{-1, 1\}$, with counting measure and uniform self reference. All source priors have self mass $1/5$ and mass $2/5$ for each peer.

Fix $\epsilon > 0$, set $J = \epsilon^3$ and $\lambda = \epsilon$, and choose the presentation-independent evaluator

$$G_i(w; z, \phi, \psi) = \frac{e^{Jz\cos\phi}}{I_0(J)}. \tag{13.7}$$

The circle integral proves its normalization for either value of z. A peer content channel copies z and draws both circles independently from H; a peer presentation channel is the identity. The circle ϕ is marked by the evaluator direction $e_1 = (1, 0)$. The circle ψ carries another physical orientation but has no such role in the communication protocol.

The transmitted bit has log odds

$$L_i = l_i + h(A_i + C_i) - h(A_i - C_i). \tag{13.8}$$

Thus l_i alone is not the public state. The peer density against the uniform bit reference is $\nu_j(z) = e^{L_j z/2}/\cosh(L_j/2)$; its probability mass is $\nu_j(z)/2$. Using that probability mass in the density KL would add $\log 2$ to peer costs alone and change the rows.

A record is $(o, \omega) \in \{-1, 1\} \times S^1$, with counting measure for o and Haar measure for ω. Choose

$$\ell(o, \omega \mid z, \phi, \psi, w) = \frac{e^{\epsilon o z/2}}{2\cosh(\epsilon/2)} \frac{e^{\lambda \cos(\omega - \phi)}}{I_0(\lambda)}. \tag{13.9}$$

Both factors normalize over their recorded variable. The first records the bit imperfectly; the second records a noisy direction on the marked circle. Neither reads ψ. Their observed values remain in the free energy throughout the ensuing relaxation.

13.3 Exact motion in the original time

Put $m_i = \tanh(L_i/2)$, $t_i = \mathbb{E}_{q_i}[Z \cos \Phi]$, and $S_i = \mathbb{E}_{q_i} \log q_i$. The content costs are

$$D_{ii} = S_i - J t_i + \log I_0(J), \qquad D_{ij} = S_i - L_j m_i/2 + \log\cosh(L_j/2). \tag{13.10}$$

The common entropy S_i cancels from the full row. Its unnormalized weights are

$$w_{ii} = \frac{e^{J t_i}}{5 I_0(J)}, \qquad w_{ij} = \frac{2 e^{L_j m_i/2}}{5\cosh(L_j/2)}, \qquad \beta_{ij} = \frac{w_{ij}}{\sum_r w_{ir}}. \tag{13.11}$$

These weights read the marked circle through L_i, L_j, t_i, but read neither b_i nor θ_i. For example, when $A_i = \alpha e_1$ and $C_i = c e_1$ with $c > 0$, the derivative of L_i with respect to α is $h_1'(\alpha + c) - h_1'(\alpha - c) > 0$, where $h_1(r) = \log I_0(r)$ and $h_1''(r) = \mathrm{Var}_r(\cos \Phi) > 0$.

Modulo a normalization constant, the own belief force is $\log q_i - \beta_{ii} \log G_i - \sum_{j \neq i} \beta_{ij} \log \nu_j - \log \ell_i$. Its centered negative is the log-density velocity in eq. (13.4). Matching the coefficients of the sufficient statistics in eq. (13.6) therefore gives

$$\begin{aligned}
\dot{l}_i &= -l_i + \sum_{j \neq i} \beta_{ij} L_j + \epsilon o_i, \\
\dot{A}_i &= -A_i + \epsilon e_{\omega_i}, & \dot{C}_i &= -C_i + \beta_{ii} \epsilon^3 e_1, \\
\dot{b}_i &= -b_i, & \dot{\theta}_i &= 0, \\
\dot{n}_i &= -n_i + \sum_{j \neq i} \gamma_{ij} n_j.
\end{aligned} \tag{13.12}$$

The apparent linearity of several parameter equations comes from the Fisher metric, not a Euclidean gradient assumption. The sufficient-statistic covariance is positive definite in the Cartesian chart $B_i = b_i e_{\theta_i}$: no nonzero combination of the distinct Fourier and bit-parity modes is constant. Hence this is an exact invariant family of the own-law flow. Finite parameters give positive analytic densities,

local uniqueness, and smooth parameter dependence. The displayed equations bound A_i, C_i, B_i directly, while $|L_i - l_i| \le 2|C_i|$ gives a linear growth bound for l_i; these bounds prevent finite-time escape. No clock has been rescaled.

13.4 How the records produce a persistent group

Retain binary outcomes $o = (1, 1, -1)$ and arbitrary angular outcomes ω_i. Start with l/ϵ near $(4, 1/2, -1)$, small $|A_i|, b_i < 1/20$, $b_i > 0$, and $|C_i| < \epsilon^3$. These strict conditions admit a relatively open set in eq. (13.6). Before either evolution, connect two agents exactly when their nonzero public odds have the same sign and

$$\frac{\min(|L_i|, |L_j|)}{\max(|L_i|, |L_j|)} > \frac{1}{4}. \tag{13.13}$$

The selected blocks are the connected components. This rule treats labels symmetrically. Initially the only same-sign pair has ratio close to $1/8$, so the partition consists of singletons.

Write $u_i = l_i/\epsilon$ and $c_i = C_i/\epsilon^3$ for the proof, retaining the original time. On bounded parameter sets, eqs. (13.8) and (13.11) give $L_i/\epsilon = u_i + O(\epsilon^2)$ and $\beta_{ij} = \pi_{ij} + O(\epsilon^2)$, including first derivatives in these scaled variables. The limiting public equation is

$$\dot{u} = Mu + o, \qquad M = -\frac{7}{5}\mathrm{id} + \frac{2}{5}\mathbf{1}\mathbf{1}^\top, \qquad u_* = -M^{-1}o = (15/7, 15/7, 5/7). \tag{13.14}$$

The common mode has eigenvalue $-1/5$ and the two difference modes have eigenvalue $-7/5$. Thus a positive shared component survives the conflicting third record. At equilibrium every sign is positive and the smallest pair ratio is $1/3$, strictly above the selection threshold.

This limiting calculation gives an exact small-parameter result. The other limiting equations are $\dot{c}_i = -c_i + e_1/5$ and $\dot{A}_i = -A_i$. Their combined derivative is negative definite. The implicit function theorem gives a nearby exact equilibrium x_ϵ for $x = (u, c, A)$, and the symmetric part of the derivative remains at most $-\mathrm{id}/10$ in a fixed small neighborhood. Consequently

$$\frac{d}{dt}|x - x_\epsilon|^2 \le -\frac{1}{5}|x - x_\epsilon|^2. \tag{13.15}$$

Choose that neighborhood inside the strict all-positive selection region. The limiting trajectories enter it at one finite time uniformly over a compact initial patch. Finite-time continuous dependence puts the exact trajectories there for sufficiently small fixed $\epsilon > 0$, after which eq. (13.15) proves persistence for all later times. The argument is uniform in $\omega \in (S^1)^3$. It assumes that no additional record is assimilated during this episode.

The null has a different field, obtained by deleting the full nonself costs before row reduction:

$$\begin{aligned} \beta^0_{ii} &= \frac{e^{-D_{ii}}}{e^{-D_{ii}} + 4}, & \dot{l}^0_i &= -\beta^0_{ii} l^0_i + \epsilon o_i, \\ \dot{A}^0_i &= -\beta^0_{ii} A^0_i + \epsilon e_{\omega_i}, & \dot{C}^0_i &= \beta^0_{ii}(\epsilon^3 e_1 - C^0_i), \\ \dot{b}^0_i &= -\beta^0_{ii} b^0_i, & \dot{n}^0_i &= -\gamma^0_{ii} n^0_i. \end{aligned} \tag{13.16}$$

Its phase is also constant. Here the hidden concentration contributes to D_{ii} and no longer cancels from the row. An all-time sign bound needs this dependence.

For $0 < \epsilon \le 1/100$, consider

$$\frac{1}{4} \le \frac{o_i l_i}{\epsilon} \le 12, \qquad |A_i| \le \frac{1}{10}, \qquad |C_i| \le 2\epsilon^3, \qquad b_i \le \frac{1}{10}. \tag{13.17}$$

The sufficient-statistic vector has squared norm four, so the log-partition Hessian has operator norm at most four. Its integral Taylor remainder bounds the self divergence by $D_{ii} \le 72\epsilon^2 + 1/25 + 18\epsilon^6 < \log 2$. Hence $1/9 \le \beta^0_{ii} \le 1/5$. For $r_i = o_i l_i/\epsilon$, the equation $\dot r_i = 1 - \beta^0_{ii} r_i$ points inward at both endpoints of eq. (13.17). The radial derivatives of A_i and C_i also point inward, and b_i decreases. This proves invariance of the whole region.

Finally $\|\nabla^2 h\| \le 1$ gives

$$|L_i - l_i| \le 2|C_i|(|A_i| + |C_i|) \le \frac{2}{5}\epsilon^3 + 8\epsilon^6 < \frac{\epsilon}{8}, \qquad o_i L_i \ge \frac{\epsilon}{8}. \tag{13.18}$$

The third null agent stays negative for every finite time. The full flow eventually selects the three-agent block, whereas the null never does. Deleting event times cannot remove this difference of partition histories.

13.5 The parent and its complete admitted histories

For a selected block I, replace each hidden factor $e^{b_i \cos(\psi - \theta_i)}/I_0(b_i)$ by $e^{b_i \cos\psi}/I_0(b_i)$, and take the product of the resulting normalized belief laws and the retained presentation laws. The parent keeps l_i, A_i, C_i, b_i, n_i, all primitive ports and source rows, constitutive data, the complete ordered record history, and membership events and their times. Its retained vector field is exactly eq. (13.12) with θ_i omitted. Every retained derivative is independent of the omitted phases, so this retention intertwines the flows in their original time.

Compression is physical relative to the marked comparison frame. For $I_1(b) = dI_0(b)/db$, a separate child diagnostic gives

$$\mathbb{E}_{q_i} e^{\mathrm{i}\Psi} = \frac{I_1(b_i)}{I_0(b_i)} e^{\mathrm{i}\theta_i}. \tag{13.19}$$

It distinguishes the phase fibers whenever $b_i > 0$, which holds at every finite full-flow time because $b_i(t) = e^{-t} b_i(0)$. The marked-frame group acts only on ϕ, so changing θ_i is not its gauge transformation. The parent retains b_i, including the information carried by variations of the hidden shape.

Declare the boundary protocol before selection. It may query retained public and marked coordinates, primitive source labels, and transmitted bits, with actions and observation times chosen from the retained history. Assimilated records use eq. (13.9), with declared strengths; each new outcome appends its likelihood and actual value without instantaneously resetting the recognition law. Their natural parameters add, so the same finite family remains invariant. Fresh predictive draws are not assimilated as new evidence. A retained source-label trace does not reveal that latent label to the inferring agent; revealing it would change the optimized-row conditioning convention. Reused

samples must have their queried coordinates stored. No hidden absolute-angle query is in this protocol, before or after a handoff.

One must also specify how records are sampled. The normalized generative predictive kernel integrates eq. (13.1), whereas a recognition-predictive kernel integrates the likelihood against $q_i \otimes s_i$. These generally differ. Either declared kernel depends only on retained data here; the geometric construction below uses the former for its initial records. If x is the complete configuration and R its retention, the joint next-output and retained-update kernel has the form

$$K_a(x, dy, dx'_c) = \widehat{K}_a(Rx, dy, dx'_c), \qquad x'_c = U(Rx, a, y), \tag{13.20}$$

where a is the retained action label and U appends the output and any persistent likelihood coefficients. Equal parent states give equal waiting-time flows, equal next outputs, and equal next retained states conditional on those outputs. Successive conditioning proves equality of every finite adaptive joint history, including its dependence with future states. A ψ-reading sensor would violate this identity immediately by eq. (13.19).

The parent therefore carries exact directed equations on all its child ports. It is not a fixed-size agent with one new source row. Nor is its field the Fisher gradient of $\sum_{i\in I} F_i$: differentiating that sum would add the source-role forces that the per-agent dynamics excludes.

13.6 Geometry of histories and current states

Use a common marked S^1 comparison frame for a block. A passive change through angle g sends $\phi \mapsto \phi + g$, $\omega \mapsto \omega + g$, $A_i \mapsto R_g A_i$, and $C_i \mapsto R_g C_i$. It also sends the evaluator direction $e_1 \mapsto R_g e_1$. The hidden variables ψ, θ_i and the presentation sector do not transform. Evaluators, record kernels, and recognition laws are then expressed in the same new coordinates. Rotating the beliefs while holding numerical evaluator and record directions fixed would instead change the model.

For a contextual section c, draw the initial records independently from the agents' normalized generative predictive kernels, conditional on their prepared laws. Let Π_c be the law of the resulting history H. Condition on the positive-probability binary event $E = \{o = (1, 1, -1)\}$, retain its label, and leave the angular records random. If $Q^h_{c,t}$ is the product of the selected children's current recognition laws and their Gibbs-label laws after history h, the actual experiment is

$$\mathcal{P}_{c,t}(dh, dy) = \Pi_c(dh \mid E) Q^h_{c,t}(dy). \tag{13.21}$$

This experimental product introduces no global population free energy. Its score and Fisher metric satisfy

$$\begin{aligned} s_X &= \partial_X \log \Pi_c(H \mid E) + \partial_X \log Q^H_{c,t}(Y), \\ g^{\text{joint}} &= g^{\text{history}} + \mathbb{E}_H g^{\text{conditional}}. \end{aligned} \tag{13.22}$$

The cross term vanishes because each conditional density is normalized. The first score includes $-\partial_X \log \Pr_c(E)$, and the second includes the source-row label scores. A metric computed only at one fixed realized history would omit the first contribution.

Restrict now to sections whose erased phases are constant in context. Since they are also constant in time and history, the product shift $\psi_i \mapsto \psi_i - \theta_i$ is one fixed change of sample coordinates. It

carries eq. (13.21) to the parent's actual experiment, preserves Haar measure, and commutes with the marked-frame action. Consequently, for every contextual tangent X and common angular comparison velocity ξ,

$$\mathcal{C}_{\mathrm{fine}}(X,\xi) = \mathbb{E}_{\mathcal{P}}(s_X + \xi s_{\mathrm{orb}})^2 = \mathcal{C}_{\mathrm{parent}}(X,\xi), \tag{13.23}$$

where s_{orb} is the marked-orbit score. Equality before minimization gives the same minimizing connection and contextual tensor. This is the mechanical-connection construction for the declared statistical metric [44]; the result here is its exact preservation by this parent. Context-varying erased phases would contribute positive terms $[b_i I_1(b_i)/I_0(b_i)](d\theta_i)^2$ that the parent loses, so that broader equality is false.

Nonzero curvature occurs in the same model. Choose a compact contextual patch $c = (a,\chi)$ with $A_1(0) = ae_\chi$, $A_2(0) = de_1$, and $a, d > 0$ fixed independently of ϵ. Keep $C_i(0) = 0$, hidden phases and concentrations constant over context, and initial presentation laws uniform. Other selected children may have fixed small positive marked amplitudes. At $\epsilon = 0$, the initial history is uniform and the row-label laws equal their context-independent priors, so both scores vanish; meanwhile $A_i(t) = e^{-t}A_i(0)$. Define

$$\mathcal{I}(v) = v\frac{I_1(v)}{I_0(v)}, \qquad s = e^{-t}a, \qquad B(t) = \sum_{i\in I\setminus\{1\}} \mathcal{I}(e^{-t}|A_i(0)|) > 0. \tag{13.24}$$

The angular part of the cost is $\mathcal{I}(s)(d\chi+\xi)^2 + B(t)\xi^2$. Minimization, followed by differentiation in context, gives

$$\begin{aligned}
\mathcal{A}_{0,t} &= -\frac{\mathcal{I}(s)}{\mathcal{I}(s)+B(t)}d\chi,\\
\mathcal{F}_{0,t} &= -\frac{B(t)e^{-t}\mathcal{I}'(s)}{[\mathcal{I}(s)+B(t)]^2}da\wedge d\chi,\\
h^{0,t}_{aa} &= e^{-2t}\,\mathrm{Var}_s(\cos\Phi) > 0,\\
h^{0,t}_{\chi\chi} &= \frac{\mathcal{I}(s)B(t)}{\mathcal{I}(s)+B(t)} > 0, \qquad h^{0,t}_{a\chi} = 0.
\end{aligned} \tag{13.25}$$

Here $\mathcal{I}'(s) = I_1(s)/I_0(s) + s\,\mathrm{Var}_s(\cos\Phi) > 0$. The phase inertia identity follows by integrating the derivative of $\sin\phi e^{s\cos\phi}$ around the circle. Thus the tensor has rank two and the connection has nonzero curvature on this patch.

Fix a finite time window containing the common grouping entrance time, and compare sections at the same physical time. The positive inertia, tensor eigenvalues, and curvature magnitude have positive lower bounds on the compact patch throughout that window. Analytic finite-time dependence of the flow and its contextual derivatives, together with positivity of the normalized history density, preserves those strict bounds for sufficiently small fixed $\epsilon > 0$. Equation (13.23) then gives exact equality with the compressed parent's geometry at that same positive parameter. No connection enters eq. (13.12); it diagnoses the evolved statistical sections. Nonflatness is asserted on the finite window, whereas group persistence concerns the entire fixed-record episode. Later admitted records preserve exact quotient closure but may change the group and its geometry.

The remaining formation obligation is concrete. Every erased phase was already invisible to the admitted interface before any group was selected. The interaction chooses membership, but this construction does not make the sufficient interface appear. Joining already canonical parents concatenates their retained ports and histories and erases no new phase, so it gives neither a second compressive handoff nor the required counterfactual dependence of that second stage. A construction deriving such an interface and a genuinely new inherited loss remains open.

Chapter 14 treats a different transmitted-angle model whose loss is collective rather than an invisible primitive factor. It gives conditional local realizations and scoped locality obstructions; the general local-parent and second-stage formation problems remain open.

Chapter 14

A Collective Phase and Its Local Realization

Three circular laws can hide one common orientation without any child hiding an independent orientation. Each child communicates its angle law, and a query compares two sampled angles. Rotating one child changes such a response; rotating all three preserves every joint difference. A parent for the whole group can therefore erase one physical angle even though no proper block can do so with its exterior retained. The symmetry belongs to the declared interface before grouping. The interaction does not create it.

This quotient admits exact local evolution with a finite reference configuration and the complete history. A finite preparation generator gives another exact realization on a restricted family. A local memory can instead accumulate the physical phase change from the actual preparation time. It preserves the geometry when the initial collective direction is coherent across context. That condition is assumed, not created by waiting or recording. The unrestricted causal-local-parent question remains inconclusive; the obstructions below concern specified memory classes.

14.1 The transmitted-angle model

Each child's law has a public sign z, marked angle ϕ and physical angle ψ:

$$q_i = Z_i^{-1} \exp\{l_i z/2 + A_i \cdot e_\phi + zC_i \cdot e_\phi + B_i \cdot e_\psi\}, \qquad e_\theta = (\cos\theta, \sin\theta). \tag{14.1}$$

Relative to a uniform sign and two Haar circles its normalizer is

$$Z_i = I_0(|B_i|)\frac{e^{l_i/2} I_0(|A_i + C_i|) + e^{-l_i/2} I_0(|A_i - C_i|)}{2}. \tag{14.2}$$

Its presentation law is Bernoulli with log odds n_i. The self evaluator is $e^{\epsilon^3 z \cos\phi}/I_0(\epsilon^3)$, uniform in ψ. Peer content channels copy the sign and physical angle and resample the marked angle. The transmitted sign has log odds

$$L_i = l_i + \log I_0(|A_i + C_i|) - \log I_0(|A_i - C_i|). \tag{14.3}$$

All sources remain normalized. The per-agent model and own-block Fisher update are those of chapter 13, with self prior $1/5$, each peer prior $2/5$ and exact unit-temperature Gibbs rows.

The positive normalized record likelihood is

$$\ell_{r,\lambda}(o,\omega \mid z,\phi) = \frac{e^{roz/2}}{2\cosh(r/2)} \frac{e^{\lambda\cos(\omega-\phi)}}{I_0(\lambda)}. \tag{14.4}$$

Evidence appends its actual outcome and keeps its likelihood in the objective. Its coefficients are $H_i = \sum ro$ and $\Lambda_i = \sum \lambda e_\omega$, with the entire ordered word also retained. A diagnostic prediction appends its result without becoming a likelihood. These operations remain distinct.

With $m_i = \tanh(L_i/2)$, $M(B) = \nabla \log I_0(|B|)$ and $t_i = \mathbb{E}[Z\cos\Phi]$, the content row is proportional to

$$w_{ii} = \frac{e^{\epsilon^3 t_i}}{5I_0(\epsilon^3)}, \qquad w_{ij} = \frac{2e^{L_j m_i/2 + B_j\cdot M(B_i)}}{5\cosh(L_j/2)I_0(|B_j|)}. \tag{14.5}$$

The exact own-block field is

$$\begin{aligned} \dot l_i &= -l_i + \sum_{j\neq i} \beta_{ij} L_j + H_i, \quad & \dot A_i &= -A_i + \Lambda_i, \\ \dot C_i &= -C_i + \beta_{ii}\epsilon^3 e_1, & \dot B_i &= -B_i + \sum_{j\neq i} \beta_{ij} B_j, \\ \dot n_i &= -n_i + \sum_{j\neq i} \gamma_{ij} n_j. \end{aligned} \tag{14.6}$$

These are Fisher equations on an invariant exponential family. The relative orientations affect source competition through $B_j \cdot M(B_i)$, and every derivative of the field retains that dependence. No source-role derivative of another agent's objective is inserted.

For $B_i = \delta b_i$ with small positive δ and noncollinear cone vectors, the added row terms are $O(\delta^2)$. The scaled public field remains a small smooth perturbation of the stable system with equilibrium $(15/7, 15/7, 5/7)$. Connect same-sign public odds when the smaller-to-larger magnitude ratio exceeds $1/4$. Initial odds near $\epsilon(4, 1/2, -1)$ give singleton blocks; a common finite census gives the persistent full block with the initial $(+,+,-)$ episode fixed. In the prescribed null, deleting peer costs retains source labels, priors and entropy. Its self weight lies between $1/9$ and $1/5$ on the invariant box, and the third public sign stays negative. Both fixed census times and labels are retained. Later evidence can change grouping while still having exact inherited dynamics.

14.2 Why the loss is collective

The fixed protocol admits public, marked-lab, presentation and source-label queries, normalized evidence transactions and relative-angle responses on the complete graph. For a joint relative query, sample each involved child's ψ once and return all requested differences. Shared vertices share samples. The first pair moment is

$$c_{ij} = \mathbb{E}e^{\mathrm{i}(\Psi_i - \Psi_j)} = \zeta_i\overline{\zeta_j}, \qquad \zeta_i = \frac{I_1(|B_i|)}{I_0(|B_i|)} e^{\mathrm{i}\arg B_i}. \tag{14.7}$$

For three nonzero laws, $|\zeta_i|^2 = |c_{ij}||c_{ik}|/|c_{jk}|$. The Bessel ratio is strictly increasing because its derivative is a positive circular variance. Thus all concentrations and relative directions are identified,

leaving one common orientation. With an exterior law fixed, $c_{ij}/\overline{\zeta_j}$ instead identifies the interior law itself. This is identifiability from exact response laws, not a finite-sample reconstruction method.

A triangle query uses three primitive samples, not independently sampled edge outputs. Its differences retain their cycle constraint. Every accessible label, wait, record and response/update dependence is kept. Absolute ψ responses are excluded before and after selection. Adding one would defeat the quotient. No existing probe is deleted at handoff.

14.3 Finite reference memory closes the local laws

Context c indexes a prepared statistical family and is distinct from inference time t. Fix c_0 before retention. Let $x(c) = (l_i, A_i, C_i, B_i, n_i)_i$ and $x_\circ = x(c_0)$, with $S = \sum_i B_i(x_\circ)$. Rotate only the physical B blocks by

$$U(S) = \frac{1}{|S|}\begin{pmatrix} S_x & S_y \\ -S_y & S_x \end{pmatrix}, \qquad y = \mathsf{R}_U x(c), \qquad \widehat{x}_\circ = \mathsf{R}_U x_\circ. \tag{14.8}$$

The same matrix is used at every evaluation context. Cone projections obey $v\cdot B_i(t) \geq e^{-t} v\cdot B_i(0) > 0$, so the anchor stays nonzero on every finite admitted branch. No angle branch is chosen.

The reference sum or angle alone need not close its own motion. Two reference tuples can share S and have different public laws, hence different Gibbs rows and angular velocities. Keeping the full native reference tuple is a sufficient finite choice, not a proved minimal one. Let V denote eq. (14.6) and let $\mathcal{J}$ rotate each B block through $\pi/2$. Exact differentiation gives

$$\begin{aligned} \Omega &= \frac{\det(S, \dot{S})}{|S|^2}, \\ \dot{y} &= V(y, c, h) - \Omega \mathcal{J} y, \\ \dot{\widehat{x}}_\circ &= V(\widehat{x}_\circ, c_0, h) - \Omega \mathcal{J} \widehat{x}_\circ. \end{aligned} \tag{14.9}$$

The time is unchanged. The subtraction is the velocity of the chosen representative, not a new parent force. This is the exact-lumping chain rule [24].

Retain the local two-jet, reference tuple, persistent coefficients and full history. The reference at fixed c_0 is held fixed under evaluation differentiation. With $W = V - \Omega\mathcal{J}y$,

$$\begin{aligned} \dot{y}_r &= W_y y_r + W_r, \\ \dot{y}_{rs} &= W_y y_{rs} + W_{yy}[y_r, y_s] + W_{yr} y_s + W_{ys} y_r + W_{rs}. \end{aligned} \tag{14.10}$$

The explicit partials hold the local value, reference and word fixed. No third jet enters because the native field reads no contextual derivatives or neighboring contexts. Evidence updates the same likelihood at both evaluations; diagnostics only append outcomes. For a two-dimensional base the local law two-jet has 144 coordinates, plus 24 reference coordinates before constraints. The record word remains unbounded in length.

This is exact local evolution with finite reference memory, not initialization from the germ at c alone. One local two-jet also cannot recover an unknown function at all other contexts. A quadratic initial

natural-parameter polynomial realizes the actual local law germ and its derivatives; evolving it exactly preserves that jet at the evaluation point, without asserting equality away from it.

14.4 The normalized laws carry the geometry

The parent samples the canonical normalized recognition laws, presentations and original source-label factors. The reference is unsampled deterministic memory. Sampling it as another independent factor would generally change orbit inertia. To include the past, initialize the history law from the original normalized kernels and a compatible preparation seed; an arbitrary snapshot plus an observed word does not specify that word's probability. Retain the public-pattern conditioning normalizer.

At fixed t, h, rotating fresh ψ samples by the reference matrix is independent of evaluation c. The actual experiment is

$$\mathcal{P}_{c,t}(dh, dY) = \Pi_c(dh) Q^h_{c,t}(dY). \tag{14.11}$$

A passive marked frame rotates fresh ϕ and local embeddings of evaluator and record directions. Stored numerical lab labels stay fixed; presentations have the trivial representation of the same marked group. This action commutes with the physical reference rotation. Thus the actual contextual and marked orbit scores are carried isometrically, giving

$$\mathbb{E}_{\mathcal{P}}(s_X + \ell_\xi)^2 = \mathbb{E}_{\overline{\mathcal{P}}}(\overline{s}_X + \overline{\ell}_\xi)^2. \tag{14.12}$$

The minimizing connection class, isotropy, minimized tensor and locally free curvature agree. The same invariant joint transaction kernels prove every admitted finite adaptive response/next-state history by successive conditioning. There is no replacement by separate marginals or independent edge outputs.

For $B_i(c) = b_i e_{\chi(c)}$, the actual canonical law contains $b_i e_{\chi(c)-\chi(c_0)}$. It keeps the private phase information

$$\sum_i b_i \frac{I_1(b_i)}{I_0(b_i)} (d\chi)^2 > 0. \tag{14.13}$$

Erasing phase separately at each context would delete this term. Reference memory works because that dependence remains in the normalized density, not because a tensor is stored separately. The erased global angle cannot be recovered from the canonical reference and seed, since a simultaneous original rotation changes none of them.

The common nonflat window uses $A_1 = a e_\chi$, $A_2 = d e_1$ and the same finite grouping entrance time. At $\epsilon = \delta = 0$, its marked connection, curvature and positive rank-two tensor are those in eq. (13.25). Their compact-window margins persist for sufficiently small positive ϵ, δ by smooth dependence of the fine laws, jets and normalized history experiment. The private anchor is applied only at positive δ, and eq. (14.12) transfers the geometry exactly. Later evidence can change that window's geometry.

14.5 Memory can accumulate a change of phase

A rotating coordinate frame and the physical law it describes must be distinguished. At each context let $S_{\mathrm{loc}}(x) = \sum_i B_i(x)$ and put $\bar{x} = \mathsf{R}_{U(S_{\mathrm{loc}}(x))}x$, using the same matrix as in eq. (14.8) but evaluated locally. Its sum points along the first axis. The local laws determine their own angular velocity,

$$\omega_{\mathrm{loc}}(\bar{x}, c, h) = \frac{\det(S_{\mathrm{loc}}(\bar{x}), \sum_i V_{B_i}(\bar{x}, c, h))}{|S_{\mathrm{loc}}(\bar{x})|^2}. \tag{14.14}$$

The cone bound keeps the denominator nonzero. The public and marked coordinates enter through the original Gibbs rows, so the collective sum alone still does not determine this velocity. No absolute-angle query is added.

Start a scalar memory at the actual initialization time t_* and use

$$\dot{\bar{x}} = V(\bar{x}, c, h) - \omega_{\mathrm{loc}}\mathcal{J}\bar{x}, \qquad \dot{\vartheta} = \omega_{\mathrm{loc}}, \qquad \vartheta(t_*) = 0. \tag{14.15}$$

The parent samples the normalized native laws at $\widetilde{x} = \mathsf{R}_{R_\vartheta}\bar{x}$. Their derivative is

$$\dot{\widetilde{x}} = \mathsf{R}_{R_\vartheta}(\dot{\bar{x}} + \dot{\vartheta}\mathcal{J}\bar{x}) = V(\widetilde{x}, c, h). \tag{14.16}$$

The accumulated phase cancels the rotating-coordinate velocity. The actual laws obey the original own-block Fisher equations in the original time; no parent force is fitted. The angle memory records a change, and can be represented by a rotation matrix if a real angle branch is inconvenient.

The native flow and its reconstruction give

$$\widetilde{x}(c, t) = \mathsf{R}_{U(S_{\mathrm{loc}}(x(c,t_*)))}x(c, t). \tag{14.17}$$

This is a frozen local initial frame implemented causally, rather than a continuously supplied reference frame. At a fixed context its rotation is constant throughout the trajectory. Consequently the same joint relative samples, source labels, public and marked-lab responses, evidence updates and waits generate every original adaptive history. Each primitive angle is still sampled once per joint query. Persistent likelihoods and both census records are retained. A diagnostic result stays distinct from assimilated evidence.

For geometry retain the local two-jet, the corresponding phase jet, the known coefficient jets, and the complete chronological word. Unlike a reference fixed at c_0, ω_{loc} depends on evaluation context. Its derivative contributes

$$\dot{\vartheta}_a = D\omega_{\mathrm{loc}}\bar{x}_a + (\omega_{\mathrm{loc}})_a. \tag{14.18}$$

Differentiating once more closes at order two because both fields in eq. (14.15) read only the local law value and fixed data. These derivatives enter the sampled $B_i(\widetilde{x})$; a phase stored only beside the slice density would not restore its Fisher score. The memory is unsampled internal data, not an extra independent experiment.

The precise sufficient preparation condition is

$$\frac{S_{\mathrm{loc}}(x(c, t_*))}{|S_{\mathrm{loc}}(x(c, t_*))|} = e_* \in S^1 \qquad \text{for all } c. \tag{14.19}$$

Its absolute direction is not retained. Each site then applies the same initial matrix without consulting another context. In eq. (14.17) the sample shift is independent of c, t, h, commutes with the marked action and preserves Haar measure. This proves the complete actual-law cost equality eq. (14.12), including history and conditioning scores. The parent therefore has the same minimizing connection, tensor and curvature. Context-constant initial noncollinear cone vectors $B_i = \delta b_i$, together with $A_1 = ae_\chi$ and $A_2 = de_1$, give the same rank-two nonflat finite window already described. Arbitrary later evidence preserves exact inheritance but need not preserve nonflatness or membership.

All canonical initial data, phase differences, replay seed and records are invariant under one common original rotation. Within eq. (14.19), equal canonical initial fields differ by only that constant angle. The lost physical orientation is not recovered through memory, while every proper block remains exposed to its exterior. This introduces no independently invisible child factor.

The start time matters. Carry the actual accumulator from t_* through the selected census T. Setting it to zero anew at T freezes the frame at T, whose direction need not remain context-constant even from a coherent initial preparation. No old record or memory is reset. To reproduce the probability law of the past, initialize the original normalized chronological experiment, or retain its canonical local preparation seed and replay its kernels. One observed word is not its own probability law. This remains a local field with an unbounded record, rather than one finite state storing an arbitrary contextual function.

For arbitrary preparations let $\alpha_0(c)$ be the initial direction angle. The actual score at corresponding samples is

$$\widetilde{s}_X(T_0 Y) = s_X(Y) - (X\alpha_0)\ell_{\mathrm{phys}}(Y). \tag{14.20}$$

The moving sample map has deleted a physical contextual score. Temporal integration supplies only

$$d_c\alpha(c,t) = d_c\alpha_0(c) + d_c \int_{t_*}^{t} \omega_{\mathrm{loc}}(c,s)ds. \tag{14.21}$$

The missing datum is $d_c\alpha_0$, which eq. (14.19) sets to zero. Causal accumulation does not establish that condition. This exact physical-phase derivative concerns ψ; it is different from the nonflat marked connection acting on ϕ.

14.6 Finite generators and scoped obstructions

Freeze the reference frame once at census T, and the canonical local law thereafter follows the original field without a moving-frame term. This removes the live reference from future local evolution while keeping its coherent handoff information. It does not make initialization pointwise.

Alternatively, suppose the prepared section has a known finite generator $x_0(c) = f(c;p)$. Retain its physical-rotation-canonical coefficients, the full chronological history and original time. Compose the original native flows and likelihood appends to evaluate the current law at any context. The normalized response kernels generate the same path law, and the same sample-coordinate argument proves eq. (14.12). No evolved section is projected back onto a finite ansatz. The marked patch above, constant initial public odds and context-constant noncollinear b_i give an explicit family inside the grouping window. It is a restricted preparation family, not an open set of arbitrary contextual functions. Replay work and word length may grow.

A derivative-only repair faces a stronger obstruction than one failed truncation. Suppose a smooth finite-order rule maps $j^k x$ and history to an actual parent law, with no separate phase-bearing reference memory. Require equality of both the minimizing connection classes and tensors for every section in a rich smooth open class. Hold the k-jet fixed and vary the next jet. The fine first score is fixed, while the parent tangent becomes $u + \tau w$. A fixed common minimizing representative would require

$$\|u + \overline{J} A_f + \tau w\|^2 = h_f^{\min} \tag{14.22}$$

on an interval. Its quadratic coefficient is the positive norm of w, hence $w = 0$. Repeating by descending order removes all derivative dependence from the actual law. For random histories the coefficient is $\mathbb{E}_H \|w_H\|^2$ in the joint L^2 experiment; conditional equality is not assumed from averaged equality. The remaining order-zero phase quotient loses eq. (14.13). The proof uses independent smooth jets and equal connection classes. It does not exclude reference memory or constrained generator families.

A different rank argument excludes one smooth finite state encoding every direct response of an arbitrary open section class over the entire patch. More disjoint public-law perturbations than state dimensions give a response Jacobian of rank too large to factor through that state. This is about one global encoding of an indexed experiment, not an added context-relocation action. It excludes neither a field of local states nor a restricted finite generator.

Keeping more temporal history cannot by itself supply an arbitrary initial contextual phase. To see this, rotate the whole initial block at each c by a smooth angle $\kappa(c)$ held constant in time. The native local trajectory rotates by the same angle, while every original history law, persistent coefficient and census result is unchanged. A strictly local retention of the value path that erases constant rotations must therefore give the same complete retained state at each c. This remains true with the entire temporal path available and no bound on memory dimension. Derivatives, neighboring contexts and an initialized contextual phase field are excluded from this particular statement.

At a point with $\kappa = 0$ and $X\kappa = \lambda$, the fine contextual score gains $\lambda \ell_{\mathrm{phys}}$. Conditional private independence makes this score orthogonal to the marked orbit, and

$$\|\ell_{\mathrm{phys}}\|^2 = \mathbb{E}_H \sum_i |B_i^h| \frac{I_1(|B_i^h|)}{I_0(|B_i^h|)} > 0. \tag{14.23}$$

The fine minimized cost therefore contains a positive λ^2 coefficient, whereas the parent actual family is unchanged. They cannot agree on an interval of phase-gradient variations. Small variations preserve the original strict nonflat window. This excludes rotation-invariant local value-path memory for the rich open preparation class, even with unlimited temporal storage. Regular stochastic retention kernels with the same complete invariance and phase-independent randomization obey the same argument.

There is also a class result for smooth finite-dimensional deterministic memories reading a uniformly bounded contextual jet. Suppose initialization, evolution, record updates and actual-law readout use $j^k x$ for one finite k, with no supplied reference or contextual phase field. On a fixed finite record branch the augmented jet-memory ODE closes. Its memory is a smooth function of the initial jet. The native fixed-record flow and its jet prolongation are locally invertible on their reached regular image, so the parent actual law is a smooth function $F_{t,h}(c, j^k x(t))$. The positive-norm argument in

eq. (14.22) applies when both the original minimizing affine classes and tensors must agree. Thus adding arbitrarily many temporal recurrence steps within this smooth bounded-jet class does not evade that obstruction. The past-law initialization and joint-L^2 qualifications remain necessary.

Initial coherence restricts the phase-gradient variations, so its causal construction does not contradict either obstruction. A causal geometrically faithful parent on arbitrary preparations remains inconclusive. Unbounded contextual-order memories, differently specified infinite-dimensional memories and stochastic mechanisms outside the value-path theorem have not all been classified. Any proposed calibration must come from information allowed by the original protocol, which has no context-relocation action. Interaction-created invisibility and a second compressive handoff remain open.

Chapter 15

Coupling Statistical Change Across Contexts

A context-local law flow can change a statistical section without comparing its values at neighboring contexts. Giving the connection a dynamical role requires an energy that reads those comparisons. The construction here penalizes the change of collective statistical shape after a common frame motion has been removed. Its gradient then adds a neighboring-context force to the reciprocal law dynamics. On a finite contextual graph this force is a sum of aligned geodesic displacements. On a continuum base it becomes a diffusion of shape coupled to local frame and routing equations.

15.1 The statistical material and its local forces

At each context, agent i has normalized laws q_i and s_i on standard Borel state and model spaces. Restrict its joint law to $\zeta_i = q_i \otimes s_i$, with each factor in a finite-dimensional smooth statistical family. Assume quadratic-mean differentiability, centered square-integrable directional scores ℓ_U, and a smooth positive-definite Fisher tensor after redundant coordinates have been removed. The family need not be Gaussian. Fixed bimeasurable comparisons place the agent laws in compatible target spaces, giving $p = (p_i)_i \in \mathcal{N}$. Use the actual normalized product experiment and its metric:

$$P_p = \bigotimes_i p_i, \qquad \mathsf{G}_p(U,W) = \sum_i g^F_{p_i}(U_i, W_i), \qquad g^F_p(U,W) = \mathbb{E}_p[\ell_U \ell_W]. \tag{15.1}$$

Product scores add and their cross expectations vanish. Within each agent, the declared product restriction gives the sum of belief and model Fisher metrics. An unrestricted joint law would also carry dependence directions. The restriction is part of this dynamical model; the unrestricted law flow need not preserve it.

Choose proper references $r_i(dm)$, normalized evaluators $G_i(m, dk)$, and normalized channels $\mathcal{K}^q_{ij}[p]$, $\mathcal{K}^s_{ij}[p]$. The channel rules may depend on all current laws but not on source rows. Put $R_i = r_i G_i$, $D^q_{ij} = D_{\mathrm{KL}}(q_i \| q_j \mathcal{K}^q_{ij}[p])$, and $D^s_{ij} = D_{\mathrm{KL}}(s_i \| s_j \mathcal{K}^s_{ij}[p])$. For finite nonempty source sets,

positive normalized prior rows π_i^q, π_i^s, and temperatures $T_q, T_s > 0$, the local potential is

$$\begin{aligned} V(p,\beta,\gamma) = &\sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) \\ &+ \sum_i \left[\sum_j \beta_{ij} D_{ij}^q + T_q D_{\mathrm{KL}}(\beta_i \| \pi_i^q) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ij} D_{ij}^s + T_s D_{\mathrm{KL}}(\gamma_i \| \pi_i^s) \right]. \end{aligned} \tag{15.2}$$

All displayed terms must be finite and sufficiently smooth on the chosen chart. Smooth Fisher geometry alone does not imply smooth KL costs. Data are fixed in evolution time; every declared law dependence is differentiated. In a justified density calculation this includes

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b) da - \int (a/b) db, \qquad d(q_j \mathcal{K}[p]) = (dq_j)\mathcal{K}[p] + q_j D\mathcal{K}[p](dp). \tag{15.3}$$

Thus a source moves in response to its receivers as well as its own costs. This is the reciprocal law potential. The product experiment in eq. (15.1) specifies its mobility; it does not make this potential an ELBO of a fixed population model. Own-block per-agent updates omit some of eq. (15.3) and have no general total-energy dissipation identity.

One finite-dimensional Lie group G acts diagonally on $\mathcal{N}$ by sample pushforward, with its respective actions on the two sectors. Assume the family is stable under this action. Sample bijections preserve Fisher inner products, so the action is isometric. Write $J_p\xi = \partial_a|_{a=0} \exp(a\xi) \cdot p$ for its fundamental tangent. Comparisons remain fixed in the evolution. Making them state dependent would change both the metric pullback and the variation.

15.2 A finite graph of contextual comparisons

Let the contextual vertices have masses $m_v > 0$ and the unoriented edges have weights $w_e > 0$. Choose one orientation $e : u \to v$ and an auxiliary group element U_e carrying target coordinates from u to v. Define

$$c_e(p_u, p_v, U_e) = \frac{1}{2} d_{\mathcal{N}}(p_v, U_e \cdot p_u)^2, \qquad \mathcal{C} = \sum_e w_e c_e. \tag{15.4}$$

The distance is intrinsic to the admitted Fisher family. Reversing this geometric edge uses U_e^{-1} and leaves its cost unchanged. Independent attending links between agents at one context remain separate data; their opposite directions need not be inverses. Finite edge transports are group elements rather than connection one-forms [29].

Hypothesis 15.1 (A regular finite alignment branch). For every edge, choose a stationary transport where the compared endpoints have a unique minimizing geodesic, squared distance is smooth, and the transport Hessian is positive definite in group coordinates. Continue only the resulting local minimizing branch $U_e^*(p_u, p_v)$ while these conditions hold.

The first variation of half squared distance at an endpoint is minus the geodesic logarithm toward the other endpoint. Varying U_e as $\exp(a\xi)U_e$ therefore gives

$$\mathsf{G}_{U_e p_u}\big(\mathrm{Log}_{U_e p_u} p_v, J_{U_e p_u}\xi\big) = 0. \tag{15.5}$$

The residual is orthogonal to a common frame motion. The implicit-function theorem supplies the smooth branch in hypothesis 15.1. Its transport derivative vanishes, so differentiation of $\mathcal{C}_*(p) = \mathcal{C}(p, U^*(p))$ keeps only endpoint derivatives. A local minimizing branch need not realize the global infimum over G.

Give each interior source row its categorical Fisher metric and use

$$\begin{aligned} \mathbb{G} &= \sum_v m_v \left(\mathsf{G}_{p_v} + \sum_i g^F_{\beta_{iv}} + \sum_i g^F_{\gamma_{iv}} \right), \\ \mathcal{F}_* &= \sum_v m_v V_v + \lambda \mathcal{C}_*, \qquad \lambda \geq 0. \end{aligned} \tag{15.6}$$

Up to the constant $\sum_v m_v$, this metric is the Fisher metric of an experiment that first samples the context with probability proportional to m_v, then samples its product laws and independent source labels. Its natural-gradient velocity [1] is

$$\dot{p}_v = -\operatorname{grad}_{\mathsf{G}} V_v + \frac{\lambda}{m_v} \sum_{e=\{u,v\}} w_e \,\mathrm{Log}_{p_v}(U^*_{v\leftarrow u} \cdot p_u). \tag{15.7}$$

Here the reversed orientation uses the inverse transport. The contextual force points toward aligned neighbors. The masses cancel from the local force, so zero coupling recovers the chosen unit-mobility reciprocal flow.

For either sector, write its row as η, its temperature as T, and $h_j = D_j + T\log(\eta_j/\pi_j)$. Its equation and the total dissipation are

$$\dot{\eta}_j = -\eta_j \left(h_j - \sum_l \eta_l h_l \right), \qquad \frac{d\mathcal{F}_*}{dt} = -\|\dot{p}, \dot{\beta}, \dot{\gamma}\|^2_{\mathbb{G}}. \tag{15.8}$$

Pairing the centered row force with a mass-zero row variation gives minus its energy derivative. Pairing the full velocity with $d\mathcal{F}_*$ proves the second identity. With C^3 local potentials, the reduced field is C^1 in ordinary law and interior-row coordinates, so the integral equation is a contraction on a sufficiently short interval. This gives a unique local solution and its maximal continuation in the regular domain. Energy decrease does not prevent escape from that domain.

Free action alone does not supply the branch hypothesis. For two planar Gaussian location means, let $SO(2)$ act on both columns and take $X_\epsilon = (e_1, (1+\epsilon)e_2)$ and $Y = (e_1, -e_2)$. Both configurations have trivial stabilizer near zero, but direct rotation gives

$$\min_R \frac{1}{2}\|X_\epsilon - RY\|^2_F = 2 + \epsilon + \frac{1}{2}\epsilon^2 - |\epsilon|. \tag{15.9}$$

The minimizer switches from I to $-I$, with every rotation minimizing at zero. The reduced energy is not differentiable there. Conversely, for compact G acting freely at p_0, agreement of the contextual endpoints at (p_0, p_0) has a unique minimizer and positive Hessian $\mathsf{G}(J\cdot, J\cdot)$; regular alignments exist near that agreement.

15.3 A continuum energy and its auxiliary connection

Supply the base $\mathcal{C} = \mathbb{T}^d = \mathbb{R}^d/(2\pi\mathbb{Z})^d$ with its flat metric, normalized measure $d\mu = (2\pi)^{-d}d^dx$, and periodic data. Work in the trivial principal bundle, with comparisons independent of x in this trivialization. For a periodic Lie-algebra-valued one-form $A = A_\mu dx^\mu$, set

$$\begin{aligned} D^A_\mu p &= \partial_\mu p + J_p A_\mu, \\ \mathcal{F}[p, b, A] &= \int_{\mathcal{C}} \left[V(x, p, b) + \frac{\lambda}{2} \sum_\mu \|D^A_\mu p\|^2_{\mathsf{G}} \right] d\mu. \end{aligned} \tag{15.10}$$

where b collects row log odds and $\lambda > 0$. The connection has no separate kinetic term or evolution law. Assume a free proper isometric action on an invariant open region $\mathcal{N}_0$. Then the locked inertia $\mathfrak{k}_p(\xi, \eta) = \mathsf{G}_p(J_p\xi, J_p\eta)$ is positive, and the auxiliary equation is the strictly convex quadratic minimization

$$\mathfrak{k}_p(A^*_\mu, \xi) = -\mathsf{G}_p(\partial_\mu p, J_p\xi), \qquad D^{A^*}_\mu p = P_H \partial_\mu p. \tag{15.11}$$

Here P_H is Fisher-orthogonal projection off the group orbit. This is the mechanical connection, familiar from separating rotations and internal shape [32]. Expanding a finite alignment as $U^*(\epsilon) = \exp[-\epsilon A(X) + O(\epsilon^2)]$ gives the residual $\epsilon(dp(X) + J_pA(X)) + O(\epsilon^2)$, and eq. (15.5) reduces to eq. (15.11). This local expansion establishes consistency, without supplying a graph-refinement theorem.

Use one adapted chart $p = \Psi(u, r)$, where u parametrizes the quotient $\mathcal{N}_0/G$ and r the group fiber. The metric has the form

$$\mathsf{G} = h_{ab}(u)du^a du^b + k_{\alpha\beta}(u, r)(dr^\alpha + C^\alpha_a du^a)(dr^\beta + C^\beta_b du^b). \tag{15.12}$$

Both h and k are positive. The horizontal vectors are $H_a = \partial_{u^a} - C^\alpha_a \partial_{r^\alpha}$, and h is the quotient metric of the Riemannian submersion [38]. Auxiliary elimination gives

$$\mathcal{F}_*[u, r, b] = \int_{\mathcal{C}} \left[V(x, u, r, b) + \frac{\lambda}{2} h_{ab}(u)\partial_\mu u^a \partial_\mu u^b \right] d\mu. \tag{15.13}$$

Thus contextual energy sees shape variation. Pure common frame variation costs zero, and a transitive group action leaves no shape directions to regularize.

15.4 Diffusion of shape coupled to frame and row motion

For a row with l entries, take $b_j = \log(\beta_j/\beta_l)$, $j < l$. Its Fisher matrix is $B_{jk} = \beta_j\delta_{jk} - \beta_j\beta_k$; collect these positive blocks for all rows. The field metric is the integral of eq. (15.12) and $db^T B db$. It is the Fisher metric of a normalized experiment that samples $x \sim \mu$ and then the local product laws and

source labels. Define the quotient tension

$$\tau_h(u)^a = \Delta u^a + \Gamma^a_{bc}(u)\partial_\mu u^b \partial_\mu u^c, \tag{15.14}$$

with Γ the Levi-Civita coefficients of h. Periodic integration by parts gives the energy derivatives $V_u - \lambda h \tau_h$, V_r, and V_b. The metric-gradient equations are

$$\begin{aligned} h u_t + C^T k(r_t + C u_t) &= -V_u + \lambda h \tau_h, \\ k(r_t + C u_t) &= -V_r, \qquad B b_t = -V_b. \end{aligned} \tag{15.15}$$

Solving these equations yields

$$\begin{aligned} u_t &= \lambda \Delta u + \lambda \Gamma(u)(\partial_\mu u, \partial_\mu u) - h^{-1}(V_u - C^T V_r), \\ r_t &= -C(u,r) u_t - k(u,r)^{-1} V_r, \qquad b_t = -B(b)^{-1} V_b. \end{aligned} \tag{15.16}$$

Fixed references may distinguish a frame direction, so V_r need not vanish. The $C^T V_r$ term is then required by the Fisher metric. Transforming references, evaluators, and channels together with the laws is a passive covariance statement; it supplies no active symmetry with those data held fixed. A smooth frame change fixed in evolution time obeys $A' = gAg^{-1} - (dg)g^{-1}$ and carries the equations covariantly. A time-dependent change adds its coordinate velocity.

The shape operator is the harmonic-map heat operator [14]. The complete law equation is degenerate in frame directions: its frame equation contains u_t rather than an independent frame Laplacian. For each source row, eq. (15.16) reads

$$\partial_t b_j = -(D_j - D_l) - T\left(b_j - \log \frac{\pi_j}{\pi_l}\right). \tag{15.17}$$

Bounded costs on a finite interval bound the log odds and keep rows positive. They do not provide a separate spatial smoothing estimate for rows. Instantaneous Gibbs rows $\beta^*_j \propto \pi_j e^{-D_j/T}$ instead replace their energy by $-T \log \sum_j \pi_j e^{-D_j/T}$, producing a different law-only flow.

Theorem 15.2 (Local continuum evolution). Assume the periodic geometry above, the declared regular finite-dimensional product family, a free proper action, and one adapted chart with smooth metric coefficients and finite smooth $V(x,u,r,b)$ on a neighborhood of the initial image. Fix $\lambda > 0$, an integer $s > d/2 + 2$, and $m = s - 1$. If $u_0 \in H^s(\mathbb{T}^d)$, $r_0, b_0 \in H^m(\mathbb{T}^d)$, and their pointwise image lies in a compact subset of the regular interior chart, then a unique local solution satisfies

$$\begin{aligned} u &\in C_t H^s_x \cap L^2_t H^{s+1}_x \cap H^1_t H^m_x, \\ r, b &\in C_t H^m_x \cap H^1_t H^m_x, \qquad b \in C^1_t H^m_x. \end{aligned} \tag{15.18}$$

It depends locally Lipschitz-continuously on the initial data in these norms on a common short interval, and continues while $\|u\|_{H^s} + \|r\|_{H^m} + \|b\|_{H^m}$ stays bounded and its image remains in a fixed compact subset of the chart.

Proof. Write the first equation as $u_t - \lambda \Delta u = F(z)$ with $z = (u, r, b)$, and the others as $r_t = -C(z) u_t + Q(z)$ and $b_t = L(z)$. Since $m > d/2$, products are bounded in H^m. Smooth composition on bounded sets with compact image is locally Lipschitz. The gradient product in F also lies in

H^m, because $u \in H^s$ gives $\partial_\mu u \in H^m$. Thus F, C, Q, L are bounded and locally Lipschitz from $H^s \times H^m \times H^m$ into H^m.

Fourier expansion of the forced heat equation gives, for $T \le 1$,

$$\|v\|_{C_t H^s_x} + \|v\|_{L^2_t H^{s+1}_x} + \|v_t\|_{L^2_t H^m_x} \le K_\lambda\big(\|v_0\|_{H^s} + \|f\|_{L^2_t H^m_x}\big). \tag{15.19}$$

For each nonzero Fourier mode, multiply its equation by the H^s weight times its conjugate and apply Young's inequality; $|k|^2(1+|k|^2)^s$ controls the H^{s+1} weight. The zero mode is bounded by integrating its forcing. The equation itself bounds v_t, proving the estimate. Fourier approximation gives continuity at the initial time.

On continuous paths in $H^s \times H^m \times H^m$, first solve the heat equation with forcing $F(z)$ to obtain u^+. Then define

$$r^+(t) = r_0 + \int_0^t [-C(z)u_t^+ + Q(z)]da, \qquad b^+(t) = b_0 + \int_0^t L(z)da. \tag{15.20}$$

The new heat derivative appears here, so the input path needs no time derivative. The heat estimate bounds $\|u_t^+\|_{L^2 H^m}$ by a constant A_T bounded as $T \downarrow 0$. For two inputs at distance D in the continuous-path norm, the zero-initial heat estimate bounds their output heat difference and derivative difference by $K_\lambda L_F \sqrt{T} D$. Splitting the fiber difference as $C(z)(u_t^+ - \widetilde{u}_t^+) + (C(z) - C(\widetilde{z}))\widetilde{u}_t^+$ then bounds it by a constant times $(T + \sqrt{T}A_T)D$; the row difference is bounded by $TL_L D$. The map is therefore a contraction for small T.

It preserves a bounded path set with small displacement in H^m: the homogeneous heat displacement is at most $\sqrt{\lambda T}\|u_0\|_{H^s}$, the forced displacement is $O(T)$, and the fiber displacement is $O(\sqrt{T}A_T + T)$. Sobolev embedding preserves the interior chart. The fixed point has eq. (15.18) by the heat estimate and the two integral equations. Keeping initial-data differences proves the stated dependence and uniqueness. Bounded norms and a fixed compact image make all these constants uniform, giving a uniform restart interval and hence the continuation assertion. □

The energy chain rule is valid in eq. (15.18). Pairing the gradient equations with their velocities gives, almost everywhere in time,

$$\frac{d\mathcal{F}_*}{dt} = -\int_{\mathcal{C}} \left[u_t^T h u_t + V_r^T k^{-1} V_r + V_b^T B^{-1} V_b\right] d\mu. \tag{15.21}$$

The energy is absolutely continuous and the integrated identity holds from the initial time. Its decrease controls the total energy, without supplying the higher Sobolev bounds in the continuation criterion. Setting $\lambda = 0$ instead gives a locally Lipschitz ODE in H^m for all coordinates. The extra derivative assigned to u need not persist in that ODE, and the positive-coupling heat estimates are not uniform as $\lambda \downarrow 0$.

15.5 An exact amplitude that diffuses across contexts

Take two agents on $\mathcal{C} = S^1$, with $q_i = \mathcal{N}(\mu_i, I_2)$, $s_i = \mathcal{N}(\nu_i, I_2)$, $r_i = \mathcal{N}(0, I_2)$, and $G_i(m, dk) = \mathcal{N}(m, I_2)(dk)$. Each agent attends only to the other through an identity channel in each sector.

The rows are therefore singleton rows. Direct integration gives the local contribution $\frac{1}{2}\|\mu_i - \nu_i\|^2 + \frac{1}{2}\|\nu_i\|^2 + 1$, so both sectors participate. With $P = (\mu_1, \nu_1, \mu_2, \nu_2)$ the full reciprocal potential is

$$V(P) = 2 + \frac{1}{2}\langle P, PH\rangle_F, \qquad H = \begin{pmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & 0 & -2 \\ -2 & 0 & 3 & -1 \\ 0 & -2 & -1 & 4 \end{pmatrix}. \tag{15.22}$$

The Fisher metric is Frobenius. A common $SO(2)$ rotation acts on all columns and is free whenever $P \neq 0$. For $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, write $A_x = \alpha J$. The auxiliary equation and law flow become

$$\alpha_* = -\frac{\langle P_x, JP\rangle_F}{\|P\|_F^2}, \qquad P_t = -PH + \lambda(\partial_x + \alpha_* J)^2 P. \tag{15.23}$$

The derivative in the last expression acts on α_* as well as P. It follows by first varying with α fixed, using skew adjointness of $\partial_x + \alpha J$, and then substituting its auxiliary minimizer.

Put $\tau = (\sqrt{5} - 1)/2$, $\kappa = 5 - \tau$, $v = (1, \tau, -1, -\tau)^T$, and $P_0 = e_1 v^T$. Let $R_\varphi = \exp(\varphi J)$. Since $Hv = \kappa v$, the field $P(t, x) = R_{\varphi(x)} a(t, x) P_0$, with smooth periodic phase $|\varphi| < \pi/4$ and positive amplitude a, obeys

$$\alpha_* = -\varphi_x, \qquad a_t = \lambda a_{xx} - \kappa a, \qquad \varphi_t = 0. \tag{15.24}$$

The connection removes phase variation from the spatial derivative. The remaining amplitude is gauge invariant: $a = \|P\|_F / \|P_0\|_F$. For $n \geq 1$ and $a_0 > |a_1| > 0$, an exact solution is

$$a(t, x) = a_0 e^{-\kappa t} + a_1 e^{-(\kappa + \lambda n^2)t} \cos(nx). \tag{15.25}$$

It remains positive at every finite time. Relative to the constant mode, the oscillatory mode decays by $e^{-\lambda n^2 t}$; without contextual coupling that ratio stays fixed. Initial profiles with frequencies one and two have identical laws and first derivatives at $x = 0$, but their local velocities differ by $-3\lambda a_1 R_{\varphi(0)} P_0$. Neighboring contextual variation has become a force. The statistical shape approaches the singular zero-mean orbit only at infinite time in this explicit solution.

This example uses a restricted Gaussian location family and singleton rows; it supplies no invariant Gaussian family for the full density flow and no test of nontrivial row dynamics. The continuum theorem concerns a supplied torus, one regular chart, fixed comparisons, and a fixed agent census. It leaves unrestricted density families, singular orbit changes, nontrivial base bundles, boundary problems, graph refinement, and global continuation open. Independent rows and communication links acquire no separate diffusion, and pure common frame variation remains unpenalized. Chapter 16 separately constructs contextual displacement transports from regular law, interface, and communication-link data. Its comparison rule need not minimize the Fisher energy used here, so its substitution would require a new energy variation and evolution analysis.

Chapter 16

From Agent Links to Base Transport

Agent frames can select a connection over the contextual base if the model says how to compare them at neighboring contexts. Declaring one marked frame parallel gives flat local transport. A collection of law-derived vectors and linked messages can instead give curvature, because its members need not fit into one parallel frame. A second construction aligns entire link configurations and can be curved even when every directed pair is reciprocal. In both cases the comparison rule is part of the model.

16.1 What a frame supplies

Let $\mathcal{C}$ be a smooth contextual base and let $V = \mathbb{R}^K$ be one content fiber in reference coordinates. Agent i presents it through $U_i(c) : V \to E_i$. A passive change in the agent's displayed coordinates sends U_i to $g_i U_i$. A change of common reference instead sends U_i to $U_i h$ and every reference vector v to $h^{-1}v$. All these maps are assumed twice continuously differentiable on the patch considered.

A marked frame is a matrix $Q(c)$ whose columns are a basis of actual retained vectors. Comparing the same columns at two contexts gives

$$T_Q(c', c) = Q(c')Q(c)^{-1}, \qquad a_Q = -dQQ^{-1}. \tag{16.1}$$

Here the reference derivative is $D^a = d + a$. The map equals identity at zero displacement, and differentiating the first endpoint gives $T_Q(c + \delta c, c) = I - a_Q(\delta c) + O(|\delta c|^2)$. It also composes exactly: the intermediate $Q(c')^{-1}Q(c')$ cancels. Since $D^{a_Q}Q = 0$, every marked column is parallel. Differentiating Q^{-1} gives $da_Q + a_Q \wedge a_Q = 0$.

Under reference rechoice $Q' = h^{-1}Q$, this construction gives

$$a' = h^{-1}ah + h^{-1}dh, \qquad B_i = U_i a U_i^{-1} - dU_i U_i^{-1}. \tag{16.2}$$

The displayed connection B_i is independent of the common reference and transforms as $B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}$ under passive agent re-expression. Using only a coordinate presentation and setting $B_i = -dU_i U_i^{-1}$ is the special choice $a = 0$. Resetting a to zero after a context-dependent reference change changes the connection. A choice of coordinates alone therefore supplies no reference-independent comparison.

Several marked frames can disagree about parallelism. If a_1, a_2 are flat connections on the same bundle, $\Delta = a_1 - a_2$, and λ is a smooth scalar weight, the affine average has

$$a = a_2 + \lambda\Delta, \qquad F^a = d\lambda \wedge \Delta - \lambda(1 - \lambda)\Delta \wedge \Delta. \tag{16.3}$$

Weight variation and noncommuting disagreements can produce curvature. This uses a declared comparison of already defined connections; averaging square frames and applying eq. (16.1) still gives a flat connection.

16.2 Turning messages into a displacement map

At a context, write an invertible directed agent link as $\Theta_{ij} = U_i K_{ij} U_j^{-1}$, where $K'_{ij} = h^{-1} K_{ij} h$. It maps the source agent's fiber coordinates to the receiver's at the same context. Contextual transport needs a map from the reference fiber at c to the reference fiber at c'. The following construction supplies one from smoothly varying contextual messages.

Choose a fixed labeled message space $H = \mathbb{R}^r$ with a stated Euclidean metric. Its labels persist across contexts. For example, if a normalized law ν_j has a finite mean μ_j, its reference mean is $\bar{\mu}_j = U_j^{-1} \mu_j$. The vectors $\bar{\mu}_j$, $K_{ij}\bar{\mu}_j$, and columns of receiving maps $O_i : H_i \to V$ can be assembled into a matrix

$$X(c) : H \longrightarrow V, \qquad X' = h^{-1} X. \tag{16.4}$$

Column selection, weights, and message metric are constitutive data. Smooth law-dependent weights are allowed, but their derivatives then contribute to dX. Finite moments and smooth moment fields do not require Gaussian laws. They also do not follow merely from normalization.

Assume X is C^2 and has row rank K on an open region. Define

$$H_X = X X^T, \qquad Q_X = X^T H_X^{-1}, \qquad P_X = Q_X X. \tag{16.5}$$

The map Q_X lifts a vector v into its minimum-norm message coordinates, and $X Q_X v = v$. The projector P_X selects the part of message space that can be read in the content fiber. Lift at the source context and decode at the target context:

$$C(c', c) = X(c') Q_X(c), \qquad C(c, c) = I. \tag{16.6}$$

This identity is exact. Continuity makes C invertible near each diagonal point. Since $Q'_X = Q_X h$, its endpoint law is

$$C'(c', c) = h(c')^{-1} C(c', c) h(c). \tag{16.7}$$

Taylor expansion now derives the reference connection:

$$C(c + \delta c, c) = I - a(\delta c) + O(|\delta c|^2), \qquad a = -dX Q_X. \tag{16.8}$$

The first derivative is linear in the displacement. Inserting $h^{-1} X$ in this formula produces the inhomogeneous term $h^{-1} dh$ in eq. (16.2). Compatible reference expressions on overlaps therefore give a connection on the bundle. A global surjection from a trivial message bundle can exist even when the content bundle is nontrivial; redundancy $r > K$ leaves room for a nontrivial moving subspace. A global square frame would trivialize it.

The minimum-norm choice in eq. (16.5) uses the fixed message metric. If message coordinates are changed by $w' = s(c) w$, the metric must transform too. A message-bundle connection Γ then

gives the intrinsic form

$$a = -(dX - X\Gamma)Q_X, \qquad \Gamma' = s\Gamma s^{-1} - dss^{-1}. \tag{16.9}$$

Fixed labels with $\Gamma = 0$ recover eq. (16.8). Setting Γ to zero again after a varying coordinate change introduces a different comparison rule.

16.3 Why redundant messages can be curved

The residual $dX + aX = dX(I - P_X)$ is the part of the changing message matrix that cannot be absorbed by moving the content frame. Direct differentiation gives

$$F^a = da + a \wedge a = dX \wedge (I - P_X)dX^T H_X^{-1}. \tag{16.10}$$

To see the projection, differentiate $XQ_X = I$ and $P_X = Q_X X$. The resulting identity $dP_X Q_X = dQ_X + Q_X dX Q_X$ combines da and $a \wedge a$. The component of dQ_X proportional to X^T is killed by $I - P_X$, leaving eq. (16.10). This is induced subbundle geometry, related to the classical universal connection construction [34, 37]; the agent-specific choice is how laws and links supply X.

For a square invertible X, $P_X = I$ and curvature vanishes. With redundant columns, changing the readable subspace can give curvature. On a two-dimensional base, take

$$X = \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \end{pmatrix}, \qquad v = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad s = 1 + x^2 + y^2. \tag{16.11}$$

Then

$$a = -\frac{(dv)v^T}{s}, \qquad F^a = \frac{1}{s^2}\begin{pmatrix} -xy & 1 + x^2 \\ -(1 + y^2) & xy \end{pmatrix} dx \wedge dy. \tag{16.12}$$

At the origin the connection coefficients vanish while the curvature is a nonzero rotation generator. The value of a connection at one point does not determine its curvature there.

A law-and-link realization uses receiving columns e_1, e_2, a product Laplace source law with unit coordinate variances and mean $(x, 1)^T$, and the reciprocal shear pair

$$K_{ij} = \exp(yE_{21}) = \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix}, \qquad K_{ji} = K_{ij}^{-1}. \tag{16.13}$$

Its linked mean is $(x, 1 + xy)^T$. Use this as the third column of X. The map $(x, y) \mapsto (x, 1 + xy)$ has Jacobian determinant x, so the pulled-back curvature above is nonzero wherever $x \neq 0$. Receiving axes, labels, and link rules are declared data. The example uses general normalized laws without a Gaussian closure assumption.

This selector preserves the induced fiber metric H_X^{-1}, because

$$dH_X + aH_X + H_X a^T = 0, \qquad \operatorname{tr} a = -\frac{1}{2} d \log\det H_X, \qquad \operatorname{tr} F^a = 0. \tag{16.14}$$

Consequently the explicit rule does not generate every general-linear connection. Its induced metric is not automatically the Fisher metric of the agent's normalized statistical experiment.

16.4 Reversal and information in directed pairs

Independent attending edges need not be inverses. Where the spectrum of $N_{ij,j} = \Theta_{ji}\Theta_{ij}$ avoids $(-\infty, 0]$, the principal matrix square root defines

$$S_{ij,j} = N_{ij,j}^{1/2}, \qquad \Omega_{ij} = \Theta_{ij} S_{ij,j}^{-1}, \qquad \Theta_{ij} = \Omega_{ij} S_{ij,j}. \tag{16.15}$$

Similarity covariance of the principal root gives $\Omega_{ji} = \Omega_{ij}^{-1}$ [22]. Keeping Ω and S retains the original directed pair. A message matrix can include both $U_i^{-1}\Omega_{ij}\mu_j$ and $U_i^{-1}\Theta_{ij}\mu_j$. Raw links can also be used directly, without a square-root branch.

For example, let the scalar links satisfy $\Theta_{ij} = \Theta_{ji} = s(c) > 0$ and take unit seed and source means. Then $\Omega_{ij} = 1$ while the backtrack is s^2. Raw messages give $X = [1, s]$ and $a = -sds/(1 + s^2)$; reciprocal-only messages give $X = [1, 1]$ and $a = 0$. Both connections are flat, but their transports differ when s varies. A backtrack can affect the derived connection, although one connection need not encode every distinction in the input links.

Reversal of a contextual path has a different meaning. The finite cross-map C in eq. (16.6) need not obey reversal or composition. Near the diagonal set

$$N(c', c) = C(c, c')C(c', c), \qquad T(c', c) = C(c', c)N(c', c)^{-1/2}. \tag{16.16}$$

The same root calculation gives $T(c, c') = T(c', c)^{-1}$ and the endpoint covariance law. The linear terms in N cancel, so $N = I + O(|\delta c|^2)$ and $T = I - a(\delta c) + O(|\delta c|^2)$. This contextual reconstruction defect is not the agent backtrack in eq. (16.15).

For a piecewise C^2 path γ, exact transport solves

$$\dot{\mathcal{T}} = -a(\dot{\gamma})\mathcal{T}, \qquad \mathcal{T}(0) = I. \tag{16.17}$$

Continuity gives a unique invertible solution, and uniqueness gives concatenation and inverse reversal. Ordered products of either C or T converge to it: each factor differs from the exact short propagator by $O(\Delta t^2)$, while bounded coefficients control product norms. The sum of the local errors is bounded by a constant times the largest partition step. This is the local relation between discrete and continuous connections [29]. An unrefined endpoint map need not already be exact transport along a chosen path.

16.5 One structure group for both sectors

The formula $a = -dXQ_X$ takes values in $\mathfrak{gl}(K)$. A prescribed smaller structure group G requires a different minimization. Let M be a positive fiber metric with $M' = h^T M h$. For a tangent direction Y, choose $a(Y) \in \mathfrak{g}$ to minimize

$$\frac{1}{2}\operatorname{tr}\left[M(dX(Y) + \xi X)(dX(Y) + \xi X)^T\right], \qquad \xi \in \mathfrak{g}. \tag{16.18}$$

Full row rank makes its Hessian positive, because $\xi X = 0$ implies $\xi = 0$. Its unique solution is linear in Y and obeys the affine transformation law. For the unrestricted algebra the normal equation is $(dX + aX)X^T = 0$, recovering eq. (16.8). For a proper subalgebra, the explicit curvature and induced-metric formulas above need not remain valid.

Belief and model sectors carry representations ϱ_q, ϱ_s of one principal group. Sum their comparison costs and minimize over one $\xi \in \mathfrak{g}$, using $d\varrho_q(\xi)X^q$ and $d\varrho_s(\xi)X^s$. The required condition is that their common infinitesimal kernel is zero. Separate minimization over unrelated sector generators would not automatically define one principal connection. Belief and model interaction links may remain distinct; tying them is a separate assumption.

16.6 A curved connection from reciprocal links alone

One can align a target of complete link data without first choosing spanning vector columns. Let $Z(c)$ lie in a finite-dimensional Riemannian target with smooth metric $\mathcal{H}$ and an isometric left G-action. Reference changes act by $Z' = h^{-1} \cdot Z$. Write $J_Z\xi$ for its infinitesimal action and assume J_Z is injective. In normal neighborhoods, align nearby contexts by the local minimizing branch through identity of

$$\frac{1}{2} d_{\mathcal{H}}(Z(c'), g \cdot Z(c))^2. \tag{16.19}$$

At coincident endpoints the group Hessian is the positive form $\mathcal{H}(J_Z\xi, J_Z\eta)$. The implicit-function theorem supplies the local branch. Isometry gives endpoint covariance and, together with symmetry of distance, inversion under reversal. Its first derivative is the mechanical connection [32]:

$$\mathcal{H}(J_Z a(Y), J_Z\eta) = -\mathcal{H}(dZ(Y), J_Z\eta). \tag{16.20}$$

The metric may depend on coordinates of the augmented target, but it must be one smooth invariant metric on that target. Injectivity identifies the identity branch; it excludes neither distant competing minima nor later singularities.

Take $G = SO(3)$ and two reference links on a three-agent star,

$$Q_1 = R_x(x), \qquad Q_2 = R_z(y)R_x(\beta)R_z(-y), \qquad (x, y) \in (0, \pi)^2, \tag{16.21}$$

with fixed $\beta \in (0, \pi)$. Set opposite links to their inverses, so every agent backtrack is identity. Use the product bi-invariant metric on $SO(3)^2$. Under common reference change the pair transforms by simultaneous conjugation. Identify $\mathfrak{so}(3)$ with vectors by $\widehat{v}w = v \times w$ and normalize the metric by $\langle \widehat{v}, \widehat{w} \rangle = v \cdot w$.

For $C_r = I - \mathrm{Ad}_{Q_r}$ and $V_r = dQ_r Q_r^{-1}$, the infinitesimal conjugation is $C_r\xi$. Thus eq. (16.20) becomes

$$\mathcal{K}a = -\sum_{r=1}^{2} C_r^* V_r, \qquad \mathcal{K} = \sum_{r=1}^{2} C_r^* C_r. \tag{16.22}$$

The two rotation axes are e_1 and $(\cos y, \sin y, 0)^T$. They are nonparallel, so no nonzero infinitesimal rotation fixes both links and $\mathcal{K}$ is positive definite. Let $w_x = 4\sin^2(x/2)$ and $w_\beta = 4\sin^2(\beta/2)$. Since

$C_1^* V_1 = 0$, $C_2^* V_2 = w_\beta \widehat{e}_3 dy$, and $\mathcal{K}\widehat{e}_3 = (w_x + w_\beta)\widehat{e}_3$, one obtains

$$a = -\frac{w_\beta}{w_x + w_\beta}\widehat{e}_3 dy, \qquad F^a = \frac{2w_\beta \sin x}{(w_x + w_\beta)^2}\widehat{e}_3 dx \wedge dy \neq 0. \tag{16.23}$$

This curvature concerns changes of the link configuration across the contextual base. It exists with trivial graph backtracks and no agent cycle of length three. Normalized rotationally invariant laws can be attached without changing the calculation, because their infinitesimal orbit terms vanish. The declared link metric, rather than those invariant laws, determines the comparison.

16.7 What remains to be selected

Conjugation cannot reveal a continuous gauge subgroup that fixes every input link. If a nonzero $Z \in \mathfrak{g}$ commutes with all admitted link fields, $h(c) = \exp(f(c)Z)$ leaves those fields and all their contextual derivatives unchanged. At a point where $f = 0$ but $df \neq 0$, affine covariance would nevertheless require $a' = a + Z df$. No selector of those unchanged inputs alone can satisfy that equation. Scalar gauge motion is therefore invisible to defining general-linear conjugation links, while the two nonparallel $SO(3)$ axes above have no common infinitesimal stabilizer. Laws or labeled interfaces can remove an ambiguity only when their combined action identifies it.

Even on a regular region, different column sets and target metrics can select different connections. These choices require a declared agent model. They also determine whether the connection equals the actual-law Fisher selector of chapter 3. The contextual-flow theorem in chapter 15 does not apply to a substituted selector merely because both obey the connection transformation law: the full dependence on laws, rows, interfaces, and links must be included in the energy variation. Rank preservation, singular continuation, inheritance of the link sector, and selection of the comparison rule by dynamics remain separate problems. The base and its topology and dimension are supplied throughout.

Chapter 17

Communication Codes and Learned Alignment

A receiver can infer more variables than a message directly describes because its own model relates the measured and unmeasured variables. That additional structure belongs to the receiver's model. It need not reproduce the sender's structure. A second uncertainty concerns alignment: even when the agents use content spaces of the same dimension, the receiver must learn how the sender uses those coordinates. A code describes what is expressed, a dictionary describes the proposed alignment, and a conditional law describes how a receiver reconstructs what was left unsaid.

All comparisons here occur at one context. There is one principal frame group G, acting on belief and model contents through two representations ϱ_q and ϱ_s. The matrix examples use the belief representation $V = \mathbb{R}^K$ and suppress its sector superscript. The model representation can have a different dimension. One frame change acts in both representations, but a shared dictionary between the sectors is an additional tied-link assumption. This common group does not make the two laws or their communication rules identical.

17.1 What the receiver supplies

Let q be the sender's law in coordinates already aligned with the receiver, and let p be the receiver's prior law. Both are probability measures on a standard Borel content space E. A noiseless code is a measurable map $T : E \to H$. The message law is the pushforward $\mu = T_{\#}q$, meaning that $\mu(A) = q(T^{-1}A)$ for a measurable set A of messages. If the receiver knows this whole message law, it can construct

$$\widehat{q}(dy) = \int p(dy \mid h)\mu(dh). \tag{17.1}$$

The conditional law $p(dy \mid h)$ comes from the receiver. Regular conditional laws exist on the stated spaces [25]; if $q \ll p$, their versions agree at all message values needed, up to a μ-null set.

The reconstruction has the sender's message marginal and the receiver's conditional distribution inside each message fiber. For finite $D_{\mathrm{KL}}(q\|p)$, the chain rule gives

$$D_{\mathrm{KL}}(q\|\widehat{q}) = D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(T_{\#}q\|T_{\#}p). \tag{17.2}$$

To read this identity, factor each law into its message marginal and its conditional law given the message. The logarithm of a density ratio becomes the sum of a marginal logarithm and a conditional logarithm. The two laws q and $\widehat{q}$ share the marginal, so their divergence contains only

the conditional part. This is the information-projection form of Jeffrey conditioning [11, 13]. Nothing in the argument assumes Gaussian contents.

A small Gaussian example makes the missing information visible. Take

$$p = \mathcal{N}(0, I_2), \qquad q = \mathcal{N}((a, b)^T, I_2), \qquad T(y_1, y_2) = y_1. \tag{17.3}$$

The message law is $\mathcal{N}(a, 1)$. Under p, the second coordinate is independent of the first and has law $\mathcal{N}(0, 1)$, so

$$\widehat{q} = \mathcal{N}((a, 0)^T, I_2), \qquad D_{\mathrm{KL}}(q\|\widehat{q}) = \frac{b^2}{2}. \tag{17.4}$$

The full prior disagreement is $(a^2 + b^2)/2$ and the visible disagreement is $a^2/2$. The receiver obtains the first shift exactly and has no message evidence for the second.

Now replace the receiver covariance by

$$\Sigma_p = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad |r| < 1. \tag{17.5}$$

Its conditional model says $y_2 \mid y_1 = h \sim \mathcal{N}(rh, 1 - r^2)$. Receiving the same message law $\mathcal{N}(a, 1)$ produces a reconstructed mean $(a, ra)^T$ and covariance Σ_p. The second mean has changed, although only the first coordinate was sent. It changed because the receiver supplied a correlation. The sender's actual second mean remains b, and these agree only when $b = ra$.

Receiving a single value h is a different experiment from receiving its entire law. With a linear noisy observation $h = Ay + \varepsilon$, a Gaussian receiver prior $\mathcal{N}(m, \Sigma_p)$, and independent noise $\varepsilon \sim \mathcal{N}(0, R)$ with $R > 0$, the posterior mean is

$$m^+ = m + O(h - Am), \qquad O = \Sigma_p A^T (A\Sigma_p A^T + R)^{-1}. \tag{17.6}$$

The posterior covariance is $\Sigma_p - OA\Sigma_p$, and its precision is $\Sigma_p^{-1} + A^T R^{-1} A$. This is Gaussian conditioning, the same gain mechanism used in the Kalman filter [26]. The gain depends on the receiver's correlations and on its model of the code. Its linearity and the absence of the observed value from the posterior covariance are Gaussian features.

For a general noisy Markov kernel $\mathcal{K} : E \rightsquigarrow H$, reconstructing from the message law gives $\widehat{q} = \int p^h (q\mathcal{K})(dh)$, where p^h is the receiver's posterior given h. The corresponding finite-divergence statement is

$$D_{\mathrm{KL}}(q\|\widehat{q}) \le D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}). \tag{17.7}$$

The chain rule first computes the mean conditional discrepancy. Mixing the conditional laws can reduce it, which explains why the noiseless identity becomes an inequality.

17.2 An invertible link does something different

In a linear realization, let $U_i \in G$ be agent i's displayed frame. Sender j has a reference code $\bar{T}_j : V \to H$. Receiver i assigns it a reference dictionary $\bar{\theta}_{ij} \in G$, with displayed link and effective

code

$$\theta_{ij} = U_i \bar{\theta}_{ij} U_j^{-1}, \qquad A_{ij} = \bar{T}_j \bar{\theta}_{ij}^{-1}. \tag{17.8}$$

The first map takes sender coordinates to receiver coordinates. The second asks what message the sender would emit if the content had the receiver's coordinates. This accounts for the inverse in A_{ij}.

Under a passive change $U_i \mapsto g_i U_i$, the displayed link changes as $\theta_{ij} \mapsto g_i \theta_{ij} g_j^{-1}$ and the displayed code as $T_j \mapsto T_j g_j^{-1}$. The message value is unchanged. Changing the common reference requires re-expressing $\bar{T}_j$ and $\bar{\theta}_{ij}$ together. Endpoint covariance alone does not choose a dictionary or prove that its data are independent of that common reference choice.

The loss in a code is not reproduced by exponentiating a code-shaped generator. For the two-dimensional maps

$$T = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad O = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad P = OT = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{17.9}$$

the direct map sends (y_1, y_2) to $(y_1, 0)$. In contrast,

$$e^P = \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix}. \tag{17.10}$$

It dilates the first coordinate and preserves the second. It is invertible. More generally,

$$e^{OT} = I + O\varphi_1(TO)T, \qquad \varphi_1(z) = \frac{e^z - 1}{z}, \qquad \varphi_1(0) = 1, \tag{17.11}$$

so the exponential acts as identity on $\ker T$ [22]. In the earlier reconstruction example, the mean update was instead $(I - P)m + Pa$, combining sender and receiver data through a noninvertible projector.

An unrestricted dictionary is a group element. Writing it as e^{ω} is an extra parameterization, useful near identity. A single real matrix exponential does not cover every element of $\mathrm{GL}^+(K, \mathbb{R})$ [12]. The exact dictionary-identification statements use their declared group domain. Restricting them to an exponential image means restricting their solution sets accordingly.

17.3 What a grounded message identifies

To learn an alignment, suppose the receiver knows a referent vector y and observes a message

$$h = \bar{T}\bar{\theta}^{-1} y + \varepsilon. \tag{17.12}$$

The noise has a fixed law independent of the referent. Referents explore the space with finite positive-definite second moment $M = \mathbb{E}[yy^T]$. This experiment differs from observing one average content repeatedly.

Two dictionaries give the same conditional law for every referent precisely when their effective codes agree. Equivalently,

$$\bar{\theta}'' = \bar{\theta}' s, \qquad s \in \mathrm{Stab}_G(\bar{T}) = \{s \in G : \bar{T}s = \bar{T}\}. \tag{17.13}$$

This statement holds for every additive-noise law on $\mathbb{R}^r$: a probability law has no nonzero translation symmetry. Its characteristic function is nonzero near the origin, where invariance under translation by a would force $e^{\mathrm{i}\langle t,a\rangle} = 1$ for all small t, hence $a = 0$.

For $G = \mathrm{GL}(K, \mathbb{R})$ and a rank-r code, the stabilizer has dimension $K(K-r)$. Indeed, $\bar{T}(s - I) = 0$ says that every column of $s - I$ lies in the $(K-r)$-dimensional kernel of the code, and invertibility is an open condition. The remaining Kr parameters are identifiable. In $K = 2$ with $\bar{T} = (1, 0)$,

$$s = \begin{pmatrix} 1 & 0 \\ u & v \end{pmatrix}, \qquad v \neq 0. \tag{17.14}$$

No number of messages from this code can distinguish $\bar{\theta}$ from $\bar{\theta}s$ under the declared experiment.

Gaussian noise with covariance $R > 0$ gives an explicit local information form. For the perturbation $\bar{\theta}e^{\delta}$,

$$F(\delta, \delta) = \mathrm{tr}(R^{-1}\bar{T}\delta C \delta^T \bar{T}^T), \qquad C = \bar{\theta}^{-1} M \bar{\theta}^{-T}. \tag{17.15}$$

Because R and C are positive definite, this vanishes exactly when $\bar{T}\delta = 0$. For regular non-Gaussian location noise, the same formula uses its location Fisher matrix instead of R^{-1}, with the same kernel when that matrix is positive definite.

The record experiment matters. Let $T = I_2$, let every observed referent equal $c = e_1$, and perturb by $\delta = E_{22}$, the matrix with only its lower-right entry nonzero. Then $\delta c = 0$, so that one referent sees no change. But $T\delta \neq 0$, and referents with positive-definite second moment detect it. A direction invisible at one operating content is not necessarily invisible to a grounded code.

17.4 Several modalities and one shared dictionary

Suppose speech, diagrams, or other modalities have codes $\bar{T}^c$ and use the same dictionary. The unresolved subgroup is the intersection of their stabilizers. If r_c is the rank of code c, their identifiable dimension in $\mathrm{GL}(K, \mathbb{R})$ is

$$K \dim \left(\sum_c \mathrm{row}\, \bar{T}^c \right) \leq K \min \left(K, \sum_c r_c \right). \tag{17.16}$$

When the total width is at most K, reaching this bound requires independent row spaces. When the total width is at least K, it requires only that the combined rows span V^*. Two identical full-rank codes already saturate the dimension K^2, although their row spaces coincide.

Learning a separate reading S_c for every code does not always produce one shared dictionary. Write T_* and S_* for the stacked matrices. There is an invertible θ with $T_*\theta^{-1} = S_*$ precisely when

$$\ker(T_*^T) = \ker(S_*^T). \tag{17.17}$$

These kernels contain the linear relations among rows. An invertible change of content coordinates preserves every such relation. Conversely, matching relations define an isomorphism of the row spaces that can be extended to the full dual space.

For example, $T_1 = T_2 = (1, 0)$ requires identical readings. The choice $S_1 = (1, 0)$ and $S_2 = (2, 0)$ has the same stacked rank, one, but violates that requirement. Equal rank is therefore insufficient. If both stacks have full row rank with total width at most K, their relation spaces are both zero and a shared dictionary does exist.

Routing also controls what is learned. If modality c listens to sender j with probability $\beta_{ij}^{(c)}$, observed source labels give information $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$. Conditional independence of messages makes the information add. A modality with zero source weight contributes nothing about that sender. If source labels are hidden, their uncertainty reduces information in the positive-semidefinite order, by the missing-information identity [33]. Extra modalities help only through the distinctions their actual observations resolve.

17.5 Loops, agreement, and the chosen observable

The dictionary $\bar{\theta}_{ij}$ belongs to receiver i and the opposite dictionary $\bar{\theta}_{ji}$ belongs to receiver j. They can be learned independently. Traversing the opposite attending edge therefore need not undo the first dictionary. Geometric reversal of the first map would use its inverse instead.

For a directed loop $\gamma = (i_0 \to i_1 \to \cdots \to i_k = i_0)$, its ordered product is

$$H_\gamma = \bar{\theta}_{i_k i_{k-1}} \cdots \bar{\theta}_{i_1 i_0}. \tag{17.18}$$

It compares the interpretation obtained after returning along the loop with the starting interpretation. Such loops belong to the communication graph at one context; they do not by themselves define a connection between contexts.

If the directed graph is strongly connected, a transported consensus law obeying $q_i = (\bar{\theta}_{ij})_\# q_j$ on every edge is determined by a base law invariant under every loop based there. To see why, transport that law along a directed path to each vertex. Two paths give the same result because a return path closes each into a loop, and transport along the return path is invertible. Strong connectivity supplies the required return paths. For means, the allowed base values are the common fixed vectors of the matrices H_γ. This is the same fixed-section mechanism underlying connection Laplacians [3, 43].

There are two distinct ways to measure a loop. The law divergence $D_{\mathrm{KL}}(q \| (H_\gamma)_\# q)$ vanishes when the loop stabilizes the law. A link observable dressed by a positive-definite covariance is instead

$$Q(H; \Sigma) = \frac{1}{2} \left\| \Sigma^{-1/2} (I - H) \Sigma^{1/2} \right\|_F^2. \tag{17.19}$$

It vanishes only at $H = I$ and remains unchanged when H and Σ are re-expressed together. For the quarter-turn

$$H = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad q = \mathcal{N}(0, I_2), \tag{17.20}$$

the transported law equals q, so the law divergence is zero. But $\|I - H\|_F^2 = 4$, giving $Q(H; I_2) = 2$. Stabilizer blindness belongs to the law comparison, not to this dressed link defect.

Grounded edge records can identify the dictionaries, and hence their loop product, without relaying one realized sample around the entire loop. Identifiability from these records, comparison of transported marginal laws, and observation of one sample along several routes are different experiments.

17.6 Message count and source confidence

Let J be a source label with prior probabilities $\pi_j > 0$. Conditional on $J = j$, suppose the receiver models n messages as independent draws with density ν_j. Bayes' rule gives

$$\mathbb{P}(J = j \mid h_{1:n}) = \frac{\pi_j e^{-nC_{j,n}}}{\sum_k \pi_k e^{-nC_{k,n}}}, \qquad C_{j,n} = -\frac{1}{n}\sum_{t=1}^{n} \log \nu_j(h_t). \tag{17.21}$$

The temperature $1/n$ is an exact rewriting of this likelihood model. It counts conditionally independent messages; the width of a real vector alone is not a bit rate.

If the actual messages are independent with law ν and every candidate log density is integrable under ν, the strong law gives

$$\frac{1}{n} \log \frac{\mathbb{P}(J = j \mid h_{1:n})}{\mathbb{P}(J = k \mid h_{1:n})} \longrightarrow \mathbb{E}_\nu[\log \nu_j - \log \nu_k]. \tag{17.22}$$

Only when both candidate KL divergences are finite may this be written as $D_{\mathrm{KL}}(\nu\|\nu_k) - D_{\mathrm{KL}}(\nu\|\nu_j)$. The expected log-likelihood ratio remains meaningful under the stated integrability even when a difference of two infinite divergences would be undefined. This is the relative-rate aspect of Bayesian concentration under possible model misspecification [4].

For a concrete example, take two candidate Bernoulli message laws with probabilities $3/4$ and $1/4$ of message 1, and equal source priors. If k of n messages are ones, the log posterior odds in favor of the first source are $(2k - n)\log 3$. If the true message probability is p, the normalized log odds approach $(2p - 1)\log 3$. Repetition sharpens confidence about the message law; it supplies no evidence about content distinctions the code never expresses.

17.7 What a loop prior fills in

Dictionary learning in a Gaussian tangent model can be solved exactly. Write $\bar{\theta}_e = e^{\omega_e}$ on edge e and linearize at $\omega_e = 0$. Grounded messages then depend on $\bar{T}_e\omega_e$. Stack all edge generators into ω and define the invisible space

$$N = \{\nu : \bar{T}_e\nu_e = 0 \text{ on every edge}\}. \tag{17.23}$$

Let $B\omega$ collect the sums of generators around a declared finite set of directed loops. It is the first-order loop generator. With a positive-definite anchor precision Π_0 and positive-definite loop

metric W, choose the normalized Gaussian prior

$$p_0(\omega) \propto \exp\left[-\frac{1}{2}\omega^T \Lambda_0 \omega\right], \qquad \Lambda_0 = \Pi_0 + \lambda B^T W B, \qquad \lambda \geq 0. \tag{17.24}$$

The anchor is needed to control directions the loop term does not penalize. This is a declared prior over dictionaries.

As grounded observations accumulate on every edge, with positive limiting sampling rates and the stated referent condition, they identify the affine set $\omega^\star + N$, where $\omega^\star$ is the true tangent generator. They cannot distinguish points within it. If the columns of Z form an orthonormal basis of N, define $S = Z^T \Lambda_0 Z$. The limiting mean and covariance are

$$\omega_\infty = \omega^\star - Z S^{-1} Z^T \Lambda_0 \omega^\star, \qquad \mathrm{Cov}_\infty = Z S^{-1} Z^T. \tag{17.25}$$

These are the prior mean and covariance conditioned on the identified affine set. In the observed directions, covariance falls as $1/n$. In invisible directions, uncertainty is controlled by the prior.

A one-coordinate example isolates the mechanism. Suppose the observed data fix one contribution a to a loop, while an invisible coordinate z supplies the other contribution. With isotropic anchor variance σ^2, the conditional density of z is proportional to

$$\exp\left[-\frac{z^2}{2\sigma^2} - \frac{\lambda}{2}(a+z)^2\right]. \tag{17.26}$$

Completing the square gives

$$\mathbb{E}[z \mid a] = -\frac{\lambda\sigma^2}{1+\lambda\sigma^2} a, \qquad \mathrm{Var}(z \mid a) = \frac{\sigma^2}{1+\lambda\sigma^2}. \tag{17.27}$$

If the truth is flat, so $z^\star = -a$, increasing the loop weight recovers that truth. If the actual invisible coordinate is different, the same prior still drives the estimate toward $-a$. The messages cannot decide between these cases. A recovered flat interpretation and evidence for flatness are different conclusions.

In the full tangent model, changing the invisible coordinates changes the loop value only through $B(N)$. If Q is the W-orthogonal projection onto the complement of that subspace, the message-identifiable residual is

$$\mathcal{D}(\omega) = \min_{\nu \in N} \|B(\omega+\nu)\|_W^2 = \|QB\omega\|_W^2. \tag{17.28}$$

A strong loop prior removes the component it can change within the identified affine set. The remaining residual is the part the messages force it to keep.

17.8 Reciprocity, inaccessible directions, and exact scope

For a reciprocal pair, write $\mathcal{K}_0, \mathcal{K}_1$ for the kernels of its two codes. The invisible changes in the loop have matrix columns in $\mathcal{K}_0 + \mathcal{K}_1$. Invisible changes that preserve the loop have opposite edge generators with columns in $\mathcal{K}_0 \cap \mathcal{K}_1$. With the Frobenius loop metric, the detectable residual lies in the intersection of the two code row spaces.

For example, in $K = 2$ take the two codes $(1, 0)$ and $(0, 1)$. Their kernel intersection is zero, so a flat truth has no invisible flat freedom. Their kernel sum is all of $\mathbb{R}^2$, so every loop discrepancy can nevertheless be canceled by invisible changes. The flat-truth hypothesis permits complete recovery in the strong-prior limit, while the grounded data alone cannot reject flatness. If both codes instead read only the first coordinate, opposite changes along the unread second coordinate remain inaccessible to both messages and the loop prior.

For a larger connected graph, the same question is expressed by

$$H^0 = \{u = (u_i)_i : \bar{T}_e(u_i - u_j) = 0 \text{ for every } e : j \to i\}. \tag{17.29}$$

These are compatible assignments of content vectors through the codes, the global sections of a cellular sheaf [21]. If the chosen loop chains span the signed cycle space of the underlying multigraph, the invisible flat-generator space for $\mathfrak{gl}(K, \mathbb{R})$ has dimension $K(\dim H^0 - K)$. The subtracted K is the space of constant vector assignments. Mere connectedness does not ensure the cycle-spanning hypothesis for directed loops. One must supply enough directed return paths or explicitly use signed geometric reversals, keeping them distinct from independent opposite attending edges.

Some identification statements survive without linearization. For two unrestricted $\mathrm{GL}(K, \mathbb{R})$ dictionaries whose truth is reciprocal, exact message laws and reciprocity leave

$$\bar{\theta}_{01} = \bar{\theta}^{\star}_{01} s, \qquad s \in \mathrm{Stab}(\bar{T}_1) \cap \mathrm{Stab}(\bar{T}_0 \bar{\theta}^{\star}_{01}). \tag{17.30}$$

The stacked map $(\bar{T}_1; \bar{T}_0 \bar{\theta}^{\star}_{01})$ is injective exactly when that residual set is a singleton. This is a global algebraic statement about the declared dictionary group. It does not imply a Gaussian posterior, convergence of a nonlinear learning flow, or exact agreement among the global, per-agent, and reciprocal objectives.

The remaining learning problem is to determine conditional limits and dynamics for nonlinear dictionaries, non-Gaussian priors, latent source labels, and codes beyond location families. Those extensions must keep their observable experiment explicit. The exact Gaussian completion above supplies a model of what a prior can infer within an unresolved dictionary class; it supplies no theorem of endogenous parent formation.

Chapter 18

Learning Directed Links under Three Objectives

A message can change a receiver's belief about the subject or its understanding of the speaker. These are separate variables. If one agent reports a higher temperature than another expects, the receiver may revise its temperature estimate, revise the conversion between their scales, or share the correction between the two. A comparison record alone may not distinguish those responses. The distinction becomes mathematical once the dictionary is included among the uncertain quantities.

18.1 What changes when a dictionary changes

Work at one context in reference coordinates. Agent i represents content $y_i \in \mathbb{R}^K$. On a directed edge $e : j \to i$, the sender has a fixed code $T_e : \mathbb{R}^K \to \mathbb{R}^{r_e}$ of full row rank. The receiver owns an invertible dictionary $\theta_e = e^{\omega_e}$, where ω_e is a $K \times K$ matrix. According to this dictionary, the receiver's content y_i would be expressed by the sender as $\theta_e^{-1} y_i$. A comparison record therefore reads

$$o_e = T_e(\theta_e^{-1} y_i - y_j) + \varepsilon_e. \tag{18.1}$$

The code can discard content coordinates even though the dictionary is invertible. The independently owned dictionary on the opposite attending edge need not be θ_e^{-1}.

Fix an operating content c_i and expand at the identity dictionary. Dropping terms quadratic in ω_e and $y_i - c_i$ gives

$$o_e = T_e(y_i - y_j - \kappa_i \omega_e) + \varepsilon_e, \qquad \kappa_i \omega_e = \omega_e c_i, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e), \tag{18.2}$$

with R_e positive definite. The Gaussian noise and tangent approximation make the following calculations explicit; the theory of normalized communication kernels does not require them.

Suppose $c_i \neq 0$. For arbitrary content shifts, choose

$$\delta\omega_e = (\delta y_i - \delta y_j) c_i^T / |c_i|^2. \tag{18.3}$$

Then $\kappa_i \delta\omega_e = \delta y_i - \delta y_j$, so the comparison record has exactly the same law. This is an observational ambiguity of the declared record. Other records and priors can distinguish its two explanations. At $c_i = 0$ this first-order dictionary response vanishes, which will matter in the ring example.

The ambiguity is not confined to Gaussian noise. If a record kernel depends only on $T_e(\theta_e^{-1} y_i - y_j)$, every change preserving that discrepancy preserves its law. For $K \geq 2$ and dictionaries in

$\mathrm{GL}^+(K,\mathbb{R})$, put $v_e = \theta_e^{-1} y_i + y'_j - y_j$. Whenever v_e and y'_i are nonzero, an orientation-preserving invertible map can send v_e to y'_i. Taking that map as θ'_e preserves the discrepancy. This is a pointwise statement; it does not prove a global nonlinear learning flow or its convergence.

18.2 How one record divides the adjustment

Take independent Gaussian priors for y_i, y_j, and ω_e, with means m_i, m_j, a_e and covariances Σ_i^0, Σ_j^0, V_e. Matrix coordinates of ω_e use the Frobenius inner product. The uncertainty in its action on the receiver's operating content is $C_\rho = \kappa_i V_e \kappa_i^T$. Define the record residual and its covariance by

$$\begin{aligned} r &= o_e - T_e(m_i - m_j - \kappa_i a_e), \\ S &= R_e + T_e(\Sigma_i^0 + \Sigma_j^0 + C_\rho)T_e^T. \end{aligned} \tag{18.4}$$

Gaussian conditioning multiplies each variable's covariance with the record by $S^{-1}r$. These cross-covariances have opposite signs for receiver and sender, giving

$$\begin{aligned} \delta y_i &= \Sigma_i^0 T_e^T S^{-1} r, \qquad \delta y_j = -\Sigma_j^0 T_e^T S^{-1} r, \\ \kappa_i \delta\omega_e &= -C_\rho T_e^T S^{-1} r. \end{aligned} \tag{18.5}$$

The receiver adjusts its content, the sender adjusts its content, and the dictionary changes the receiver's predicted message. The noise covariance accounts for the remaining residual, so the three adjustments need not fit the record exactly. With isotropic dictionary uncertainty $V_e = \sigma^2 I$, one has $C_\rho = \sigma^2 |c_i|^2 I$. Reinterpretation can then absorb more of the discrepancy farther from the reference origin.

In a per-agent model, the sender's law is data. Integrating y_j out of eq. (18.2) adds $T_e \Sigma_j^0 T_e^T$ to the noise covariance. The receiver and dictionary updates in eq. (18.5) remain the same for this one-record calculation, but the sender is not updated by the receiver's inference. This exact agreement of two conditional means in the isolated Gaussian example does not equate population dynamics.

Grounded messages provide information about the dictionary by pairing a message with an independently specified referent. If the referents span the content space, their unobserved dictionary directions are $N_e = \{\delta : T_e \delta = 0\}$. Give the observed directions precision gG_e with $\ker G_e = N_e$, on top of a fixed positive-definite prior precision. Let Π_e project onto $N_e^\perp$. Block inversion then gives

$$\Pi_e V_e \Pi_e = O(g^{-1}), \qquad T_e C_\rho T_e^T = T_e \kappa_i \Pi_e V_e \Pi_e \kappa_i^T T_e^T = O(g^{-1}). \tag{18.6}$$

Grounding suppresses reinterpretation through the identified coordinates. A direction satisfying only $T_e \delta c_i = 0$ is invisible to the single operating content, but it can be visible to these varied grounded referents. The two experiments must be kept distinct.

18.3 Objective derivatives and actual velocities

A population model also needs self information and a prior on dictionaries. Give each content a self precision $P_i > 0$ about m_i, each dictionary an anchor precision $s > 0$, and the dictionaries a loop

penalty. If $B\omega$ collects sums of generators around a declared finite set of simple directed loops, take the prior precision to be $sI + \lambda B^T W B$, where W is positive definite and $\lambda \geq 0$. Grounding adds its edgewise quadratic terms. Combining this proper prior with the Gaussian record likelihoods gives a proper global posterior

$$p(x \mid o) \propto \exp\{-\tfrac{1}{2}x^T J x + \eta^T x\}, \qquad x = (y, \omega), \quad J > 0. \tag{18.7}$$

The prior is a declared model of loop consistency, not evidence that conversation has measured every loop coordinate.

Group the variables into receiver blocks $x_i = (y_i, \omega_{\mathrm{in}(i)})$ and use product recognition $Q = \bigotimes_i \mathcal{N}(u_i, S_i)$. The global free energy, up to a constant, is

$$\mathcal{F}_G = \tfrac{1}{2}(u - u^*)^T J (u - u^*) + \tfrac{1}{2}\sum_i [\mathrm{tr}(J_{ii} S_i) - \log\det S_i], \qquad u^* = J^{-1}\eta. \tag{18.8}$$

Thus $S_i = J_{ii}^{-1}$ at its covariance optimum, and the factor means equal the exact posterior mean. The exact marginal covariance is larger because integrating other blocks subtracts a positive-semidefinite term from the marginal precision. Factorization has removed correlations, not made the underlying variables better measured.

The Fisher metric for a Gaussian mean is its inverse covariance. With these optimized covariances held fixed, put $D = \bigoplus_i J_{ii}$. Then

$$\dot{u} = -D^{-1} J (u - u^*). \tag{18.9}$$

The relaxation matrix is similar to the symmetric positive-definite matrix $D^{-1/2} J D^{-1/2}$. Every mode therefore decays with a real rate. Each record contributes derivatives at the receiver, the sender, and the receiver's dictionary.

The per-agent model contains received records only. In its plug-in version, peers are replaced by their current means. Its dictionary derivative equals the corresponding global derivative because the same edge likelihood, anchor, grounding, and loop factors occur in the owner's model. Its content derivative omits records for which that agent is the sender. Integrating peer laws instead gives the law-level version. For a record and a loop through receiver i, its effective covariances and precisions are

$$\widetilde{R}_e = R_e + T_e \Sigma_j T_e^T, \qquad \Omega_{i\gamma} = \left[(\lambda W_\gamma)^{-1} + \sum_{f \in \gamma \setminus \mathrm{in}(i)} V_f \right]^{-1}. \tag{18.10}$$

The loop expression assumes $\lambda > 0$ and the declared product of peer dictionary marginals; at $\lambda = 0$ the loop term is absent. This widening changes dictionary derivatives as well. With one scalar edge, $T = c = R = \Sigma_j = 1$, zero record, unit content difference, and zero dictionary mean, the global record derivative in the dictionary is -1, whereas the law-level derivative is $-1/2$.

Even the plug-in equality is an equality of covectors. If A_i is the own-block Hessian of the per-agent model, its optimized Gaussian recognition has covariance A_i^{-1}, and its velocity multiplies the derivative by that covariance. The global covariance J_{ii}^{-1} need not equal A_i^{-1}; both can mix content and dictionary components. Matching derivative rows alone therefore does not match dictionary velocities.

18.4 A ring that cannot be a gradient flow

Consider $\ell \geq 3$ scalar agents, each receiving from its successor. Let the self precision be $p > 0$, the effective record precision be $q > 0$, the dictionary anchor be $s > 0$, and the common operating content be c. Omit grounding and loop penalties in this example. The receiver's quadratic energy is

$$E_i = \tfrac{1}{2}p(y_i - m_i)^2 + \tfrac{1}{2}s\omega_i^2 + \tfrac{1}{2}q(o_i - y_i + y_{i+1} + c\omega_i)^2. \tag{18.11}$$

With the Gaussian own-block covariance at its optimum, the Hessian and the off-diagonal coupling are

$$A = \begin{pmatrix} p+q & -qc \\ -qc & s+qc^2 \end{pmatrix}, \qquad C = \begin{pmatrix} -q & 0 \\ qc & 0 \end{pmatrix}. \tag{18.12}$$

For a Fourier mode $x_i = \zeta^i v$ with $\zeta^\ell = 1$, the linear flow is $-(I + \zeta A^{-1}C)v$. The second column of $A^{-1}C$ vanishes, and its first diagonal entry is $-w$, where

$$w = \frac{qs}{ps + pqc^2 + qs} = \frac{q_{\text{eff}}}{p + q_{\text{eff}}}, \qquad q_{\text{eff}} = (q^{-1} + c^2/s)^{-1}. \tag{18.13}$$

The two rates in each Fourier sector are -1 and $-(1 - \zeta w)$. Since $0 < w < 1$, all modes decay, but nonreal roots give damped oscillation. The Jacobian of a Riemannian gradient flow at an equilibrium is self-adjoint in the metric there and has real spectrum. This ring therefore supplies a stable per-agent dynamics that cannot be made a gradient flow by choosing another metric, a familiar effect of non-reciprocal coupling [16].

The formula also quantifies reinterpretation. For $c \neq 0$, dictionary uncertainty adds effective variance c^2/s, and $w \to 0$ as $s \downarrow 0$. For $c = 0$, however, the dictionary is absent from the record and $w = q/(p+q)$ for every positive s. The weak-anchor limit is taken from proper models with $s > 0$; it does not assert normalization of a zero-anchor dictionary prior. In a strongly grounded population, the global and plug-in equations instead approach their known-dictionary limits. One retains sender and receiver derivatives and the other retains receiver derivatives only, so their limiting content estimates need not agree. Coupled law-level covariance updates require a separate limit argument.

18.5 Why an unread loop can look certain

Let one unread scalar dictionary coordinate z_e live on each of $\ell \geq 2$ edges with different owners. Give each an anchor variance σ^2 and use one factor $\exp[-\lambda(\sum_e z_e)^2/2]$, with $\lambda > 0$. The exact precision and its marginal variance are

$$J = \sigma^{-2}I + \lambda \mathbf{1}\mathbf{1}^T, \qquad v_{\text{exact}} = \sigma^2 \frac{1 + (\ell-1)\lambda\sigma^2}{1 + \ell\lambda\sigma^2}. \tag{18.14}$$

This follows by inverting the rank-one perturbation. As $\lambda \to \infty$, the sum is pinned but $\ell - 1$ independent directions remain, leaving marginal variance $\sigma^2(\ell-1)/\ell$. The constraint permits correlated fluctuations of opposite sign.

Product recognition instead gives variance $(\sigma^{-2} + \lambda)^{-1}$, which tends to zero. Its independent factors cannot carry the compensating fluctuations. In a symmetric law-level per-agent fixed point, each

receiver convolves the loop factor with $\ell - 1$ peer laws of variance v, obtaining

$$v^{-1} = \sigma^{-2} + [\lambda^{-1} + (\ell - 1)v]^{-1}. \tag{18.15}$$

Writing $\epsilon = \lambda^{-1}$ turns this into $(\ell - 1)\sigma^{-2}v^2 + (\epsilon\sigma^{-2} + 2 - \ell)v - \epsilon = 0$. Its roots have opposite signs, so it has one positive symmetric solution. For $\ell \geq 3$ that solution tends to $\sigma^2(\ell - 2)/(\ell - 1)$. For two edges it satisfies $v^2 + \epsilon v - \epsilon\sigma^2 = 0$ and hence $v \sim \sigma\lambda^{-1/2}$. Each owner has recycled information that previously reached the partner through the same constraint.

A cavity message excludes the factor to which it is being sent. There is only one factor here, so the peer cavity laws are their anchors. The resulting variance is

$$v_{\text{cavity}} = \left[\sigma^{-2} + (\lambda^{-1} + (\ell - 1)\sigma^2)^{-1}\right]^{-1} = v_{\text{exact}}. \tag{18.16}$$

This exactness is the tree property of the factor graph with one central factor and ℓ variable leaves [39]. It is consistent with a loop in the communication graph: the communication graph and factor graph are different objects. Overlapping loop factors need their own analysis. All three approximate schemes here agree on the zero mean, so a point-estimate comparison would miss their disagreement about uncertainty.

18.6 The reciprocal potential and source uncertainty

Now treat the content and dictionary laws as configuration variables of a reciprocal potential. Let $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and let the dictionary laws be independent Gaussians $\mathcal{N}(a_e, V_e)$. For edge $e : j \to i$, compare

$$m_e^{\text{rec}}(\omega_e) = \mathcal{N}(T_e(\mu_i - \kappa_i\omega_e), S_{i,e}), \qquad m_e^{\text{src}} = \mathcal{N}(T_e\mu_j, S_{j,e}), \tag{18.17}$$

where $S_{k,e} = R_e + T_e\Sigma_k T_e^T$. Add their expected divergence with weight $\beta_e > 0$ to content self divergences and the dictionary divergence from the declared prior. There is no registered comparison record in this potential.

At fixed covariances and fixed weights, its mean-dependent part is a quadratic global energy with zero comparison values and record covariances $S_{j,e}/\beta_e$. Its Gaussian mean flow consequently has real decay rates. This equality of mean quadratics does not equate the complete objectives or their covariance dynamics. Gibbs-optimized rows give the same pointwise derivative with the current weights, but moving weights do not yield the same constant-coefficient flow. A dictionary enters only its edge cost and its prior. A source content enters every edge on which it is heard, producing reciprocal source-role derivatives.

The covariance derivative exposes the distinction. Put $d_e = T_e(\mu_i - \mu_j - \kappa_i a_e)$ and $U_e = T_e\kappa_i V_e \kappa_i^T T_e^T$. The unweighted edge cost has sender-covariance derivative

$$\frac{1}{2}T_e^T S_{j,e}^{-1}[S_{j,e} - S_{i,e} - d_e d_e^T - U_e]S_{j,e}^{-1}T_e. \tag{18.18}$$

Indeed, its relevant terms are $[\operatorname{tr}(S_{j,e}^{-1}(S_{i,e} + d_e d_e^T + U_e)) + \log\det S_{j,e}]/2$. Differentiation gives eq. (18.18). The source covariance is compared with the receiver's predicted second moment about the source mean, which includes their disagreement. More generally, for an exponential-family

source q_η with statistic t, one has $\partial_\eta D_{\mathrm{KL}}(p\|q_\eta) = \mathbb{E}_{q_\eta} t - \mathbb{E}_p t$ under regularity and finite moments. Source stationarity matches those expected statistics.

There is a precise scalar conclusion. Freeze the means, the receiver variance v_k, the noise $R > 0$, and a deterministic dictionary. Let the sender have only its self term of precision p_j and one outgoing edge of unit weight. Its stationary variance satisfies

$$f(v, d^2) = p_j - v^{-1} + (v + R)^{-1} - (v_k + R + d^2)(v + R)^{-2} = 0. \tag{18.19}$$

At a nondegenerate minimum, $\partial_v f > 0$, whereas $\partial_{d^2} f = -(v + R)^{-2} < 0$. Implicit differentiation gives $dv/d(d^2) > 0$. This establishes an increasing stationary uncertainty under these frozen conditions. It does not establish a monotone covariance increase along a flow in which means, other edge costs, and source rows all move. In linear-Gaussian inference the optimizing covariance is record-independent because the precision matrix is fixed; non-Gaussian Bayesian uncertainty can depend on the observed data. The coupled Gaussian calculation below allows source selection and all uncertainties to move, while retaining fixed codes, operating contents, and proper references.

18.7 When content, interpretation, and attention all move

The expectation defining an edge cost has a general-law meaning. For a dictionary law r_e and normalized message kernels, write $C_e = \mathbb{E}_{r_e} D_{\mathrm{KL}}(A_e(\omega_e)\|B_e)$, where A_e is the receiver's conditional message law and B_e is the source's law. Changing content changes either argument of the divergence, and changing the dictionary law changes the expectation. Averaging message laws before computing their divergence would erase part of this uncertainty cost. These definitions do not require Gaussian laws; the differentiable dynamics requires regular families and justified differentiation.

Take one content sector, with singleton model-presentation spaces, and set $F_0 = \sum_i D_{\mathrm{KL}}(q_i\|p_i) + D_{\mathrm{KL}}(r\|\pi_\omega)$, where $r = \bigotimes_e r_e$ is product recognition and π_ω is a fixed proper joint dictionary prior. For each nonempty source set $\mathcal{I}_i$ choose a positive normalized prior π_i and temperature $T_i > 0$. The complete row objective is

$$\begin{aligned} \mathcal{E}_T &= F_0 + \sum_i \left[\sum_{e\in\mathcal{I}_i} \beta_e C_e + T_i D_{\mathrm{KL}}(\beta_i\|\pi_i) \right], \\ \beta_e^* &= \frac{\pi_e e^{-C_e/T_i}}{\sum_{f\in\mathcal{I}_i} \pi_f e^{-C_f/T_i}}, \qquad E_T = F_0 - \sum_i T_i \log \sum_{e\in\mathcal{I}_i} \pi_e e^{-C_e/T_i}. \end{aligned} \tag{18.20}$$

The row prior and entropy oppose concentrating all attention on the cheapest source. Substituting $\log(\beta_e^*/\pi_e) = -C_e/T_i - \log\sum_f \pi_f e^{-C_f/T_i}$ gives $\mathcal{E}_T - E_T = \sum_i T_i D_{\mathrm{KL}}(\beta_i\|\beta_i^*)$ and $dE_T = dF_0 + \sum_e \beta_e^* dC_e$. The row entropy is therefore part of the reduced scalar potential, even though its optimized derivative takes the weighted-cost form.

For the Gaussian tangent family, put $K_e = T_e\kappa_i$, $J_e = S_{j,e}^{-1}$, $d_e = T_e(\mu_i - \mu_j) - K_e a_e$, and $Q_e = S_{i,e} + d_e d_e^T + K_e V_e K_e^T$. Then

$$C_e = \tfrac{1}{2}[\mathrm{tr}(J_e Q_e) - n_e + \log\det S_{j,e} - \log\det S_{i,e}], \tag{18.21}$$

where n_e is the message dimension. The uncertainty of the dictionary contributes to the receiver's second moment through $K_e V_e K_e^T$; it does not enter the receiver covariance's log determinant because the conditional divergence was computed first.

Write the fixed joint dictionary prior as $\mathcal{N}(a_0, A_\omega^{-1})$ with $A_\omega \succ 0$, and set $b_\omega = A_\omega a_0$. It may correlate different dictionaries even though recognition factors over edges. With $\mathrm{in}(i)$ and $\mathrm{out}(i)$ denoting receiver and source roles, differentiation gives the complete forces

$$\begin{aligned} g_{\mu_i} &= P_i(\mu_i - m_i) + \sum_{e\in\mathrm{in}(i)} \beta_e T_e^T J_e d_e - \sum_{e\in\mathrm{out}(i)} \beta_e T_e^T J_e d_e, \\ g_{a_e} &= (A_\omega a - b_\omega)_e - \beta_e K_e^T J_e d_e, \\ 2G_{\Sigma_i} &= P_i - \Sigma_i^{-1} + \sum_{e\in\mathrm{in}(i)} \beta_e T_e^T (J_e - S_{i,e}^{-1}) T_e \\ &\quad + \sum_{e\in\mathrm{out}(i)} \beta_e T_e^T (J_e - J_e Q_e J_e) T_e, \\ 2G_{V_e} &= (A_\omega)_{ee} - V_e^{-1} + \beta_e K_e^T J_e K_e. \end{aligned} \tag{18.22}$$

The G symbols are symmetric trace-pairing covariance derivatives. The source term follows from $dJ_e = -J_e(dS_{j,e})J_e$; it includes receiver uncertainty, mean disagreement, and dictionary uncertainty together. The prior's full matrix acts on the dictionary means, while only its diagonal block enters a product factor's covariance derivative.

The Gaussian score metric has mean block Σ_i^{-1} and covariance quadratic form $\mathrm{tr}(\Sigma_i^{-1}\delta\Sigma_i\Sigma_i^{-1}\delta\Sigma_i)/2$. Its inverse converts the forces to velocities:

$$\begin{aligned} \dot\mu_i &= -\Sigma_i g_{\mu_i}, \quad \dot\Sigma_i = -2\Sigma_i G_{\Sigma_i}\Sigma_i, \\ \dot a_e &= -V_e g_{a_e}, \quad \dot V_e = -2V_e G_{V_e} V_e. \end{aligned} \tag{18.23}$$

This is the natural-gradient rule applied to the product of the declared Gaussian laws [1]. At full row relaxation one also evolves

$$\dot\beta_e = -\beta_e(h_e - \mathbb{E}_{\beta_i} h), \qquad h_e = C_e + T_i(\log(\beta_e/\pi_e) + 1). \tag{18.24}$$

Instantaneous Gibbs reduction instead reevaluates β^* at every state and uses eq. (18.23) with the same matter metric. Pulling back the full metric to the graph of the Gibbs row would add $\sum_e (d\beta_e^*)^2/\beta_e^*$ and give a different mobility. Finite-rate row relaxation and this reduced convention generally follow different paths.

At a stationary point every force in eq. (18.22) vanishes, and the full row is Gibbs. For example,

$$V_e^{-1} = (A_\omega)_{ee} + \beta_e K_e^T J_e K_e. \tag{18.25}$$

It is a coupled condition: source covariance determines J_e, and the row depends on costs that themselves contain V_e. Freezing either input changes the stationary problem.

18.8 Why the Gaussian flow stays in its regular domain

For a finite graph with fixed codes and operating contents, positive-definite content reference precisions and joint dictionary precision, positive-definite message noises, and positive fixed source priors and temperatures, both conventions have unique solutions for all forward time from every interior initial state. The mechanism is the proper reference cost. Contracting eq. (18.22) with eq. (18.23) gives a negative squared Fisher-gradient norm; full row relaxation adds $-\sum_i \mathrm{Var}_{\beta_i}(h)$. Every self and dictionary KL remains bounded by the initial energy, since the full row terms and the reduced terms $-T_i \log Z_i$ are nonnegative.

For a content covariance let λ_k be the eigenvalues of $P_i^{1/2}\Sigma_i P_i^{1/2}$. Its self KL is

$$\tfrac{1}{2}\|\mu_i - m_i\|_{P_i}^2 + \tfrac{1}{2}\sum_k (\lambda_k - \log\lambda_k - 1). \tag{18.26}$$

The scalar function diverges when an eigenvalue collapses to zero or grows without bound. The mean quadratic also diverges at large displacement. The joint dictionary KL applies the same argument to $a - a_0$ and $A_\omega^{1/2} V A_\omega^{1/2}$, where $V = \mathrm{diag}(V_e)$. Thus all matter coordinates stay in a compact subset of the Gaussian domain. This conclusion does not assert a monotone change of any covariance.

Costs are bounded there, say $0 \le C_e \le M$. Gibbs rows satisfy $\beta_e^* \ge \pi_e e^{-M/T_i}$. For relaxing rows, the adjusted log-ratio $\eta_{ef} = \log(\beta_e/\pi_e) - \log(\beta_f/\pi_f)$ obeys

$$\dot\eta_{ef} = -T_i \eta_{ef} - (C_e - C_f). \tag{18.27}$$

Variation of constants bounds its magnitude by $L_i = \max\{M/T_i, \max_{e,f} |\eta_{ef}(0)|\}$. Normalization gives $\beta_e \ge \pi_e e^{-L_i}$, so no row reaches the boundary. The vector field is smooth near the resulting compact trajectory closure and can therefore be continued for every forward time.

The reduced energy attains an interior minimum on a compact sublevel; its Gibbs row gives a full stationary point. Integrated dissipation is finite. On the compact trajectory closure its time derivative is bounded, so dissipation is uniformly continuous and tends to zero: sustained positive pulses would otherwise have infinite total area. Every accumulation point is consequently stationary. These conclusions require neither a unique equilibrium nor a claim of convergence to one particular point. They apply to this fixed Gaussian restriction and provide no uniform limit at zero temperature or singular reference precision.

18.9 Uncertainty can change the stability verdict

Source selection adds a feedback to the curvature of the potential. If $H_{\mathrm{fixed}} = \nabla^2 F_0 + \sum_e \beta_e^* \nabla^2 C_e$ is the Hessian with the current weights fixed, differentiating the Gibbs formula gives

$$H_{\mathrm{red}} = H_{\mathrm{fixed}} - \sum_i T_i^{-1} \mathrm{Cov}_{\beta_i^*}(\nabla C_e). \tag{18.28}$$

A perturbation that makes competing costs change differently causes attention to shift toward the decreasing cost. The shift subtracts a positive-semidefinite form from the fixed-row curvature.

This compares the same state and weights; it does not order arbitrary stationary branches by temperature.

An explicit example makes the difference measurable without freezing uncertainty. There are three scalar content laws and two dictionaries on edges $1 \to 0$ and $2 \to 0$. Take unit codes and message noises, fixed $\kappa_0 = 1$, no loop penalty, content reference precisions $(1, 5/4, 5/4)$, content reference means $(0, -23/15, 23/15)$, and dictionary references $\mathcal{N}(0, 4/3)$. Receiver 0 chooses exactly between these two peers with prior $(1/2, 1/2)$ and temperature $T > 0$. Its row has no identity self candidate. The two sources need no nontrivial rows; singleton identity self rows contribute zero.

Write $q_i = \mathcal{N}(\mu_i, v_i)$ and $r_j = \mathcal{N}(a_j, V_j)$, and order the ten matter coordinates as

$$x = (\mu_0, \mu_1, \mu_2, a_1, a_2, v_0, v_1, v_2, V_1, V_2). \tag{18.29}$$

Every mean and variance varies. With $S_i = v_i + 1$ and $d_j = \mu_0 - \mu_j - a_j$, the costs are $C_j = [(S_0 + d_j^2 + V_j)/S_j - 1 + \log S_j - \log S_0]/2$. For every temperature,

$$x_* = (0, -4/3, 4/3, 1/3, -1/3, 1, 1, 1, 1, 1), \qquad b_* = 1/2 \tag{18.30}$$

is stationary. Here b is the weight of source 1. Both costs equal $1/2$. Each source mean and dictionary mean has a self force canceled by its weighted edge force. The source variance forces cancel as $1/8 - 1/8$, the dictionary variance forces as $-1/8 + 1/8$, and receiver forces cancel or vanish.

Put $H = \nabla^2(F_0 + (C_1 + C_2)/2)(x_*)$ and $g = \nabla(C_1 - C_2)(x_*)$. The complete ten-dimensional positive-definiteness check can be written as a sum of squares. For $\xi = (u_0, u_1, u_2, b_1, b_2, h_0, h_1, h_2, k_1, k_2)$, let $z_j = u_0 - u_j - b_j$ and $\epsilon_1 = 1$, $\epsilon_2 = -1$. Differentiation of the preceding cost gives

$$\begin{aligned} \xi^T H \xi = {} & u_0^2 + \tfrac{5}{4}(u_1^2 + u_2^2) + \tfrac{3}{4}(b_1^2 + b_2^2) + \tfrac{1}{2}h_0^2 \\ & + \sum_{j=1}^{2} \left[\tfrac{1}{4}(z_j - \epsilon_j h_j/2)^2 + \tfrac{1}{16}(h_0 - h_j)^2 + \tfrac{71}{128}h_j^2 + \tfrac{1}{2}(k_j - h_j/8)^2 \right]. \end{aligned} \tag{18.31}$$

Every nonzero perturbation has positive cost in this quadratic form. With a semicolon separating the mean and variance blocks,

$$g = (1, -1/2, -1/2, -1/2, -1/2; 0, -1/4, 1/4, 1/4, -1/4)^T. \tag{18.32}$$

Solving the linear equations from eq. (18.31) gives

$$\begin{aligned} (H^{-1}g)_{\mu,a} &= (2400, -960, -960, -1600, -1600)^T/5059, \\ (H^{-1}g)_{v,V} &= (0, -716, 716, 2440, -2440)^T/5059. \end{aligned} \tag{18.33}$$

Thus $g^T H^{-1} g = 6538/5059$. At the equal-cost point eq. (18.28) is the rank-one correction $H_T = H - gg^T/(4T)$. Changing tangent coordinates by $H^{-1/2}$ leaves nine positive directions and one sign determined by $1 - g^T H^{-1} g/(4T)$. The exact critical temperature is

$$T_c = \frac{3269}{10118}. \tag{18.34}$$

The stationary Fisher Jacobian is similar to a symmetric matrix with the opposite Hessian inertia. Hence the equilibrium is exponentially stable for $T > T_c$ and has exactly one unstable direction for $0 < T < T_c$.

Full row relaxation adds the Hessian row block $4T$ and mixed block g. Its quadratic form completes to

$$\xi^T H_T \xi + 4T \left(\delta b + \frac{g^T \xi}{4T} \right)^2 . \tag{18.35}$$

It has the same threshold and one extra positive direction, although the two flows have different rates. At T_c the reduced null vector is $H^{-1}g$, while the full vector is $(H^{-1}g, -1)$. The covariance components in eq. (18.33) show that the neutral displacement includes changes of source and dictionary uncertainty.

If all five variances are fixed at one, only the first five coordinates remain. At fixed $t = g_{\mathrm{mean}}^T \xi_{\mathrm{mean}}$, exchange symmetry and convexity reduce the minimum of their quadratic form to $u_1 = u_2 = y$, $b_1 = b_2 = z$. Its minimizing values are $(u_0, y, z) = (15, -6, -10)t/31$ and its value is $61t^2/62$. Thus $g_{\mathrm{mean}}^T H_{\mathrm{mean}}^{-1} g_{\mathrm{mean}} = 62/61$, giving the lower threshold $T_{\mathrm{mean}} = 31/122$. Consequently

$$\frac{31}{122} < \frac{3}{10} < \frac{3269}{10118}. \tag{18.36}$$

At $T = 3/10$ the frozen-variance model is stable and the complete model is unstable. Allowing uncertainty to respond changes the available perturbations and the response to source selection. This is a local statement for the declared Gaussian tangent potential. The nonlinear behavior at the critical point, the eventual destinations below it, and any formation of an inherited parent remain unresolved.

Chapter 19

Curvature Costs and Supplied Hierarchies

An inconsistent set of dictionaries can make a conversation difficult to explain even when nobody measures a loop directly. The participants can adjust their contents to remove some discrepancies. The discrepancies that remain after this adjustment produce an effective cost for the dictionaries. This cost is determined by the records and the content model; it need not prefer flat links. A supplied hierarchy changes which contents can move independently, but that observation alone does not explain how a hierarchy forms.

19.1 Evidence as an effective dictionary action

Let y denote all contents, ω all dictionary variables, and o a registered record. For a content law $p(dy)$, dictionary prior $\pi(d\omega)$, and likelihood density $\ell(o \mid y, \omega)$, integration gives

$$Z(\omega) = \int \ell(o \mid y, \omega) p(dy), \qquad \pi(d\omega \mid o) \propto Z(\omega)\pi(d\omega). \tag{19.1}$$

The proportionality defines a probability law when $\int Z(\omega)\pi(d\omega)$ is finite and positive. Thus $-\log Z$ is the action induced by the contents wherever Z is finite and positive. This statement holds for general probability laws. The following Gaussian calculation makes its geometric content explicit.

Take a finite directed graph. Agent i has reference content $y_i \in \mathbb{R}^K$ and self law $\mathcal{N}(m_i, P_i^{-1})$, with positive-definite precision P_i. On an edge $e : j \to i$, the receiver has a matrix dictionary generator ω_e, a code $\bar{T}_e : \mathbb{R}^K \to H_e$, and independent Gaussian record noise of covariance $R_e > 0$. Fix an operating content c_i. The tangent comparison record is

$$o_e = \bar{T}_e(y_i - y_j - \omega_e c_i) + \varepsilon_e, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e). \tag{19.2}$$

Here $\omega_e c_i$ is the reinterpretation of one fixed operating content. The model drops products of ω_e with deviations of y_i from c_i. Give each matrix an independent isotropic Gaussian anchor of variance σ^2, using a declared reference inner product.

Collect the contents and records into vectors, and define $(\delta y)_e = \bar{T}_e(y_i - y_j)$, $(\mathcal{V}\omega)_e = \bar{T}_e\omega_e c_i$, $P = \bigoplus_i P_i$, and $R = \bigoplus_e R_e$. The self law makes $\delta y + \varepsilon$ Gaussian with mean δm and covariance $C_P = R + \delta P^{-1}\delta^T$. Consequently the exact marginal action in this tangent model is

$$S_{\mathrm{marg}}(\omega) = \frac{\|\omega\|^2}{2\sigma^2} + \frac{1}{2}\|o + \mathcal{V}\omega - \delta m\|^2_{C_P^{-1}} + \mathrm{const}, \tag{19.3}$$

where $\|v\|_A^2 = v^T A v$. This is Gaussian elimination, equivalently a Schur complement of the joint precision. The extra dictionary precision is $\mathcal{V}^T C_P^{-1} \mathcal{V}$: dictionaries become statistically coupled because they share uncertain contents.

The entire affine residual matters. Nonzero records can be better explained by nonzero dictionary loop sums. A positive quadratic penalty for perturbing dictionaries therefore does not establish a flat preferred state. In this tangent model C_P is independent of ω, so its determinant adds no dictionary force. Nonlinear models can have dictionary-dependent content precision and a corresponding log-determinant contribution; its effect requires its own calculation.

19.2 The part that changing contents cannot explain

Weaken every self precision by setting $P = \epsilon \bar{P}$ and taking $\epsilon \downarrow 0$. Contents can then move almost freely. For a coded reinterpretation $v = \mathcal{V}\omega$, its remaining quadratic cost is

$$v^T \Pi_H v = \min_a \|v - \delta a\|_{R^{-1}}^2, \qquad \Pi_H = R^{-1} - R^{-1}\delta(\delta^T R^{-1} \delta)^+ \delta^T R^{-1}. \tag{19.4}$$

The plus sign denotes the Moore–Penrose inverse. The vector a is a proposed change of everyone's contents. Its coded effect is δa. Weighted least squares removes the part of v that such a change can explain and retains the orthogonal residual.

This residual has a cohomological description. The vertex spaces are $\mathbb{R}^K$, the edge spaces are H_e, and the two maps into each edge space are its code. These data form a cellular sheaf on the graph [21]. Since the graph has no two-cells, its first cohomology is the quotient of all coded edge values by $\operatorname{im} \delta$. Equation eq. (19.4) gives the quotient norm. The term says exactly which reinterpretations cannot be absorbed by changes of contents; the terminology adds no further dynamical claim.

For injective codes and common operating content $c \neq 0$, zero residual means $\omega_e c = a_i - a_j$ on every edge. Such differences sum to zero around every signed cycle of the underlying undirected multigraph. They can be tested using directed closed walks only if the chains of those walks span that cycle space. Strong connectivity is sufficient when all directed closed walks are allowed. Opposite attending edges remain distinct variables; the graph convention does not replace one by the inverse of the other.

A directed triangle without a return route shows why this hypothesis is needed. Take edges $1 \to 2$, $2 \to 3$, and $1 \to 3$, scalar unit codes and noise, and $v = (0, 0, 1)$. There are no directed closed walks. Nevertheless,

$$\delta a = (a_2 - a_1, a_3 - a_2, a_3 - a_1), \qquad \min_a \|v - \delta a\|^2 = \frac{1}{3}. \tag{19.5}$$

The residual direction is $(1, 1, -1)$, and v has squared projection $1/3$ onto it. The two directed routes to vertex 3 disagree, even though this graph has no directed loop. At nonzero records the same projection applies to $o + \mathcal{V}\omega$, so the records can favor a nonzero residual class of $\mathcal{V}\omega$.

19.3 Invisibility belongs to an experiment

The tangent record at one operating content misses a matrix perturbation A when $\bar{T}Ac = 0$. Grounded messages that sample referents with positive-definite second moment instead test $\bar{T}A$. For instance, with $\bar{T} = I_2$, $c = (1, 0)^T$, and $A = E_{22}$, the tangent record is unchanged although grounded referents can identify the perturbation. An unpenalized direction in one conversation need not be inaccessible to every experiment.

Law comparisons and link defects are also different measurements. A quarter-turn H preserves the centered isotropic Gaussian $q = \mathcal{N}(0, I_2)$, so $D_{\mathrm{KL}}(q\|H_{\#}q) = 0$. The covariance-dressed defect

$$Q(H; \Sigma) = \frac{1}{2}\|\Sigma^{-1/2}(I - H)\Sigma^{1/2}\|_F^2 \tag{19.6}$$

equals 2 for that rotation with $\Sigma = I_2$. For any positive-definite Σ, it vanishes only at $H = I$. The invertible similarity transform preserves whether $I - H$ is zero, regardless of the stabilizer of the compared law.

A loop may be probed by carrying one sample around it and comparing the return with the original. It may also be inferred by identifying its constituent dictionaries from grounded edge records. The induced action in eq. (19.1) requires neither procedure. It comes from the likelihood of the records already declared in the joint model.

19.4 One topic gives a rank-one stationary reinterpretation

In the Gaussian tangent model, suppose there is no grounded term and no explicit loop prior. With $\Delta_e = y_i - y_j$, differentiating the record cost and the isotropic anchor gives

$$\omega_e = \sigma^2 \bar{T}_e^T (R_e + \sigma^2 |c_i|^2 \bar{T}_e \bar{T}_e^T)^{-1} (\bar{T}_e \Delta_e - o_e) c_i^T. \tag{19.7}$$

The adjoint of the map $\omega \mapsto \omega c_i$ sends v to $v c_i^T$, which explains the final factor and the rank at most one. The matrix is optimized only through its action on one vector; the anchor sets its unused part to zero. This is an equation for a stationary dictionary mean, not a restriction on every sample of its recognition law.

For zero records define $F_e = \sigma^2 |c_i|^2 \bar{T}_e^T (R_e + \sigma^2 |c_i|^2 \bar{T}_e \bar{T}_e^T)^{-1} \bar{T}_e$. Then $\omega_e c_i = F_e \Delta_e$. If every operating content is the same nonzero c, a directed closed walk obeys

$$\left(\sum_{e \in \gamma} \omega_e \right) c = \sum_{e \in \gamma} F_e \Delta_e = \sum_{e \in \gamma} (F_e - \bar{F}) \Delta_e. \tag{19.8}$$

The last equality uses the telescoping sum of the content differences. Equal gains cancel the linearized loop sum. Unequal gains and persistent disagreement permit a residual, but do not guarantee one. The cancellation concerns the first-order sum of generators; it does not prove that the ordered product of exponentials is identity.

For plug-in per-agent models the same local equation uses peer means as data. A law-level per-agent model widens R_e by the sender's coded covariance. At fixed covariances the reciprocal

potential has another effective weight and registers agreement. These substitutions change the gains, so owner-local dependence is not equality of all dictionary forces. A spread content law can probe several independent directions, and the rank-one formula need not survive beyond the fixed-topic Gaussian model.

19.5 What a supplied parent inherits statically

Supply a partition into clusters and a parent content y_A for each cluster. A cross-scale Gaussian factor of precision τ penalizes $y_i - y_A - \omega_{iA}c_A$. If its dictionary is free with anchor variance σ^2, integration leaves child-parent precision

$$(\tau^{-1} + \sigma^2|c_A|^2)^{-1}I. \tag{19.9}$$

For $c_A \neq 0$ this remains bounded even as $\tau \to \infty$: the child can reinterpret its parent. For $c_A = 0$ the precision is τI. Pinning the dictionary at identity removes this reinterpretation.

Now impose rigidity, $y_i = y_A$, with internal and cross-scale dictionaries fixed at identity. For general self densities relative to one common measure μ, the proposed parent self density is

$$p'_A(z) = \frac{p_A(z)\prod_{i\in I} p_i(z)}{Z_A}, \qquad Z_A = \int p_A(z)\prod_{i\in I} p_i(z)\mu(dz). \tag{19.10}$$

This requires $0 < Z_A < \infty$, finite positive internal record factors at zero discrepancy, and a finite positive normalizer after including the remaining bridges. Individual normalization is insufficient: two densities $\frac{1}{3}z^{-2/3}$ on $(0,1)$ have a nonintegrable product. The construction defines a density on one parent coordinate per cluster using the chosen measure; it is not automatically the conditional law on a probability-zero diagonal.

For Gaussian self laws, completing the square gives

$$P'_A = P_A + \sum_{i\in I} P_i, \qquad P'_A m'_A = P_A m_A + \sum_{i\in I} P_i m_i. \tag{19.11}$$

Every bridge keeps its code, record, dictionary, and operating vector. The parent pools self information while retaining separate boundary factors. With pinned internal and cross-scale dictionaries, the Gaussian energy is a positive quadratic plus $\frac{\tau}{2}\sum_i |y_i - y_A|^2$. Splitting variables into equal-content and transverse directions shows that the transverse minimizer is $O(\tau^{-1})$, so the global minimizer tends to the rigid one. This is a Gaussian minimizer limit, not a convergence theorem for normalized general-law hierarchies.

19.6 When a wall carries one dictionary

Let two rigid clusters have identity internal dictionaries and bridges in both directions. Require internal directed paths joining the endpoints of every opposite bridge pair; strong connectivity within each cluster suffices. A walk entering through $e : J \to I$ and returning through $g : I \to J$ has holonomy $\bar{\theta}_g\bar{\theta}_e$. Flatness for every such pair forces all forward bridges to share one dictionary

$\bar{\theta}_D$ and all backward bridges its inverse. The proof fixes one bridge in each direction and solves $\bar{\theta}_g\bar{\theta}_e = I$ for all pairs.

The path requirement has content. Two disconnected reciprocal pairs can carry scalar dictionaries $1, 1$ and $2, 1/2$. Every available directed loop is flat, although there is no common forward dictionary. Parent-to-child edges alone do not create the missing return paths. Where the requirement holds, the common boundary dictionary is read through the stacked bridge codes; full grounded identification still depends on those codes and the referents.

A supplied rigid restriction can be repeated on a supplied tree when its normalizers exist. That is an algebraic construction. Formation additionally requires state-dependent selection and persistence, inherited laws and complete response and update kernels for a predeclared probe class, compression, a change of membership relative to the null flow, and a second selection that depends on the first handoff. The curvature and rigid-parent calculations leave those obligations open.

Appendix A

Notation in Physical Terms

Symbol	Meaning and role
k, m	A represented state possibility and a model presentation. Neither symbol by itself denotes a probability law.
q_i, s_i	The agent's belief about represented contents (its perception) and its belief over generative models. They can evolve in coupled equations.
ζ_i	The joint law of represented state and model contents; $q_i \otimes s_i$ is a declared product restriction.
g^F, ℓ_v	Fisher distinguishability on a regular statistical family and the score describing one infinitesimal change.
$D^\omega\zeta, h^\omega_\zeta$	Statistical change after contextual transport is accounted for, and the resulting tensor on the context base.
$A^F, \mathfrak{k}$	The collectively minimizing connection class and the metric measuring common infinitesimal frame motions.
$G_i(m, dk)$	An evaluator assigning state predictions to each model presentation. Averaging gives $s_i G_i$.
r_i, ℓ_i, P_i	Prior over presentations, record likelihood, and the normalized generative model $r_i G_i \ell_i$. A model law predicts records only through G_i and ℓ_i.
Species	Agents that share a generative model up to a change of frame. They may still hold different beliefs.
Three objectives	A global free energy of one normalized joint over all agents; per-agent free energies in which peers' laws are data; the reciprocal law potential, which moves sources toward their receivers and is not a free energy of any fixed model with the agent laws as recognition laws.
$\mathcal{F}_i$	Free energy of agent i's own generative model. In the per-agent picture its own evaluator and its peers' channelled laws compete as sources of prior.
$x \in \mathsf{S}$	A complete configuration of agents, including their laws and the memory and interfaces read by the dynamics.
$\rho_i : \mathsf{S}_i \to \mathsf{O}_i$	A readout of an internal distinction. Its value is a record.
Γ_{ij}	A normalized interaction transition on complete configurations. It induces record probabilities through the readout.
U_i^a	An invertible frame map for sector $a = q, s$. Changing frame changes the description.
T_j^a, O_i^a	Sending and receiving interfaces. Their product can discard directions.
L_{ij}^a, Θ_{ij}^a	A reference generator and its framed exponential interaction. The generator may be singular while its exponential is invertible.
β_{ij}, γ_{ij}	Belief-sector and model-sector source weights in normalized rows.
$\mathcal{E}$	The reciprocal law potential on agent laws. Its gradient dynamics requires a metric, and every edge pushes on both its receiver and its source.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model and conditioned record.
$Q_x C$	A law of compressed represented contents. It does not specify a parent update.
$R(x), F^c$	A complete retained state and its inherited update. They must satisfy $R \circ F = F^c \circ R$, with $\circ$ composing maps in function order.
Φ^0	The null flow: the same construction with every peer cost deleted. It is the control experiment for selection.

Symbol	Meaning and role
P, P^0	Partition sequences selected along the flow and along the null flow, without event times.
(F5)	Interaction necessity: $P \neq P^0$ on an open set of admitted initial data. A shift of event times alone does not count.
Two-stage selection	Short for two-stage selection with exact inheritance: selection, inheritance, and a second selection among parents, with at least one of compression, interaction necessity, or counterfactual dependence of the second stage failing. It is not a formation, and the failed clauses are stated.
$W_2(\nu, \mathsf{P})$	A Wishart law of positive-definite matrix contents, with real shape $\nu > 1/2$ and precision P. It is not a Gaussian closure.
$\alpha_i, \eta_i; A_I, B_I, n_I$	Individual belief and model shapes, their block sums, and the number of primitive children retained by a parent.
B^q_{IJ}, B^s_{IJ}	Native source masses between current blocks, averaged over receivers with primitive-count weights.
H_I, ϕ_I	Relative constant angular offsets discarded by a block, and its retained local mean phase at the contextual anchor.
$\mathcal{C}, B, F$	The context base, a connection one-form, and its curvature. These require a rule for relating neighboring contexts. The symbol F also denotes a discrete update in the inheritance chapter.

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